

Lecture Notes on Modern QFT

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Lecture notes toward the quantum field theory textbook of ~ 2046 : the book that will exist once the developments of 2004–2026 — generalized and non-invertible symmetries, the bootstrap renaissance, monotonicity theorems, the algebraic/entanglement revival, the lattice–continuum program, modern amplitudes — are digested into pedagogy the way Wilson’s revolution was digested into the textbooks of the 1990s. The notes open with an annotated table of contents (twelve parts, sixty-two chapters): each chapter comes with a paragraph explaining why it exists and pointers into the current literature (Seiberg, Witten, Shao, Córdova, Schäfer-Nameki, Rychkov, Simmons-Duffin, Komargodski, Gaiotto, Rastelli, Yin, Polyakov’s heirs all). Chapters are then being written one at a time, starting with Part I. Hyperlinked cited papers are collected in **assets/**, indexed by arXiv number; see **assets/README.md**.

Written entirely by Anthropic’s Fable 5 model on xhigh-mode.

To view most efficiently, I’d recommend using a PDF viewer and clicking through the sidebar table of contents, e.g. if you’re on Apple then Preview, or Adobe’s PDF viewer if you’re on Linux¹.

¹Or... *holds in vomit* ... Windows.

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The premise

Every generation's textbook teaches quantum field theory as that generation understood it. The canon of the 1990s–2000s (Peskin–Schroeder, Weinberg) presents QFT as a procedure: start from a classical Lagrangian, quantize, renormalize, expand around free fields. This is how one *computes* in a QFT that happens to be weakly coupled somewhere — but twenty years of work have made it untenable as a statement of what a QFT *is*. There are interacting theories with no known Lagrangian (the $(2,0)$ theory, most of class $\mathcal{S}$); single theories with several inequivalent Lagrangians (the 2+1d duality web); symmetries invisible in any Lagrangian (non-invertible defects); and lattice models whose continuum limits violate the lore the Lagrangian paradigm is built on (fractons). Meanwhile the most precise determination of the 3d Ising exponents comes not from any Lagrangian manipulation but from unitarity and crossing alone.

The premise of these notes: imagine the standard graduate text of c. 2046, written after this reorganization is complete. What follows is its table of contents — twelve parts, sixty-two chapters — with annotations explaining what each chapter contains, why it sits where it sits, and where to learn the material *today*. It is, deliberately, a reading map of the 2012–2026 formal-QFT literature disguised as front matter. For the prehistory and spirit of this point of view, see Seiberg's essay [Thoughts About Quantum Field Theory](#) and Rychkov's recent history of the bootstrap, [2509.02779](#). A note on references: hyperlinked arXiv numbers like [2308.00747](#) refer to papers downloaded into `assets/`; plain typewriter numbers like 2307.07547 are further reading left on the arXiv.

Ten organizing principles

The book is organized around ten lessons, each of which would have read as heterodox in 2000 and reads as obvious in 2046:

1. **A theory is its observables, not its Lagrangian.** Lagrangians, lattices, bootstrap data, holographic duals are *presentations*; the theory is the abstract structure they present. Some theories have many presentations (dualities), some have none we know.
2. **Defects are part of the definition.** The operator content of a QFT includes extended operators of every codimension — lines, surfaces, interfaces, boundaries — and theories with identical local correlators can differ in their extended sectors.
3. **The arena is theory space.** The renormalization group is its dynamics, conformal field theories are its landmarks, and effective field theory is the statement that physics is locally well-approximated by coordinates around a fixed point.
4. **Symmetry = topological operators.** This single shift unifies ordinary symmetries, higher-form symmetries, higher groups, and non-invertible symmetries, and makes symmetry a property of the theory rather than of its presentation. Symmetries form (higher) categories; gauging is an operation on theory space.
5. **Anomalies are the most robust observables in physics.** 't Hooft anomalies are exactly RG-invariant, are classified by topological field theories in one higher dimension, and constrain dynamics at any coupling.
6. **Consistency alone computes.** Unitarity, crossing, and symmetry carve out the possible: the conformal bootstrap solves strongly-coupled CFTs to high precision, and dispersive positivity bounds delineate which EFTs can ever be UV-completed.

7. **Duality is the rule, not the exception.** The same point of theory space typically admits several weakly-coupled descriptions valid in different regimes, related by gauging operations.
8. **Quantization is not fundamental.** Since there are QFTs that are not quantizations of anything classical, any acceptable definition of the subject must not start from a classical theory.
9. **The continuum is subtle.** Lattice symmetries *emanate* into internal continuum symmetries; Lieb–Schultz–Mattis theorems are lattice ’t Hooft anomalies; fracton models have continuum limits with UV/IR mixing; ϕ^4 in $d = 4$ is rigorously trivial. “Take a continuum limit” is a theorem in some cases, a conjecture in most, and false in interesting ones.
10. **QFT is quantum information organized by geometry.** Local algebras are Type III, entanglement proves monotonicity theorems and energy conditions, and gravity — QFT’s largest customer via holography — repays the debt by reorganizing its algebras.

How the parts fit together

Parts I–IV are the spine: the arena (I), the exact structures on it (II), its landmarks (III), and its dynamics (IV). Parts V–VIII are dynamics in the flesh: gauge theories (V), duality (VI), the exactly solvable corners that anchor everything (VII), and the lattice–continuum dictionary where the definition of the subject is stress-tested (VIII). Parts IX–XI are interfaces with the S-matrix, quantum information, and gravity. Part XII is the frontier. A reader of 2026 wanting one path: I $\rightarrow$ II.1–II.5 $\rightarrow$ III.1–III.2 $\rightarrow$ IV.1 $\rightarrow$ V $\rightarrow$ VIII, with VII and X read in parallel. (On a first pass, chapters II.6–II.8 — the SymTFT, anomalies in parameter space, the generalized Landau paradigm — may be deferred until Parts V and VI call on them.)

Three one-semester courses are embedded in the book: *symmetry-first QFT* (Parts I, II, V, VIII — the Seiberg school); *the bootstrap* (Parts I, III, IV, IX — the Polyakov program executed); *fields, information, and gravity* (Parts I, X, XI). Three lighthouse theories recur in every part and should be solved completely, once, early: the 2d Ising model with its Majorana chain (symmetry, duality, the lattice), the 3d Ising CFT (bootstrap, defects, fuzzy sphere), and Maxwell/Yang–Mills in 4d (higher-form symmetry, θ , lines). When these notes graduate from table of contents to text, the writing should start where today’s textbooks are thinnest: Parts II and VIII exist only as lecture notes and research papers.

Part I. The Arena: Quantum Field Theory Without Lagrangians

What is being defined before any method of computing it: observables, states, and the space of theories.

I.1 What is a quantum field theory?

The question, taken seriously. Why the textbook answer of 2000 (“quantize a Lagrangian”) fails to cover the known examples; the inventory of things a QFT assigns (correlators, Hilbert spaces, defects, partition functions); QFT as the universal description of many degrees of freedom with locality. A first look at the picture that organizes the whole book: theories as points, RG as flow, fixed points as landmarks.

Sections: the standard model of textbooks and its discontents; what we must assign and to what; locality as the one non-negotiable; theory space as the picture to keep.

Key reading: Dedushenko [2203.08053](#) (survey of every known definition-attempt); Rychkov [2509.02779](#) (how we got here); Seiberg, *Thoughts About QFT* (IAS essay).

I.2 Observables I: local operators

The local operator content as primary data. Correlation functions, the operator product expansion as an exact (convergent, in CFT) structure rather than an asymptotic trick; Euclidean vs. Lorentzian, Wightman functions vs. Schwinger functions, reflection positivity; the stress tensor and currents as the operators every local QFT shares.

Sections: correlators and their analytic structure; OPE; reconstruction theorems; the stress tensor; conserved currents and their products.

Key reading: Rychkov [1601.05000](#) (chs. 1–2); Pappadopulo–Rychkov–Espin–Rattazzi [1208.6449](#) (the OPE convergence rate, proved); Kontsevich–Segal [2105.10161](#) (§2, for which Euclidean backgrounds are allowed).

I.3 Observables II: defects and extended operators

The lesson of 2014–2026: the definition of a theory includes its extended operators. Order vs. disorder operators; Wilson and ’t Hooft lines; surface operators; interfaces and boundaries; defects as probes (one-point functions of bulk operators, defect-localized degrees of freedom) and as charged objects for the symmetries of Part II.

Sections: codimension as organizing principle; genuine vs. non-genuine (attached) operators; defect Hilbert spaces; line operators as phase diagnostics; boundaries as the richest defects.

Key reading: Northe [2411.03381](#) (pedagogical BCFT/DCFT); Aharony–Seiberg–Tachikawa [1305.0318](#) (why the line spectrum is part of the definition).

I.4 States, geometry, and cutting-and-gluing

The Hilbert space is assigned to a spatial slice, and the path integral is the gluing rule: the functorial (Atiyah–Segal) skeleton of QFT, stated physically. Radial quantization and the state–operator map as the CFT incarnation. Which complex metrics are allowed (Kontsevich–Segal–Witten) — the modern answer to “what does Wick rotation mean?”

Sections: Hilbert spaces from slices; cutting and gluing; the cylinder vs. the sphere; state–operator correspondence; allowable complex backgrounds; thermal states and KMS.

Key reading: Kontsevich–Segal [2105.10161](#); Dedushenko [2203.08053](#) (§§on functorial QFT).

I.5 Presentations, not definitions

One theory, many descriptions. The same point in theory space presented as: a Lagrangian flowed from a free fixed point; a lattice model at criticality; abstract CFT data $\{\Delta_i, c_{ijk}\}$; an integrable S-matrix; a holographic bulk. Worked example threaded through the book: the 3d Ising CFT as ϕ^4 , as the lattice model, as a bootstrap island, as the fuzzy-sphere spectrum.

Sections: Lagrangians as charts; lattice presentations; CFT data; S-matrices; holographic presentations; when presentations disagree about what is elementary.

Key reading: Poland–Rychkov–Vichi [1805.04405](#) (§1, the data-centric view); Zhu et al. [2210.13482](#) (the same CFT from a Landau level).

I.6 Theory space and the renormalization group

Wilson’s picture promoted to the book’s geography. RG flow on the space of theories; fixed points, relevant/irrelevant/marginal directions as the tangent space; beta functions, Callan–Symanzik equations, and operator mixing as coordinate expressions of that geometry; universality as the statement that flows converge; effective field theory as local coordinates; conformal manifolds; lines of fixed points; the questions the rest of the book answers (what is exactly transported along flows? — Part II; what are the fixed points? — Part III; which flows are allowed? — Part IV).

Sections: Wilsonian RG without a cutoff fetish; scheme-independence and what is physical; beta functions and Callan–Symanzik; local RG and operator mixing; EFT; universality classes; conformal manifolds and exactly marginal deformations.

Key reading: Rychkov [1601.05000](#) (ch. 1); Casini–Huerta [2201.13310](#) (§1 for the entropic restatement, used in Part X).

Part II. Symmetry

The single largest reorganization since 2000: symmetry as topology, anomalies as exact data. This part is the technology the rest of the book runs on.

II.1 Symmetries are topological operators

Noether’s theorem, run backwards: a symmetry *is* a topological codimension-one operator $U_g(\Sigma)$, defined with no reference to a Lagrangian. Ward identities from deforming surfaces; action on local and extended operators; spontaneous breaking restated operatorially; coupling to background gauge fields as the universal bookkeeping device.

Sections: from currents to topological surfaces; Ward identities topologically; symmetry defects and twisted sectors; SSB without order parameters; background fields.

Key reading: Gaiotto et al. [1412.5148](#) (§1–2); Shao [2308.00747](#) (ch. 1); Iqbal [2407.20815](#) (gentlest entry).

II.2 Topological field theory: the minimal examples

The auxiliary device the modern subject runs on, built honestly before it is everywhere assumed: theories with no local degrees of freedom, whose observables depend only on topology. Finite-group gauge theory (Dijkgraaf–Witten) done completely on a lattice — finite-dimensional Hilbert

spaces, partition functions that count; $\mathbb{Z}_N$ BF theory as its continuum presentation; abelian Chern–Simons and ground-state degeneracy on the torus; spin TQFTs and invertible theories, the carriers of anomaly (II.3) and SPT order. The point is not topology for its own sake: TQFTs are the bookkeeping for anomaly inflow, for the SymTFT (II.6), and for gapped phases (II.8), and the reader who can compute in $\mathbb{Z}_N$ gauge theory owns most of the toolkit.

Sections: what “topological” means and buys; finite-group gauge theory on the lattice; $\mathbb{Z}_N$ BF theory; abelian Chern–Simons; spin TQFTs; ground-state degeneracy as an observable; invertible theories and SPT phases.

Key reading: Dunne [hep-th/9902115](#) (Chern–Simons, thorough); Shao [2308.00747](#) ($\mathbb{Z}_N$ gauge theory worked in detail).

Further: Dijkgraaf–Witten, *Commun. Math. Phys.* **129** (1990) 393; Moore–Segal ([hep-th/0609042](#)) for 2d TQFT; Freed ([1406.7278](#)) for invertible theories.

II.3 Gauging and ’t Hooft anomalies

Gauging as summing over backgrounds; ’t Hooft anomalies as the obstructions to doing so, and anomaly *matching* as the resulting exact RG invariant — the sharpest nonperturbative tool in the subject. Anomaly inflow: anomalies of a d -dimensional theory are classified by the invertible/SPT phases of chapter II.2 in $d + 1$ dimensions. Perturbative chiral anomalies, global anomalies, parity and time-reversal anomalies, mixed gauge-gravitational anomalies, and the spin/Pin structures needed to state fermionic theories all appear in one framework.

Sections: gauging finite and continuous symmetries; ’t Hooft’s argument; inflow and the bulk-boundary picture; fermion parity and spin/Pin backgrounds; SPT phases and the cobordism classification (sketch); discrete, global, and mixed gravitational anomalies; examples from QCD to the honeycomb lattice.

Key reading: Shao [2308.00747](#); Schäfer-Nameki [2305.18296](#); Córdova et al. [2205.09545](#) (panorama with full bibliography).

Further: Witten–Yonekura ([1909.08775](#)) for η -invariants and the deepest version of inflow; Delmastro–Gaiotto–Gomis ([2101.02218](#)) for global anomalies on Hilbert spaces and spin TQFT data.

II.4 Higher-form symmetries and the global structure of gauge theories

Symmetries whose charged objects are extended: one-form symmetries acting on lines, etc. Maxwell theory’s electric and magnetic one-form symmetries; the center symmetry of Yang–Mills, with confinement reborn as unbroken one-form symmetry (the Landau paradigm regained for gauge theories); $SU(N)$ vs. $SU(N)/\mathbb{Z}_N$ and discrete θ -parameters — the global structure that correlators of local operators cannot see; two-groups and symmetry fusion across dimensions.

Sections: q -form symmetries; charged defects; selection rules for extended operators; the center of Yang–Mills; reading between the lines; 2-group symmetry.

Key reading: Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#); Aharony–Seiberg–Tachikawa [1305.0318](#); Iqbal [2407.20815](#) (applications incl. magnetohydrodynamics).

II.5 Non-invertible symmetries

Topological operators without inverses: fusion replaces composition, categories replace groups. The Kramers–Wannier defect of the 2d Ising model as the founding example; topological defect lines in 2d CFT; duality defects in 3+1d gauge theories at self-dual couplings; the chiral symmetry of

massless QED resurrected as a non-invertible symmetry (the ABJ anomaly tamed); selection rules and the meaning of “symmetry” once unitarity of the action is abandoned.

Sections: fusion categories in physics; Kramers–Wannier; Verlinde lines; half-gauging and duality defects in 4d; non-invertible chiral symmetry; what non-invertible symmetries forbid.

Key reading: Shao [2308.00747](#) (the standard course); Choi–Córdova–Hsin–Lam–Shao [2111.01139](#) (4d duality defects).

Further: Kaidi–Ohmori–Zheng ([2111.01141](#)); Córdova–Ohmori ([2205.06243](#)) and Choi et al. ([2205.05086](#)) for the Standard Model; Bhardwaj et al. ([2307.07547](#)) and Brennan–Hong ([2306.00912](#)) as alternative lecture notes.

II.6 The Symmetry Topological Field Theory

The organizing gadget of the 2020s: the symmetry content of a d -dimensional theory encoded in a $(d+1)$ -dimensional topological theory with two boundaries — one topological (the symmetry and its realization), one dynamical (the physics). Gauging, discrete torsion, and duality webs as changes of topological boundary condition; anomalies as obstructions visible in the bulk; the kinematics/dynamics separation made into a picture.

Sections: the sandwich; topological boundary conditions; gauging as boundary change; anomalies in the bulk; reading phase diagrams from the SymTFT.

Key reading: Schäfer-Nameki [2305.18296](#) (§SymTFT); Kapustin–Seiberg [1401.0740](#) (the seed: coupling a QFT to a TQFT).

Further: Freed–Moore–Teleman ([2209.07471](#)) for the mathematical formulation.

II.7 Anomalies in parameter space

Anomalies generalized from symmetries to families: topological terms for *background* scalars — coupling constants — and the exact statements they yield. The θ -angle as a circle-valued coupling with an anomaly in its periodicity; consequences: level crossings that cannot be avoided, walls that must carry degrees of freedom, constraints on axion physics.

Sections: families of theories as one theory with background scalars; anomalies valued in one higher dimension of parameter space; θ -periodicity; axions; defects from winding couplings.

Key reading: Córdova–Freed–Lam–Seiberg [1905.09315](#).

II.8 The generalized Landau paradigm

Phases of matter re-founded on the full symmetry category. Gapped phases as (symmetry-enriched) TQFTs; spontaneous breaking of higher-form and non-invertible symmetries; topological order as ordinary SSB of one-form symmetry; anomalies forbidding trivially gapped phases (symmetry-protected criticality); the condensed-matter dictionary.

Sections: what replaces the order parameter; gapped phases with categorical symmetry; deconfined criticality (preview); symmetric mass generation; the dictionary with condensed matter.

Key reading: McGreevy [2204.03045](#); Córdova–Dumitrescu–Intriligator–Shao [2205.09545](#).

Part III. Fixed Points: Conformal Field Theory

The landmarks of theory space, studied as intrinsic theories with no microscopic crutch. Polyakov's 1974 program, finally executed.

III.1 Conformal kinematics in $d \geq 3$

The conformal algebra and its unitary representations; primaries and descendants; radial quantization and the state-operator map (the cleanest argument for the OPE's convergence); unitarity bounds; correlators fixed by symmetry up to functions of cross-ratios; embedding space. Also settled here: scale vs. conformal invariance — when scale invariance implies the full conformal group ($d = 2$: theorem; $d = 4$: perturbative theorem; general d : open), and why the distinction matters for the fixed-point picture.

Sections: conformal algebra and representations; radial quantization; unitarity bounds; two- and three-point functions; conformal blocks; the convergent OPE; scale vs. conformal invariance.

Key reading: Rychkov [1601.05000](#); Simmons-Duffin [1602.07982](#) (the standard modern courses).

Further: Nakayama ([1302.0884](#)) for the scale-vs.-conformal question.

III.2 The conformal bootstrap

Crossing symmetry plus unitarity as a computational engine. The numerical method (positivity, linear/semidefinite programming) and its conceptual meaning: rigorous exclusion bounds on theory space, no Lagrangian anywhere. The 3d Ising island as the flagship: critical exponents to six digits from consistency alone. The frontier: navigator methods, larger systems of correlators, the stress tensor.

Sections: crossing equations; the positivity trick; islands and kinks; 3d Ising and $O(N)$; rigor of numerics; current frontiers.

Key reading: Simmons-Duffin [1602.07982](#); Poland–Rychkov–Vichi [1805.04405](#); Rychkov–Su [2311.15844](#) (state of the art); Rychkov [2509.02779](#) (the story).

Further: El-Showk et al. [1203.6064](#), the paper that started the modern era.

III.3 The analytic bootstrap

Crossing solved in expansions: large spin, lightcone limits, and the Lorentzian inversion formula — the theorem that CFT data is analytic in spin, with the Regge trajectory as the fundamental object. Dispersive sum rules; what analyticity buys that numerics cannot, and conversely.

Sections: lightcone bootstrap and large spin; the inversion formula; Regge behavior in CFT; dispersive sum rules; interplay with numerics.

Key reading: Caron-Huot [1703.00278](#); Hartman et al. [2202.11012](#) (Snowmass survey of the whole program).

Further: Caron-Huot–Mazáč–Rastelli–Simmons-Duffin [2008.04931](#) for dispersive sum rules.

III.4 Two dimensions

The exactly solvable floor of the subject, taught as a modern topic rather than folklore: Virasoro symmetry, minimal models, modular invariance and the Cardy formula; the modular bootstrap;

Liouville theory; topological defect lines as the original non-invertible symmetries (closing the loop with Part II); the irrational frontier.

Sections: Virasoro representation theory; minimal models; modular invariance and modular bootstrap; Liouville; defect lines in 2d; what 2d does and does not teach us about $d > 2$.

Key reading: Yin, *Aspects of 2d CFTs* (TASI, in [assets/](#)); Kusuki [2412.18307](#) (a genuinely modern course).

Further: Ribault (1406.4290) for the systematic treatment.

III.5 Boundaries, defects, and interfaces in CFT

Conformal defects as CFTs' richest substructure: BCFT and the Cardy condition; defect operator expansions and bulk-to-defect data; line defects as quantum mechanics coupled to a CFT (the Wilson line as a renormalization problem); the defect bootstrap; defect RG flows and the g -theorem (proved entropically in Part X).

Sections: BCFT; Cardy states; defect CFT data and crossing; magnetic lines in the 3d Ising model; defect RG.

Key reading: Northe [2411.03381](#); Billò–Gonçalves–Lauria–Meineri [1601.02883](#).

III.6 New ways to solve a CFT

The methods invented after the textbooks were written. Fuzzy-sphere regularization: 3d CFTs from a single Landau level, with the state-operator map realized in a quantum simulator's Hilbert space. The large-charge expansion: strongly-coupled CFTs made semiclassical sector by sector. Hamiltonian truncation. What “solving” means when there was never a coupling to expand in.

Sections: fuzzy sphere: setup, spectrum, defects; large charge as controlled semiclassics; Hamiltonian truncation; cross-validation among methods and with the bootstrap.

Key reading: Zhu et al. [2210.13482](#); Álvarez-Gaumé–Orlando–Reffert [2008.03308](#).

Further: Hellerman et al. [1505.01537](#), the large-charge original.

Part IV. Flows: The Renormalization Group as Dynamics

What can be said exactly about the flow between landmarks — monotonicity, matching, the strange solvable deformations — and what the expansions we compute with actually mean.

IV.1 Monotonicity theorems

Irreversibility of the RG made precise: Zamolodchikov's c -theorem in 2d; the a -theorem in 4d via the dilaton effective action (anomaly matching weaponized); the F -theorem in 3d, where no anomaly exists and quantum information takes over; defect analogues (g - and b -theorems). What these theorems say about theory space (no cycles; a partial order) and what they do not.

Sections: c ; a via dilatons; F and sphere partition functions; entropic unification (preview of Part X); defect monotonicity; uses: ruling out IR scenarios.

Key reading: Komargodski–Schwimmer [1107.3987](#); Casini–Huerta [2201.13310](#); Klebanov et al. (1105.4598).

IV.2 Constraining the deep infrared

The modern toolkit applied to the question “what happens at long distances?”: anomaly matching forcing massless modes or symmetry breaking or TQFTs; the possible IR endpoints classified (gapped/TQFT, CFT, SSB with Goldstones, free); emergent symmetries in the IR and their anomalies; worked examples where the answer is known and where it is genuinely open.

Sections: the list of possible IR fates; matching as a sieve; emergent symmetry; examples: QCD chiral symmetry breaking; adjoint QCD₂; open cases.

Key reading: Gaiotto–Kapustin–Komargodski–Seiberg [1703.00501](#); Córdova–Dumitrescu–Intriligator–Shao [2205.09545](#) (§applications).

Further: Komargodski–Ohmori–Roumpedakis–Seifnashri (2008.07567) for adjoint QCD₂ done with full modern machinery.

IV.3 Effective field theory of broken symmetries

The most useful IR coordinates are often not fields in the UV Lagrangian but Goldstone modes and background responses. Goldstone’s theorem and the coset construction; nonlinear sigma models; Wess–Zumino and WZW terms as anomaly matching written in an action; chiral perturbation theory and pions; phonons, superfluids, and spacetime-symmetry breaking; SMEFT as the symmetry-compatible deformation of the Standard Model. This is the bridge between the exact constraints of IV.2 and the practical EFTs used in Parts V and IX.

Sections: Goldstone theorem; cosets and inverse Higgs; sigma models; WZ/WZW terms and anomalies; chiral perturbation theory; spacetime symmetry breaking; SMEFT.

Key reading: Naegels (2110.14504) for Goldstone bosons and cosets; Weinberg, *The Quantum Theory of Fields II*, for the durable EFT viewpoint.

IV.4 The edge of criticality

Fixed points are born, collide, and annihilate: merger of fixed points and the loss of conformality; complex CFTs and walking behavior; weakly first-order transitions masquerading as second-order; conformal windows of gauge theories. The RG as a dynamical system, with theory space’s bifurcations as physics.

Sections: fixed-point annihilation; complex CFTs; walking; conformal windows; dangerously irrelevant operators; diagnostics in practice.

Key reading: Rychkov–Su [2311.15844](#) (§on conformal windows); Gorbenko–Rychkov–Zan (1807.11512, 1808.04380).

IV.5 Solvable deformations and the integrable corner

Flows we can follow exactly. Integrable deformations of 2d theories; the $T\bar{T}$ deformation: irrelevant yet solvable, with a universal energy-level flow and a UV that is not a fixed point — the cleanest known counterexample to “irrelevant means uncontrollable,” and a glimpse of physics beyond Wilsonian QFT (resumed in Part XII).

Sections: integrable flows; $T\bar{T}$: definition, Burgers equation for spectra, interpretations; what replaces the UV fixed point; cousins ($J\bar{T}$ etc.).

Key reading: Smirnov–Zamolodchikov [1608.05499](#); Jiang [1904.13376](#) (pedagogical).

IV.6 What perturbation theory means

The expansion everyone computes with, finally given a status report. Dyson’s argument and the factorial divergence of perturbation series; Borel summation and its ambiguities; instantons — and, in genuine QFTs, renormalons — as the singularities in the Borel plane; the resurgent trans-series as the conjectured true expansion, in which the series around each saddle encodes, through the structure of its divergence, the contributions of all the others; Lefschetz thimbles as the geometry beneath. Sets the expectations for every “weak-coupling” statement in the book: a divergent series is the beginning of a theory, not the end of one.

Sections: Dyson’s argument; large-order growth; Borel summation and its ambiguities; instantons and renormalons; trans-series and resurgence; Lefschetz thimbles; what is proven, and where.

Key reading: Aniceto–Başar–Schiappa [1802.10441](#) (the primer).

Further: Dunne–Ünsal (1511.05977); Mariño, *Instantons and Large N* (book-length).

IV.7 Hydrodynamics: the effective theory of thermal states

Effective field theory applied to a state instead of a vacuum: the universal long-distance description of any QFT at finite temperature is hydrodynamics, and the 2010s finally made that an honest EFT. Thermal states and KMS (from I.4) as input; the gradient expansion of conserved-current dynamics; the Schwinger–Keldysh doubling and effective actions for dissipation; fluctuations and long-time tails as the loop corrections; magnetohydrodynamics rebuilt as the hydrodynamics of a one-form symmetry — Part II paying off at finite temperature; bounds on transport and a first look at chaos (taken up in Part XI).

Sections: KMS and thermal correlators; the gradient expansion; Schwinger–Keldysh EFT and dissipation from an action; long-time tails; one-form-symmetry hydrodynamics (MHD); bounds on transport.

Key reading: Glorioso–Liu [1805.09331](#); Iqbal [2407.20815](#) (§on MHD).

Further: Crossley–Glorioso–Liu [1511.03646](#), the construction; Kovtun (1205.5040) for the classic view; Grozdanov–Hofman–Iqbal (1610.07392).

Part V. Gauge Dynamics

Gauge theory with the new tools: global structure, extended order parameters, anomalies at $\theta = \pi$. The questions of the 1970s, finally answerable in part.

V.1 What is a gauge theory?

Gauge “symmetry” as redundancy of presentation, not symmetry — and gauging as an operation that maps one global-symmetry-bearing theory to another. The choices hidden in “gauge it”: the global form of the group, discrete θ -angles, the spectrum of lines. Emergent gauge fields in condensed matter as proof that gauge structure is descriptive, not fundamental.

Sections: gauging as an operation (finite groups first); choices and discrete data; line spectra again; emergent gauge fields; why “gauge symmetry breaking” is a category error stated carefully.

Key reading: Kapustin–Seiberg [1401.0740](#); Aharony–Seiberg–Tachikawa [1305.0318](#).

V.2 Phases of gauge theories

The phase diagram problem organized by extended operators and one-form symmetry: Wilson loops, area vs. perimeter law, confinement as unbroken center symmetry; Higgs–confinement complementarity and where it fails; oblique phases; ’t Hooft’s classification revisited with modern eyes.

Sections: order parameters that work; one-form SSB = deconfinement; Higgs vs. confining; dyonic condensation and oblique phases; finite temperature.

Key reading: Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#) (§4); McGreevy [2204.03045](#) (condensed-matter side).

V.3 Theta

The θ -angle as the laboratory for everything in Parts II and IV: Yang–Mills at $\theta = \pi$ and its mixed time-reversal/center anomaly — an exact statement about a famously intractable theory; domain walls forced to be nontrivial; large- N θ -dependence; the strong CP problem restated with anomalies in parameter space.

Sections: θ -dependence generalities; the $\theta = \pi$ anomaly; walls and interfaces; large N ; axions and Part II.7 revisited.

Key reading: Gaiotto et al. [1703.00501](#); Córdova et al. [1905.09315](#).

V.4 Low-dimensional laboratories

Where strong coupling is tame enough to dissect: the Schwinger model as the exactly solvable confining theory; adjoint QCD₂, where non-invertible symmetries control screening vs. confinement; $\mathbb{CP}^{N-1}$ and instantons; 3d Chern–Simons-matter; deconfined quantum criticality. Each laboratory keyed to which 4d lesson it teaches.

Sections: Schwinger model; adjoint QCD₂; sigma models with theta terms; CS-matter; deconfined criticality; the laboratory-to-4d dictionary.

Key reading: Komargodski–Ohmori–Roumpedakis–Seifnashri (2008.07567); Seiberg–Senthil–Wang–Witten [1606.01989](#) (3d).

V.5 The Standard Model through modern eyes

The SM as a case study in everything above: anomaly cancellation and what would fail without it; the global form of its gauge group and the line operators that distinguish the candidates; its accidental, higher-form, and non-invertible symmetries ($B - L$, the axial $U(1)$); pions and WZW terms from anomaly matching; neutrino masses, SMEFT, and naturalness questions restated symmetry-first.

Sections: which SM? (global structures); anomaly cancellation and mixed gravitational anomalies; magnetic lines and completeness; accidental and non-invertible symmetries of the SM; pions and WZW terms from anomaly matching; neutrino masses, SMEFT, and what is not explained.

Key reading: Córdova–Dumitrescu–Intriligator–Shao [2205.09545](#) (§SM); Choi et al. (2205.05086); Córdova–Ohmori (2205.06243).

Part VI. Duality

If a theory is not its presentation, different presentations of one theory must be expected. They are.

VI.1 What duality means

Duality as equality of all observables under a dictionary; exact dualities (Kramers–Wannier, T-duality, $\mathcal{N} = 4$ S-duality) vs. IR dualities (Seiberg, the 3d web); what must match (anomalies, symmetry categories, line spectra) and how proposed dualities are stress-tested with the Part II toolkit.

Sections: definitions and gradations of duality; the matching checklist; anomalies as duality invariants; historical exemplars.

Key reading: Seiberg–Senthil–Wang–Witten [1606.01989](#) (§1–2).

VI.2 The 2+1d duality web

Bosonization in three dimensions: particle–vortex duality old and new, fermion–boson dualities, and the web generated from a seed duality by gauging — with condensed-matter incarnations (quantum Hall transitions, surface states of topological insulators) as both motivation and evidence.

Sections: particle–vortex; the seed duality and the web; flux attachment; condensed-matter applications; status of the evidence.

Key reading: Seiberg–Senthil–Wang–Witten [1606.01989](#); Senthil–Son–Wang–Xu ([1810.05174](#)).

VI.3 Duality from gauging

The mechanism beneath the webs: dualities generated by gauging discrete symmetries (orbifolding), with Kramers–Wannier as the original example; self-duality producing non-invertible symmetry (Part II.5 closed into a loop); the SymTFT of II.6 as the bookkeeping that makes whole duality webs one picture.

Sections: gauging $\mathbb{Z}_2$ in 2d done three ways; orbifolds; self-duality $\Rightarrow$ duality defects; webs from the SymTFT sandwich.

Key reading: Kapustin–Seiberg [1401.0740](#); Choi–Córdova–Hsin–Lam–Shao [2111.01139](#); Shao [2308.00747](#) (§KW).

VI.4 Duality as change of polarization

Electric–magnetic duality, T-duality, mirror symmetry, and S-duality as changes in which variables, charges, and defects are treated as elementary. Charge lattices, Dirac pairings, line operators, and boundary conditions are the invariant data; compactification turns one duality into another; duality groups act on backgrounds as well as dynamical fields. This chapter ties the 2+1d web to Seiberg–Witten theory, class $\mathcal{S}$, and the holographic dictionaries of Part XI.

Sections: electric–magnetic duality and charge lattices; T-duality and compactification; mirror symmetry; S-duality and line operators; boundary conditions as polarizations; duality groups acting on backgrounds.

Key reading: Tachikawa [1312.2684](#); Gaiotto [0904.2715](#); Kapustin–Seiberg [1401.0740](#).

Part VII. Exactly Solvable Corners: Supersymmetry and Friends

Solvable islands are not the point of the subject; they are its lighthouses — the places where proposed general principles get checked exactly.

VII.1 Holomorphy and Seiberg duality

$\mathcal{N} = 1$ theories as the minimal setting where strong-coupling questions have exact answers: holomorphy as the supersymmetric replacement for perturbation theory; moduli spaces of vacua; Seiberg duality as the prototype of the IR dualities graded in Part VI — and the historical seed of the entire duality-centric worldview.

Sections: holomorphy arguments; exact superpotentials; moduli spaces; Seiberg duality and its checks; SUSY breaking in brief.

Key reading: Seiberg [hep-th/9411149](#) (the original electric–magnetic duality); Tachikawa [1312.2684](#) (companion volume); Intriligator–Seiberg [hep-th/9509066](#), the classic lectures.

VII.2 Seiberg–Witten theory

The first complete solution of the IR of a strongly-coupled 4d gauge theory: the Coulomb branch as a fibered geometry, monodromies, monopole condensation as the mechanism of confinement made exact. Presented as the template for “solving” a QFT: identify protected data, find the geometry that encodes it.

Sections: the Coulomb branch; the SW curve; monopole condensation and confinement; breaking to $\mathcal{N} = 1$; the template abstracted.

Key reading: Seiberg–Witten [hep-th/9407087](#); Tachikawa [1312.2684](#).

VII.3 Localization and exact partition functions

Path integrals computed exactly: supersymmetric localization on spheres, indices, and the precision-duality era they enabled. What localization computes (protected data), what it cannot, and how its outputs feed monotonicity theorems (F) and bootstrap baselines.

Sections: the localization argument; sphere partition functions; superconformal indices; precision tests of duality; limits of the method.

Key reading: Pestun [0712.2824](#); the review volume *Localization Techniques in QFT* (1608.02952).

VII.4 Theories without Lagrangians

The existence proof that broke the old definition: the 6d $(2, 0)$ theory; class $\mathcal{S}$ — 4d $\mathcal{N} = 2$ theories from punctured surfaces, with duality frames as pair-of-pants decompositions; Argyres–Douglas points. Why “the space of QFTs” is strictly larger than “the space of Lagrangians,” and how one works with a theory that has no path integral.

Sections: the $(2, 0)$ theory; compactification as theory factory; class $\mathcal{S}$ and duality made geometric; Argyres–Douglas; how to compute without a Lagrangian.

Key reading: Gaiotto [0904.2715](#); Tachikawa [1312.2684](#).

VII.5 Protected subsectors and integrability

Exact substructures inside non-integrable theories: the chiral algebra (2d VOA) inside every 4d $\mathcal{N} = 2$ SCFT — Rastelli school — feeding exact CFT data into the bootstrap; topological sectors; integrability of planar $\mathcal{N} = 4$ as the maximal solvable corner; 2d factorized S-matrices (used again in Part IX).

Sections: the 4d/2d chiral-algebra map; protected operator algebras as bootstrap input; planar integrability in one chapter; factorized scattering.

Key reading: Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees [1312.5344](#).

Part VIII. The Lattice and the Continuum

If QFTs are defined by continuum limits of lattice models, the dictionary between lattice and continuum had better be understood exactly. It is subtler than anyone assumed.

VIII.1 Continuum limits, reexamined

What “taking the continuum limit” asserts; universality as the (usually unproven) theorem behind it; which structures pass to the limit (symmetries? anomalies? Hilbert spaces?) and which arise only there. The program: treat the lattice not as a regulator but as a definition, and audit everything.

Sections: lattice models as definitions; universality audited; symmetries of the lattice vs. of the limit; emergent symmetry and its breaking by irrelevant operators.

Key reading: Cheng–Seiberg [2211.12543](#) (§1); Dedushenko [2203.08053](#) (§lattice).

VIII.2 Emanant symmetries

Seiberg–Shao: continuum internal symmetries that *emanate* from lattice translations — not emergent (violated at finite spacing) but exact consequences of lattice symmetry acting differently in the limit. The Majorana chain and Ising model worked completely; anomalies of the emanant symmetry predicted from the lattice; non-invertible translation symmetry at criticality.

Sections: emergent vs. emanant; the Majorana chain in detail; Ising and its KW operator from lattice translation; anomaly inheritance; consequences for numerics.

Key reading: Seiberg–Shao [2307.02534](#).

VIII.3 Lieb–Schultz–Mattis theorems as anomalies

LSM-type theorems — lattice systems forbidden from having trivially gapped ground states — understood as ’t Hooft anomaly matching between lattice translation and internal symmetry; Luttinger’s theorem in the same frame; the non-invertible refinement on tensor-product Hilbert spaces. Condensed matter and formal QFT collapsing into one subject.

Sections: LSM classically; the anomaly interpretation; filling constraints and Luttinger; non-invertible LSM; applications to spin chains and beyond.

Key reading: Cheng–Seiberg [2211.12543](#); Seiberg–Seifnashri–Shao [2401.12281](#).

VIII.4 Finite density and Fermi surfaces

Most gapless quantum matter is not a CFT. The Fermi liquid as a fixed point with a codimension-one manifold of gapless modes; patch EFT and the RG around a Fermi surface; Luttinger’s theorem as anomaly matching; emergent symmetries acting separately on each patch; non-Fermi liquids from Fermi surfaces coupled to gauge fields; Lifshitz and other nonrelativistic scalings as local QFTs outside the relativistic axiom templates.

Sections: finite density as a deformation; Fermi liquid fixed points; patch RG; Fermi-surface anomalies and Luttinger; non-Fermi liquids and gauge fields; Lifshitz/nonrelativistic scaling.

Key reading: Shankar, *Rev. Mod. Phys.* **66** (1994) 129; Ma–Wang (2110.09492); Wang–Hickey–Ying–Burkov (2110.10692).

VIII.5 Exotica: fractons and UV/IR mixing

The systems that break the continuum lore: fracton models with subsystem symmetries, restricted mobility, and ground-state degeneracies that depend on the lattice size — continuum limits in which discontinuous field configurations dominate and rotation invariance never returns. What, exactly, must be relaxed in the axioms; the modified Villain construction as the bridge.

Sections: the X-cube and friends; subsystem symmetry; UV/IR mixing; continuum descriptions with discontinuous fields; modified Villain lattices; what fractons teach about “QFT”.

Key reading: Seiberg–Shao (2003.10466); Gorantla–Lam–Seiberg–Shao (2103.01257); both summarized in Córdova et al. 2205.09545.

VIII.6 Rigor: constructive QFT and triviality

What mathematics actually knows: constructive successes in $d = 2, 3$; the Aizenman–Duminil-Copin theorem that ϕ_4^4 and 4d Ising have (marginally) trivial scaling limits — a rigorous statement about the Standard Model’s scalar sector; Yang–Mills existence as a Millennium problem; chiral fermions on the lattice as the deepest unsolved regulator problem.

Sections: what has been constructed; triviality in $d = 4$ and its meaning; the YM problem stated honestly; Nielsen–Ninomiya and chiral lattice fermions; rigorous bootstrap bounds as a new kind of rigor.

Key reading: Aizenman–Duminil-Copin 1912.07973; Dedushenko 2203.08053.

Part IX. Scattering, the UV, and the Space of Consistent Theories

The on-shell program and the S-matrix bootstrap: carve out the space of consistent theories from locality, unitarity, analyticity, and crossing — in flat space this time.

IX.1 Modern amplitudes

The computational revolution that replaced Feynman diagrams as the native language of scattering: spinor-helicity variables, on-shell recursion, generalized unitarity, color–kinematics duality and the double copy, positive geometry where it exists. The moral is the same as elsewhere in the book: locality and unitarity leave sharper fingerprints on observables than on Lagrangians.

Sections: spinor-helicity and little group; BCFW recursion; generalized unitarity; color–kinematics and double copy; amplitudes in EFT; positive geometries and their limits.

Key reading: Elvang–Huang, *Scattering Amplitudes in Gauge Theory and Gravity*; Bern et al. (2203.13013) for color–kinematics and double copy.

IX.2 The S-matrix bootstrap

The 1960s program with 2020s technology: first, when a flat-space S-matrix is an observable at all (mass gap, asymptotic states, massless IR sectors and inclusive observables); then nonperturbative

two-to-two amplitudes constrained by unitarity + analyticity + crossing, explored numerically (primal and dual); pion physics as a target; QFT in AdS as the regulator connecting the S-matrix bootstrap to the conformal bootstrap of Part III.

Sections: when the S-matrix exists; massless IR and inclusive observables; axioms for amplitudes; numerical primal/dual methods; the pion bootstrap; QFT in AdS; relation to the conformal bootstrap.

Key reading: Kruczenski–Penedones–van Rees [2203.02421](#); Paulos et al. (1607.06109).

IX.3 Positivity and the UV–IR connection

Which low-energy EFTs admit any UV completion at all: dispersion relations, positivity of forward amplitudes, the classic IR obstruction (Adams et al.), and the modern sharpening into two-sided bounds and the EFT-hedron geometry of Wilson coefficients — the “swampland program” of pure QFT, with experimental teeth.

Sections: dispersion relations; forward positivity; two-sided bounds; the EFT-hedron; causality constraints; applications from gravity to particle physics.

Key reading: Adams et al. [hep-th/0602178](#); de Rham et al. (2203.06805); Arkani-Hamed–Huang–Huang (2012.15849).

IX.4 Toward flat-space holography

The frontier chapter: soft theorems reorganized as symmetries of the S-matrix; memory effects and the IR triangle; celestial and Carrollian proposals for a holographic description of flat space; what is established, what is aspirational, and what the S-matrix bootstrap says either way.

Sections: soft theorems as Ward identities; memory and asymptotic symmetries; celestial sphere kinematics; Carrollian proposals; status report, written skeptically.

Key reading: Kruczenski–Penedones–van Rees [2203.02421](#) (§outlook); Strominger [1703.05448](#), the lectures.

Part X. Quantum Fields and Quantum Information

The Hilbert space of a QFT is not a big collection of qubits: its entanglement structure is the theory.

X.1 Entanglement in quantum field theory

Entanglement entropy of regions: UV divergences and what is universal; replica trick; exact results in 2d and for free fields; mutual information and relative entropy as the cutoff-free quantities; entanglement as the modern order parameter (topological entanglement entropy, anomalies detected by entanglement).

Sections: regions and reduced states; the replica trick; universal terms; relative entropy and its monotonicity; entanglement diagnostics of phases.

Key reading: Casini–Huerta [2201.13310](#); Witten [1803.04993](#).

X.2 The algebraic viewpoint

Why regions of a QFT have no density matrices: local algebras, Type III, and the failure of factorization — stated as physics, not pathology; the split property as the controlled approximation

to tensor factorization; Tomita–Takesaki modular theory and what modular flow means (Bisognano–Wichmann: it is a boost); the type classification as a physical classification of how systems entangle. *Sections:* algebras instead of tensor factors; Types I/II/III with physical fingerprints; the split property; modular flow; Bisognano–Wichmann; half-sided modular inclusions (a glimpse).

Key reading: Witten [1803.04993](#); Sorce [2302.01958](#); Liu [2510.07017](#) (lectures, with the gravity payoff).

X.3 Information-theoretic theorems

The harvest: entropic proofs of the c - and F -theorems (strong subadditivity doing the work anomalies cannot in odd d); the averaged null energy condition proved from causality and from modular theory; QNEC; Bekenstein-type bounds from relative entropy. Energy conditions — once assumptions — now theorems of quantum information.

Sections: entropic c and F ; ANEC two ways; QNEC; Bekenstein from relative entropy; what remains conjectural.

Key reading: Casini–Huerta [2201.13310](#) (incl. [hep-th/0405111](#)); Hartman–Kundu–Tajdini ([1610.05308](#)); Faulkner et al. ([1605.08072](#)).

X.4 Coupling to gravity: the crossed product

What happens to the algebras when gravity is turned on, however weakly: Witten’s question “why does QFT in curved spacetime make sense?”; the crossed-product construction rendering Type III into Type II, giving finite (renormalized) entropies; observers as part of the algebra. The bridge into Part XI.

Sections: QFT in curved spacetime as an algebra problem; the crossed product; finite generalized entropies; observers; de Sitter remarks.

Key reading: Witten [2112.11614](#); Liu [2510.07017](#).

Part XI. Large N , Holography, and Gravity

Large N is a classical limit no Lagrangian advertises, and holography is its destiny: QFT as the definition of quantum gravity.

XI.1 Large N as a classical limit

Factorization of correlators at large N as the emergence of a classical theory — whose phase space is nothing like the classical limit $\hbar \rightarrow 0$; vector vs. matrix models; planarity; the master field; $1/N$ as the loop-counting parameter of an emergent weakly-coupled description (the bulk, in waiting).

Sections: vector models solved; planar counting; factorization = classicality; large- N phase transitions; what $1/N$ organizes.

Key reading: Penedones [1608.04948](#) (§large N); ’t Hooft’s original logic as retold there.

XI.2 Holography from the CFT side

AdS/CFT presented as a statement *within* QFT: which CFTs have local bulk duals (large N + large gap); Heemskerk–Penedones–Polchinski–Sully; the holographic dictionary derived from

CFT axioms; Mellin amplitudes; bootstrap bounds becoming statements about gravity (graviton couplings, higher-derivative corrections).

Sections: the HPPS criterion; the dictionary from the CFT side; Mellin space; bulk locality from crossing; gravity constrained by CFT consistency.

Key reading: Maldacena [hep-th/9711200](#) (the original conjecture); Heemskerk–Penedones–Polchinski–Sully [0907.0151](#); Penedones [1608.04948](#).

Further: the holographic correlators program (Rastelli–Zhou and successors; see Penedones’s lectures and references therein).

XI.3 Quantum field theory as quantum gravity

The strongest claim in the book: in AdS, quantum gravity is *defined* by a QFT, making everything in Parts I–X a statement about quantum gravity. What gravity has paid back: the chaos bound, energy conditions, the algebraic reorganization of Part X.4; black-hole information as a theorem about unitary QFT.

Sections: AdS/CFT as definition; information-paradox dividends; the chaos bound as a QFT theorem; gravity’s feedback into pure QFT; limits of the definition (dS, cosmology) as segue to Part XII.

Key reading: Penedones [1608.04948](#); Maldacena–Shenker–Stanford [1503.01409](#).

Part XII. Epilogue: The Next Twenty Years

What the 2046 book admits it does not know — the chapters that will be rewritten by 2066.

XII.1 The definitional frontier

The axiomatizations compared honestly: Wightman/Osterwalder–Schrader; Haag–Kastler nets; Atiyah–Segal functorial QFT and Kontsevich–Segal allowability; factorization algebras. Each captures a fragment; none captures class $\mathcal{S}$, large N , *and* fractons at once. The definition of QFT as an open research problem, not an embarrassment.

Sections: the contenders; what each proves; the counterexample stress-tests; is one definition even desirable?

Key reading: Dedushenko [2203.08053](#); Kontsevich–Segal [2105.10161](#).

XII.2 Computational frontiers

Where new answers will come from: quantum simulation of QFTs (the Jordan–Lee–Preskill program) and the regulator questions it forces (Part VIII reappears as engineering); tensor networks; bootstrap numerics at scale; machine-learned exclusion plots. Computation as experiment for formal QFT.

Sections: quantum simulation of scattering; lattice regularizations for hardware; tensor networks and criticality; the numerical bootstrap’s next decade.

Key reading: Jordan–Lee–Preskill (1111.3633); Rychkov–Su [2311.15844](#) (§outlook).

XII.3 Beyond quantum field theory

The boundary of the subject mapped from inside: little string theories (local energy density without local fields); TT -deformed theories as UV-complete non-QFTs; QFT in de Sitter and the cosmological

bootstrap — correlators at the end of inflation as the next “S-matrix”; gravity as the deformation that finally leaves theory space.

Sections: symptoms of non-QFT UV behavior; little strings; $T\bar{T}$ revisited as a beyond-QFT laboratory; dS and cosmological observables; what replaces theory space.

Key reading: Jiang [1904.13376](#); Arkani-Hamed et al. (1811.00024); Seiberg’s talks on what lies beyond QFT.

XII.4 Open problems

A concrete list, stated so that progress is checkable: (1) Yang–Mills existence and mass gap — now with anomaly-matching constraints any solution must obey; (2) a nonperturbative definition of the full Standard Model, i.e. chiral fermions on the lattice; (3) classify 3d and 4d CFTs, or even all theories with c (or a) below a given value; (4) prove the a -theorem’s gradient-flow refinement, and monotonicity in $d = 6$; (5) solve the 3d Ising model analytically; (6) determine whether the space of QFTs is connected; (7) complete the classification of symmetry categories and their gapped/gapless realizations in $d \geq 4$; (8) glueball masses from the S-matrix bootstrap; (9) a sharp nonperturbative formulation of observables in de Sitter; (10) a definition of QFT wide enough for class $\mathcal{S}$ and fractons simultaneously — or a proof that none exists.

Key reading: every entry above, and the next twenty years of [arXiv:hep-th](#).

The library

All papers below are in `assets/`, with filenames beginning with the arXiv number. The index in `assets/README.md` maps them to the parts above. Lecture notes and reviews are marked ●; foundational research papers ○.

Framing and history

- Dedushenko, *Snowmass: The Quest to Define QFT*, [2203.08053](#)
- Kontsevich–Segal, *Wick Rotation and the Positivity of Energy in QFT*, [2105.10161](#)
- Aizenman–Duminil-Copin, *Marginal Triviality of ϕ_4^4* , [1912.07973](#)
- Rychkov, *Conformal Bootstrap: from Polyakov to Our Times*, [2509.02779](#)

Symmetry and TQFT (Part II)

- Gaiotto–Kapustin–Seiberg–Willett, *Generalized Global Symmetries*, [1412.5148](#)
- Córdova–Dumitrescu–Intriligator–Shao, *Snowmass: Generalized Symmetries*, [2205.09545](#)
- Shao, *TASI Lectures on Non-Invertible Symmetries*, [2308.00747](#)
- Schäfer–Nameki, *ICTP Lectures on (Non-)Invertible Generalized Symmetries*, [2305.18296](#)
- Iqbal, *Jena Lectures on Generalized Global Symmetries*, [2407.20815](#)
- McGreevy, *Generalized Symmetries in Condensed Matter*, [2204.03045](#)
- Dunne, *Aspects of Chern–Simons Theory*, [hep-th/9902115](#)
- Aharony–Seiberg–Tachikawa, *Reading Between the Lines of 4d Gauge Theories*, [1305.0318](#)
- Kapustin–Seiberg, *Coupling a QFT to a TQFT and Duality*, [1401.0740](#)
- Choi–Córdova–Hsin–Lam–Shao, *Non-Invertible Duality Defects in 3+1d*, [2111.01139](#)
- Córdova–Freed–Lam–Seiberg, *Anomalies in the Space of Coupling Constants I*, [1905.09315](#)

Lattice ↔ continuum (Part VIII)

- Seiberg–Shao, *Majorana Chain and Ising Model: Emanant Symmetries*, [2307.02534](#)
- Cheng–Seiberg, *LSM, Luttinger, and 't Hooft*, [2211.12543](#)
- Seiberg–Seifnashri–Shao, *Non-Invertible Symmetries and LSM-Type Constraints*, [2401.12281](#)

CFT and bootstrap (Part III)

- Rychkov, *EPFL Lectures on CFT in $D \geq 3$* , [1601.05000](#)
- Simmons-Duffin, *TASI Lectures on the Conformal Bootstrap*, [1602.07982](#)
- Poland–Rychkov–Vichi, *The Conformal Bootstrap* (RMP), [1805.04405](#)
- Rychkov–Su, *New Developments in the Numerical Bootstrap*, [2311.15844](#)
- Hartman et al., *Snowmass: The Analytic Conformal Bootstrap*, [2202.11012](#)
- Caron-Huot, *Analyticity in Spin in Conformal Theories*, [1703.00278](#)
- Pappadopulo–Rychkov–Espin–Rattazzi, *OPE Convergence in CFT*, [1208.6449](#)
- Northe, *Maynooth Lectures on CFT, BCFT and DCFT*, [2411.03381](#)
- Zhu et al., *Uncovering Conformal Symmetry: the Fuzzy Sphere*, [2210.13482](#)
- Kusuki, *Modern Approach to 2D Conformal Field Theory*, [2412.18307](#)
- Yin, *Aspects of Two-Dimensional CFTs* (PoS TASI2017)
- El-Showk–Paulos–Poland–Rychkov–Simmons-Duffin–Vichi, *Solving the 3D Ising Model with the Conformal Bootstrap*, [1203.6064](#)
- Billò–Gonçalves–Lauria–Meineri, *Defects in Conformal Field Theory*, [1601.02883](#)
- Caron-Huot–Mazáč–Rastelli–Simmons-Duffin, *Dispersive CFT Sum Rules*, [2008.04931](#)

RG flows, expansions, and thermal EFT (Part IV)

- Komargodski–Schwimmer, *On RG Flows in Four Dimensions*, [1107.3987](#)
- Casini–Huerta, *Lectures on Entanglement in QFT*, [2201.13310](#)
- Jiang, *Pedagogical Review of Solvable Irrelevant Deformations*, [1904.13376](#)
- Glorioso–Liu, *Lectures on Non-Equilibrium EFTs and Fluctuating Hydrodynamics*, [1805.09331](#)
- Smirnov–Zamolodchikov, *On Space of Integrable Quantum Field Theories ($T\bar{T}$)*, [1608.05499](#)
- Crossley–Glorioso–Liu, *Effective Field Theory of Dissipative Fluids*, [1511.03646](#)

Gauge dynamics and duality (Parts V, VI)

- Gaiotto–Kapustin–Komargodski–Seiberg, *Theta, Time Reversal, and Temperature*, [1703.00501](#)
- Seiberg–Witten, *Monopole Condensation and Confinement*, [hep-th/9407087](#)
- Seiberg–Senthil–Wang–Witten, *A Duality Web in $2+1d$* , [1606.01989](#)

Exact corners (Part VII)

- Tachikawa, *$\mathcal{N} = 2$ Supersymmetric Dynamics for Pedestrians*, [1312.2684](#)
- Gaiotto, *$\mathcal{N} = 2$ Dualities*, [0904.2715](#)
- Beem et al., *Infinite Chiral Symmetry in Four Dimensions*, [1312.5344](#)
- Seiberg, *Electric–Magnetic Duality in Supersymmetric Non-Abelian Gauge Theories*, [hep-th/9411149](#)
- Intriligator–Seiberg, *Lectures on Supersymmetric Gauge Theories and Electric–Magnetic Duality*, [hep-th/9509066](#)
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Scattering and the UV (Part IX)

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Part I

The Arena: Quantum Field Theory Without Lagrangians

I.1 What is a quantum field theory?

There is a question that every course on quantum field theory postpones, and that these notes take as their subject: what, exactly, *is* a quantum field theory? Not how to compute in one — you know how to do that. You can renormalize a loop integral, extract a cross section, run a coupling. The question is what the object is that all this machinery computes *about*.

For quantum mechanics the analogous question has an answer that fits on a postcard. A quantum system is a Hilbert space $\mathcal{H}$, states are rays in it, observables are self-adjoint operators, time evolution is a unitary e^{-iHt} . Every textbook states these axioms in chapter one; nothing discovered since 1930 has dislodged them. Now try to recall the corresponding list from your quantum field theory course. There isn't one. What you were given instead was a *recipe* — pick fields, write a Lagrangian, quantize, renormalize — together with the implicit promise that the recipe defines the subject. The first purpose of this chapter is to show, by exhibiting concrete and well-established physics, that it does not: there are quantum field theories the recipe cannot produce, single theories it produces in several inequivalent ways, and recipe-outputs that turn out not to exist. The second purpose is constructive: to assemble the list of things a quantum field theory actually provides — its observables — and to draw the picture that organizes everything else in these notes, in which theories are points in a space, the renormalization group is a flow on that space, and conformal field theories are its fixed points.

Question I.1.1 — What must we specify to specify a quantum field theory? And to what mathematical and geometric objects does a quantum field theory, once specified, assign physical data?

A warning about the spirit of what follows. Nothing in this chapter is a complaint about the practical success of the standard framework, which is the most quantitatively successful structure in the history of science and which we will use constantly. The claim is about logic, not utility: the standard framework is a description of how to *present* certain quantum field theories and expand them around free fields, and the subject has outgrown the identification of theories with their presentations. Making that distinction precise is worth a whole chapter, because almost every modern development — generalized symmetry, the bootstrap, duality, the lattice–continuum program — begins from it.

I.1.1 The standard paradigm

Let us first state the inherited answer honestly, because it deserves it. The textbook definition of a quantum field theory, circa 2000, is a procedure:

1. Choose fields: a collection of functions on spacetime transforming in representations of the Poincaré group and of some internal (possibly gauge) group.

2. Write the most general local Lagrangian density $\mathcal{L}[\phi, \partial\phi]$ consistent with the symmetries, keeping terms up to some power-counting criterion (“renormalizability”).
3. Quantize: canonically, or via the path integral

$$\begin{aligned} Z &= \int \mathcal{D}\phi e^{iS[\phi]}, \\ S &= \int d^d x \mathcal{L}, \end{aligned} \tag{I.1.1}$$

defined in practice by expanding around the Gaussian (free) theory.

4. Renormalize: absorb the divergences of the expansion into the couplings, order by order.

The output of this procedure is a perturbation series for correlation functions, and through the LSZ reduction formula, for scattering amplitudes. The procedure’s triumphs are not in dispute. Its flagship number is the magnetic moment of the electron: the measured value $g/2 = 1.00115965218059(13)$ agrees with the Standard Model prediction at the level of one part in 10^{12} — the comparison is now limited by independent measurements of α , not by QFT itself.² The same procedure, run in reverse by Wilson, explained why critical exponents of fluids and magnets are computable at all. A definition this successful does not get discarded; it gets *relocated*, and understanding where it belongs is the business of Section I.1.5.

But look closely at what the procedure actually hands you. Step 3 writes down the symbol (I.1.1) and immediately retreats to an expansion around the quadratic action. Each order of that expansion, after step 4, is finite and unambiguous. The series as a whole is another matter, and here it pays to go to the one spacetime where every step can be done exactly: a single point.

Example I.1.1 (The ϕ^4 theory in zero dimensions)

Take spacetime to be a point. A “field” is then one real number ϕ , the “path integral” is an ordinary integral, and the Euclidean ϕ^4 partition function is

$$Z(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\phi e^{-\frac{1}{2}\phi^2 - \frac{\lambda}{4!}\phi^4}, \tag{I.1.2}$$

normalized so that $Z(0) = 1$. There is nothing to renormalize and nothing to gauge-fix; the entire content of “quantizing the Lagrangian” is this one integral, which manifestly exists for all $\lambda \geq 0$.

Now run the textbook procedure: expand around the free theory. Exchanging the sum with the integral,

$$Z_{\text{pert}}(\lambda) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{4!}\right)^n \frac{1}{\sqrt{2\pi}} \int d\phi \phi^{4n} e^{-\frac{1}{2}\phi^2} = \sum_{n=0}^{\infty} a_n \lambda^n, \tag{I.1.3}$$

where the Gaussian moments are $(4n-1)!! = (4n)! / (2^{2n}(2n)!)$, so that

$$a_n = \frac{(-1)^n}{(4!)^n n!} \cdot \frac{(4n)!}{2^{2n}(2n)!}. \tag{I.1.4}$$

²Fan, Myers, Sukra, Gabrielse, 2209.13084, currently the most precisely verified prediction in science. Seiberg’s essay *Thoughts About Quantum Field Theory* opens with this number for the same rhetorical purpose.

The first few terms, $Z_{\text{pert}} = 1 - \frac{\lambda}{8} + \frac{35\lambda^2}{384} - \dots$, are exactly the sums of vacuum Feynman diagrams with the usual symmetry factors. Up to this point, everything matches the textbook treatment. But apply Stirling's formula to (I.1.4):

$$|a_n| \sim \frac{1}{\pi n \sqrt{2}} n! \left(\frac{2}{3}\right)^n \text{ as } n \rightarrow \infty. \quad (\text{I.1.5})$$

The coefficients grow *factorially*: the ratio $|a_{n+1}/a_n| \approx \frac{2}{3}n$ diverges, and the series has zero radius of convergence. For *no* value of $\lambda \neq 0$, however small, does the perturbative expansion converge.

There is a beautifully physical way to see that this had to happen, due to Dyson. Suppose the series converged in some disc $|\lambda| < R$. Then $Z(\lambda)$ would be analytic at $\lambda = 0$ and in particular would exist for small *negative* λ . But at $\lambda < 0$ the integral (I.1.2) diverges — the potential is unbounded below, the theory has no ground state — so no such analytic continuation can exist, and $R = 0$.^a

The series is not useless — it is **asymptotic**. Truncating at order N , the error is controlled by the first omitted term, $|a_N \lambda^N| \sim N!(2\lambda/3)^N$, which *decreases* with N until $N_* \approx 3/2\lambda$ and grows explosively afterwards. Truncating at the optimal order leaves an irreducible error

$$\Delta Z \sim e^{-N_*} = e^{-3/2\lambda}, \quad (\text{I.1.6})$$

invisible at every finite order in λ . This is not a random number: the action $S = \frac{1}{2}\phi^2 + \frac{\lambda}{4!}\phi^4$ has, besides $\phi = 0$, the complex saddle points $\phi_*^2 = -6/\lambda$ with $S(\phi_*) = -3/2\lambda$, and it is these *instanton-like* saddles that govern both the large-order growth (I.1.5) and the error (I.1.6). Perturbation theory around one saddle knows about the others only through the divergence of its own coefficients.

^aDyson's original argument (Phys. Rev. **85**, 631 (1952)) was made for QED, where it is heuristic; in zero dimensions it is a rigorous one-line proof.

Three lessons, in increasing order of importance. First, the perturbation series was never the theory: it is a particular asymptotic expansion *of* the theory, tied to a particular saddle point. Second, the theory exists anyway — here, trivially, as a convergent integral. Third, the gap between the two is quantitative and structured: effects of order $e^{-3/2\lambda}$, the germ of everything “nonperturbative.” In zero dimensions these statements are almost embarrassing in their simplicity. Their force comes from the fact that they persist in higher dimensions: in $d > 0$ the same factorial growth afflicts the loop expansion of essentially every interacting QFT, while the analogue of “the integral exists anyway” — some construction of the theory that does not lean on the expansion — becomes the entire problem. The recipe of steps 1–4 produces the series and is silent about the object the series approximates.

Remark I.1.1 (Borel resummation and resurgence). For $\lambda > 0$ the zero-dimensional series can be completely repaired: its Borel transform $B(t) = \sum_n a_n t^n / n!$ converges for $|t| < 3/2$ (the singularity sits at $t = -3/2$, the instanton action again), and the Borel integral $\int_0^\infty dt e^{-t} B(\lambda t)$ reproduces $Z(\lambda)$ exactly (Exercise I.1.3). The program of *resurgence* systematizes this: the perturbative series around each saddle encodes, through the detailed structure of its divergence, the contributions of all the others. In genuine QFTs the story is harder (renormalons join instantons among the Borel singularities) and is an active subject; see Aniceto–Başar–Schiappa

[1802.10441](#) for a thorough introduction, and Part IV.6 for where it sits today. We will not need resurgence machinery until then, but you should know that “the series diverges” is the beginning of a theory, not the end of one.

One should be fair about what the old framework knows beyond the series. Lattice regularization defines Z nonperturbatively at finite cutoff, and constructive field theory has *proven* that interacting continuum limits exist in low dimensions — the $P(\phi)_2$ theories, ϕ_3^4 , Gross–Neveu₂, Yukawa_{2,3} all rigorously satisfy the Wightman axioms.³ Instanton calculus patches in the leading e^{-S_*} effects. The working physicist’s position circa 2000 — “the path integral means: lattice + continuum limit, and perturbation theory is its asymptotic expansion” — is coherent and, in its domain, correct. The trouble is the domain. The next section presents four pieces of settled physics that fall outside it.

Exercise I.1.1 (easy). Verify $a_1 = -1/8$ and $a_2 = 35/384$ from (I.1.4), and check them against the vacuum Feynman diagrams of ϕ^4 theory (one and two vertices; count symmetry factors). Then verify the asymptotic statement (I.1.5) numerically: compute $|a_{n+1}/a_n|$ for $n = 5, 10, 20$ and compare with $\frac{2}{3}(n + \frac{7}{4})$.

Hint: For the diagrams: at one vertex, the figure-eight has symmetry factor 8. For the ratio: use (I.1.4) directly; no Stirling needed.

Answer: $a_{n+1}/a_n = -\frac{(4n+4)(4n+3)(4n+2)(4n+1)}{24 \cdot 4 \cdot (n+1)(2n+2)(2n+1)} = -\frac{2}{3}n - \frac{7}{6} + O(1/n)$, so the quoted ratio is accurate to a few percent already at $n = 10$.

Exercise I.1.2 (classic). (Dyson in quantum mechanics.) Consider the anharmonic oscillator $H = \frac{1}{2}p^2 + \frac{1}{2}x^2 + \lambda x^4$. Adapt Dyson’s argument to show that the Rayleigh–Schrödinger series for the ground-state energy $E_0(\lambda)$ must have zero radius of convergence. (Bender and Wu computed its coefficients to high order in 1969 and found $|c_n| \sim n!3^n$, confirming this.)

Hint: What happens to the spectrum of H for $\lambda < 0$, no matter how small? Think about tunneling out of the origin: the decay rate is $\sim e^{-c/|\lambda|}$, which is non-analytic at $\lambda = 0$.

Exercise I.1.3 (medium). (Borel summation in $d = 0$.) Define $B(t) = \sum_n a_n t^n / n!$ with a_n as in (I.1.4). Show that the series for B converges for $|t| < 3/2$, and that for $\lambda > 0$

$$\int_0^\infty dt e^{-t} B(\lambda t) = Z(\lambda).$$

Hint: Insert $1 = \frac{1}{n!} \int_0^\infty dt e^{-t} t^n$ into the (divergent) series for Z and exchange sums and integrals where legal. Convergence of B follows from (I.1.5). The Borel singularity at $t = -3/2$ sits on the *negative* axis because the series alternates — this is why the integral along the positive axis is safe. For $\lambda < 0$ the singularity obstructs the contour: the ambiguity in avoiding it is $\sim e^{-3/2|\lambda|}$, the decay rate of the metastable vacuum.

I.1.2 Four cracks

If the only deficiency of “quantize a Lagrangian” were the divergence of its series, the fix would be technical and the old definition would stand. The actual situation is more interesting: as physicists explored quantum field theory away from weak coupling — through supersymmetry, through string theory, through the bootstrap, through rigorous statistical mechanics — they found that the map

³The standard reference for the constructive program is Glimm–Jaffe, *Quantum Physics: A Functional Integral Point of View*; see also the reading list at the end of this chapter.

between Lagrangians and theories is neither surjective nor injective, and in at least one textbook case not even well-defined. We present four exhibits. None of them is derived here; each is a vignette with pointers to the part of these notes where it is done properly, and to the literature. We flag this clearly: *in this section we cite, not derive* — which is itself a lesson in how this material is established.

Crack 1: a theory with no Lagrangian. There exists a six-dimensional interacting conformal field theory, the **(2, 0) theory**, with the following résumé. It was discovered in 1995 as the decoupling limit of coincident M5-branes in M-theory, equivalently of type IIB string theory on an A–D–E singularity.⁴ It is, by every available diagnostic, an honest local QFT: it has a stress tensor, local operators with an operator product expansion, and correlation functions obeying the usual axioms of conformal field theory. It is maximally supersymmetric, and upon compactification on a circle it flows to five-dimensional Yang–Mills; on a torus, to four-dimensional $\mathcal{N} = 4$ Yang–Mills, whose S-duality becomes, from six dimensions, the merely *geometric* statement that a torus has a modular group. And it has no Lagrangian — not “none found yet” as a matter of bookkeeping, but structurally: the theory has no coupling constant whatsoever, no marginal or relevant deformation preserving its symmetry, hence no dial to turn to reach weak coupling, hence nothing for a path-integral expansion to expand around. Even the count of its degrees of freedom announces the problem: anomaly and holographic computations show the number of degrees of freedom of the rank- N theory grows as N^3 ,⁵ a scaling no known collection of free fields reproduces. Every property listed above was established through string dualities and consistency arguments, not through any intrinsic construction — which is precisely the scandal. Compactifying the (2, 0) theory on Riemann surfaces generates the enormous “class $\mathcal{S}$ ” of four-dimensional theories, most of which inherit its homelessness. The space of quantum field theories is strictly larger than the space of Lagrangians. (Part VI.4; Gaiotto [0904.2715](#), Tachikawa [1312.2684](#).)

Crack 2: one theory, several Lagrangians. The opposite failure is just as instructive: the map from Lagrangians to theories is many-to-one, and not only in trivial ways (field redefinitions) but in ways that exchange weak and strong coupling, bosons and fermions, gauge and global. The cleanest examples live in 2+1 dimensions. Take the $O(2)$ Wilson–Fisher fixed point — the critical point of a complex scalar with $|\phi|^4$ interaction, the universality class of the superfluid transition in ^4He . **Particle–vortex duality** asserts that the very same conformal field theory is reached by a *different* Lagrangian: a complex scalar coupled to a dynamical $U(1)$ gauge field (the abelian Higgs model), with the global symmetry that counts particles on one side realized as the symmetry that counts magnetic flux on the other,

$$|\partial_\mu \phi|^2 - |\phi|^4 \longleftrightarrow |D_b \hat{\phi}|^2 - |\hat{\phi}|^4 + (\text{Maxwell for } b), \quad (\text{I.1.7})$$

both tuned to criticality.⁶ Neither side is “the” description: the elementary field of one is a topological soliton of the other. In 2016 this old duality was joined by a *seed* boson–fermion duality — the gauged Wilson–Fisher boson at Chern–Simons level one is a free Dirac fermion — from which a

⁴Witten, [hep-th/9507121](#). The theory is labeled by an A–D–E Lie algebra; “the” (2, 0) theory of rank N is the A_{N-1} case.

⁵Henningson–Skenderis, [hep-th/9806087](#), from holography; Harvey–Minasian–Moore from anomaly inflow. No free-field counting reproduces N^3 .

⁶Peskin (1978) and Dasgupta–Halperin (1981) originally; stated in modern form, with the whole web it seeds, in Seiberg–Senthil–Wang–Witten [1606.01989](#), whose §1 is the recommended first reading on what duality means.

whole web of equivalences follows by formal gauging operations, with experimental relevance from quantum Hall transitions to topological insulator surfaces. If a single conformal field theory can be presented as bosons, as gauged bosons, or as fermions, then asking “which Lagrangian is the theory?” is asking which coordinate chart is the manifold. (Part VI; [1606.01989](#).)

Crack 3: computation without a Lagrangian. The cracks above could be dismissed as exotica — supersymmetric, low-dimensional. The third cannot, because it concerns the most pedestrian strongly-coupled QFT in nature: the critical point of the three-dimensional Ising model, the universality class of the liquid–vapor critical point and of uniaxial magnets. The [conformal bootstrap](#) determines its low-lying spectrum to stunning precision,

$$\begin{aligned}\Delta_\sigma &= 0.5181489(10), \\ \Delta_\epsilon &= 1.412625(10).\end{aligned}\tag{I.1.8}$$

These are the scaling dimensions that fix the classic critical exponents η and ν , and they are obtained using as input *no Lagrangian, no lattice, and no expansion parameter*: only unitarity, conformal symmetry, and the associativity (crossing) of the operator algebra, implemented as a numerical exclusion problem on the space of possible CFT data.⁷ These values are more precise than any Monte Carlo simulation of any lattice realization, and incomparably more precise than the ϵ -expansion applied to the ϕ^4 Lagrangian that ostensibly “defines” the theory. Consistency conditions on the observables, taken alone, pin the theory down: the observables behave as primary data, the Lagrangian as one expendable presentation among several. How this is possible — why crossing plus unitarity is so much stronger than it looks — is the subject of Part III. (Part III.2; Rychkov’s history [2509.02779](#) tells the remarkable story, from Polyakov’s 1974 manifesto to the 2012 Ising island.)

Crack 4: the Lagrangian that defines nothing. Finally, the recipe can fail at its own game. Its standard pedagogical example — the one in which every textbook teaches renormalization — is ϕ^4 theory in $d = 4$. In 2019 Aizenman and Duminil-Copin closed a forty-year-old question by proving that the continuum limits of lattice ϕ_4^4 and of the 4d Ising model, at or near criticality, are *Gaussian*: every choice of bare couplings, however strong, yields free fields when the cutoff is removed.⁸ The interacting continuum ϕ_4^4 theory — the object steps 1–4 purport to construct — does not exist. What does exist, and what the textbook computations were always really about, is an *effective* field theory with a finite cutoff, whose couplings at low energy are insensitive to the details at the cutoff; this is also the modern understanding of the Standard Model’s scalar sector. Triviality makes the relocation unavoidable: “quantize this Lagrangian” was never a definition, but a procedure whose meaning depends on where in theory space you stand and at what resolution you look. (Part VIII.6; the $d = 4 - \epsilon$ Wilson–Fisher fixed point, which *does* exist and explains why the recipe works so well just below four dimensions, is constructed honestly in chapter I.6.)

The four cracks share a single diagnosis. In each case the failure is not of quantum field theory but of the identification of a theory with its presentation. The $(2, 0)$ theory has no presentation

⁷Numbers from Poland–Rychkov–Vichi [1805.04405](#), Table II, obtained in Simmons–Duffin [1612.08471](#); the rigorous-island determinations are by Kos–Poland–Simmons–Duffin–Vichi [1603.04436](#). Rychkov–Su [2311.15844](#) reviews the current state of the art.

⁸Aizenman–Duminil-Copin [1912.07973](#). “Marginal” triviality: $d = 4$ is the boundary case, where the proof requires a logarithmic improvement over the classic tree-diagram bounds that settled $d \geq 5$ in the 1980s. This is a theorem of mathematics, not a scenario.

of Lagrangian type; the $O(2)$ fixed point has several; the 3d Ising CFT is best accessed with no presentation at all; ϕ_4^4 is a presentation of nothing. The conclusion was stated by Polyakov already in 1974, in the introduction — it reads as a manifesto — of the paper that founded the conformal bootstrap: the description of a critical theory “should be ‘democratic’ with respect to the different fields,” i.e. should not privilege whatever variables happen to appear in someone’s bare Hamiltonian.⁹

Idea I.1.1 (A theory is its observables)

A quantum field theory is the total structure of its observables — correlation functions, spectra, partition functions, extended operators — not any particular procedure for generating them. Lagrangians, lattice models, bootstrap data sets, and holographic duals are **presentations** of a theory: coordinate descriptions, each with a domain of convenience, none privileged. Some theories have many presentations; some have none we know.

This is the first organizing principle of these notes, and everything in this part flows from taking it seriously. It immediately raises the question that the rest of this chapter answers: if not a Lagrangian, then *what*? What is the structure that all presentations of a given theory present?

Preview I.1.1 (Where the cracks are repaired)

Theories without Lagrangians are constructed (by compactification and by bootstrap) in Part VII.4 and Part III. Dualities are stress-tested and partially derived in Part VI, with the tools — symmetries, anomalies, line spectra — built in Part II. The bootstrap occupies Part III. Triviality, constructive field theory, and the lattice–continuum dictionary occupy Part VIII. The question “what would a complete definition even look like?” returns in Part XII.1.

Exercise I.1.4 (thinker). Suppose a colleague proposes that two Lagrangians, like the two sides of (I.1.7), describe the same infrared theory. Neither side is solvable. List every quantity you could in principle compare *without* solving either theory. (Symmetries? Their representations on local and extended operators? Anomalies? Partition functions on nontrivial geometries? Behavior under deformations?) Keep your list; Part II is the systematic theory of exactly these duality invariants, and Part VI grades proposals against them.

Hint: The most useful invariants are the ones that cannot flow: discrete data, quantized coefficients, anything protected under the renormalization group. The modern checklist begins with the global symmetry *category* and its ‘t Hooft anomalies.

I.1.3 What a quantum field theory assigns

If a theory is not its Lagrangian, our definition-shaped question has to be approached from the other side: not “what generates the theory?” but “what does the theory *provide*?” Think of it as defining the species by its behavior. Whatever a quantum field theory is, any presentation of it must produce a definite collection of physical data, and two presentations describe the same theory

⁹Polyakov, *Non-Hamiltonian approach to conformal quantum field theory*, ZhETF **66** (1974) 23; quoted and contextualized in Rychkov [2509.02779](#), §5, which reproduces the full passage. Seiberg’s one-slide summary at the 2011 Solvay meeting — no Lagrangian for some theories, several for others, magic in amplitudes, not mathematically rigorous — is the same diagnosis a generation later.

exactly when all of that data agrees. The modern consensus on the inventory — assembled from the axiomatic tradition and sharpened by the functorial viewpoint — is the following.¹⁰

Definition I.1.1 (The inventory: data of a QFT)

A quantum field theory $\mathcal{T}$ in d spacetime dimensions assigns:

1. to a closed d -dimensional spacetime M_d (with metric, and possibly other background fields): a **partition function** $Z_{\mathcal{T}}[M_d] \in \mathbb{C}$;
2. to points $x_1, \dots, x_n \in M_d$ decorated by labels $\mathcal{O}_1, \dots, \mathcal{O}_n$ drawn from a set of **local operators**: correlation functions $\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_{M_d}$;
3. to a closed $(d-1)$ -dimensional space Σ_{d-1} : a **Hilbert space** $\mathcal{H}_{\Sigma}$ of states, with a d -manifold bounded by Σ defining a state in $\mathcal{H}_{\Sigma}$ (cutting and gluing);
4. to closed submanifolds of intermediate dimension, decorated by labels from sets of **extended operators** (line operators, surface operators, $\dots$), and to boundaries and interfaces: correlation functions in their presence;

subject to consistency conditions — locality of the operators, compatibility of cutting and gluing, unitarity/reflection positivity, convergence structures for the operator product expansion — that subsequent chapters make precise.

Each entry of the inventory gets a chapter of its own (local operators in I.2, extended operators in I.3, states and gluing in I.4), so here we stay at the level of orientation, and the orientation is best gained by noticing what the list does *not* contain. There are no fields: “ ϕ ” appears nowhere, only the operators $\mathcal{O}_i$, and dualities like (I.1.7) show that what one presentation calls an elementary field another calls a composite or a soliton. There is no Lagrangian and no gauge group — the gauge “symmetry” of the abelian Higgs side of (I.1.7) is invisible in the inventory, a redundancy of that presentation rather than a property of the theory (a point developed carefully in Part V.1). And there are no particles: particles are states in $\mathcal{H}$ that happen to organize themselves into asymptotic excitations when the theory is gapped, an excellent emergent concept in some theories and simply absent in others (a conformal theory like 3d Ising has a continuous spectrum of scaling states and no particles at all). Item 4 — defects — would not have appeared on anyone’s list in 2000 and is the most consequential modern addition: theories with identical local correlators can differ in their extended operators, so the extended sector is part of the *definition*, not a garnish. That lesson gets chapter I.3 to itself.

The right way to internalize the inventory is to watch all of it materialize in a system so small that nothing can hide. Quantum mechanics is a quantum field theory in $d = 1$ — the “fields” are functions of time only — and one particular quantum mechanics is, for reasons that will keep paying off through Part V, an ideal specimen for the purpose.

¹⁰This is a physicist’s distillation of structures formalized, in different fragments, by the Wightman, Osterwalder–Schrader, Haag–Kastler, and Atiyah–Segal axiom systems; Section I.1.6 below compares them. The geometric (“assign data to manifolds”) formulation follows Segal, recently extended by Kontsevich–Segal [2105.10161](#).

Example I.1.2 (The inventory in $d = 1$: a particle on a circle)

Take a free particle of unit mass on a circle, $q \simeq q + 2\pi$, with Lorentzian Lagrangian

$$L = \frac{1}{2}\dot{q}^2 + \frac{\theta}{2\pi}\dot{q}. \quad (\text{I.1.9})$$

The second term is our first **topological term**: a total derivative, with no effect on the equations of motion, whose coefficient θ will nonetheless turn out to be physical — the one-dimensional ancestor of the θ -angles of Yang–Mills theory.^a

The Hamiltonian presentation. The momentum conjugate to q is $\Pi = \dot{q} + \frac{\theta}{2\pi}$, and the Hamiltonian is

$$H = \frac{1}{2} \left(\Pi - \frac{\theta}{2\pi} \right)^2. \quad (\text{I.1.10})$$

The inventory begins to fill in. *States* (item 3): space is a point, and the Hilbert space assigned to it is $\mathcal{H} = L^2(S^1)$, spanned by $\langle q|n\rangle = e^{inq}/\sqrt{2\pi}$ with $n \in \mathbb{Z}$ — single-valuedness of the wavefunction quantizes Π to integers. The spectrum is

$$E_n = \frac{1}{2} \left(n - \frac{\theta}{2\pi} \right)^2. \quad (\text{I.1.11})$$

Local operators (item 2): the well-defined operators are built from $e^{\pm iq}$ and Π (q itself is not single-valued, hence not an operator — already a hint that “the field” is not sacred), with correlators like $\langle 0|e^{iq(t)}e^{-iq(0)}|0\rangle$ computable by inserting energy eigenstates. *Partition function* (item 1): the closed 1-manifold is a circle; continuing to Euclidean time and traversing a circle of circumference β ,

$$Z(\beta, \theta) = \text{tr } e^{-\beta H} = \sum_{n \in \mathbb{Z}} \exp \left[-\frac{\beta}{2} \left(n - \frac{\theta}{2\pi} \right)^2 \right]. \quad (\text{I.1.12})$$

Two features of the spectrum (I.1.11) repay a hard look. First, θ *is an angle*: $\theta \rightarrow \theta + 2\pi$ shifts $n \rightarrow n + 1$ and leaves the spectrum — and every entry of the inventory — invariant. (Operatorially, $H_{\theta+2\pi} = e^{iq} H_{\theta} e^{-iq}$.) Second, the invariance is realized *nontrivially*: as θ runs from 0 to 2π the levels cross, and at exactly $\theta = \pi$ the ground state is two-fold degenerate, $E_0 = E_1$. A total-derivative coupling has produced physical consequences, periodic in 2π , with a level crossing at the symmetric point that — as we will see in Parts II.7 and V.3 — no deformation respecting the symmetries can remove. This is the entire conceptual skeleton of “Yang–Mills at $\theta = \pi$,” in a system solvable by inspection.

The path-integral presentation. Now compute $Z(\beta, \theta)$ the other way, as a sum over all maps from the Euclidean time circle to the target circle. Such maps fall into disconnected topological classes labeled by the **winding number** $w \in \mathbb{Z}$; in each class the topological term contributes the *phase* $e^{i\theta w}$ — topological terms acquire no factor of i under Wick rotation and remain imaginary in Euclidean signature (Appendix A; this is the single most common sign error in the subject). The action in the class w is minimized by $q_w(\tau) = 2\pi w\tau/\beta$, with $S_E[q_w] = 2\pi^2 w^2/\beta$, and the Gaussian fluctuations around q_w are w -independent, contributing the free-particle

factor $\sqrt{2\pi/\beta}$ (Exercise I.1.7). Hence

$$Z(\beta, \theta) = \sqrt{\frac{2\pi}{\beta}} \sum_{w \in \mathbb{Z}} \exp \left[-\frac{2\pi^2 w^2}{\beta} + i\theta w \right]. \quad (\text{I.1.13})$$

Expressions (I.1.12) and (I.1.13) look nothing alike — one is a sum over momenta with θ in the exponents' shifts, the other a sum over windings with θ as a phase — yet they are equal, exactly, by the Poisson resummation formula $\sum_n f(n) = \sum_w \hat{f}(2\pi w)$ applied to the Gaussian (Exercise I.1.6). They are two presentations of one theory, and they are useful in opposite regimes: at low temperature ($\beta \rightarrow \infty$) the Hamiltonian sum is dominated by its first term or two while the winding sum converges slowly, and at high temperature ($\beta \rightarrow 0$) the roles reverse. This small duality is worth remembering as a template: the same observable admits two expansions, each accurate in the regime where the other fails. Part VII is devoted to this phenomenon in quantum field theory proper.

^aThis example is worked out, as a warm-up for 4d gauge theory, in Gaiotto–Kapustin–Komargodski–Seiberg [1703.00501](#), Appendix D.1, whose conventions we follow. It reappears in Córdova–Freed–Lam–Seiberg [1905.09315](#) as the simplest anomaly in parameter space.

The example is humble, but every theme of this chapter is present in it: data assigned to geometric objects (a Hilbert space to a point, a number to a circle); operators defined by the theory rather than by the classical configuration space (e^{iq} , not q); a parameter space with nontrivial topology ($\theta \in S^1$) and physics pinned to its special points; and two presentations related by a nontrivial identity. When in later chapters the going gets abstract, return here.

There is one more thing the example quietly demonstrates: at $\theta = \pi$ the classical symmetry $C : q \rightarrow -q$ combines with the shift symmetry of the free particle into a group that is realized on $\mathcal{H}$ only *projectively* — the operators implementing it commute only up to a phase. The two-fold degeneracy is the direct consequence, and the projective phase is the simplest example of a '**t Hooft anomaly**', the exactly-conserved quantity of the renormalization group that Part II builds into the subject's sharpest tool.¹¹ We flag it and move on; it returns in II.7 with the machinery to do it justice.

Exercise I.1.5 (easy). Show directly that $U = e^{iq}$ is a well-defined unitary operator on $L^2(S^1)$, that $UH_\theta U^\dagger = H_{\theta+2\pi}$, and that U maps $|n\rangle \rightarrow |n+1\rangle$. Conclude that all inventory data is 2π -periodic in θ , and identify which states cross at $\theta = \pi$.

Hint: Work in the $|n\rangle$ basis throughout; e^{iq} is a shift operator on momentum eigenstates.

Exercise I.1.6 (classic). Prove the Poisson resummation formula $\sum_{n \in \mathbb{Z}} f(n) = \sum_{w \in \mathbb{Z}} \hat{f}(2\pi w)$, $\hat{f}(k) = \int dx f(x) e^{-ikx}$, and use it to derive (I.1.13) from (I.1.12), including the prefactor and the sign of the phase.

Hint: The Dirac comb $\sum_n \delta(x-n)$ is periodic, so it equals its own Fourier series; integrate f against both expressions. For the application: complete the square in (I.1.12) first.

Answer: With $f(x) = e^{-\frac{\beta}{2}(x-\frac{\theta}{2\pi})^2}$ one finds $\hat{f}(2\pi w) = \sqrt{2\pi/\beta} e^{-2\pi^2 w^2/\beta} e^{-i\theta w}$; relabeling $w \rightarrow -w$ gives (I.1.13).

Exercise I.1.7 (medium). Justify the prefactor in (I.1.13): show that the Gaussian integral over periodic fluctuations $\delta q(\tau)$ around any winding saddle is independent of w , and that the zero mode

¹¹GKKS [1703.00501](#), eqs. (D.5)–(D.10): the symmetry group at $\theta = \pi$ is a central extension of $O(2)$ — $\text{Pin}(2)$ — and a $\text{Pin}(2)$ that cannot be reduced to $O(2)$ forces degenerate pairs.

(the constant shift of q) produces a factor of the target-circle circumference, yielding in total the free-particle result $2\pi/\sqrt{2\pi\beta} = \sqrt{2\pi/\beta}$.

Hint: $S_E[q_w + \delta q] = S_E[q_w] + \frac{1}{2} \int \delta \dot{q}^2$ exactly — the action is quadratic and the cross term integrates to zero by periodicity. For the normalization, compare with the diagonal heat kernel of a free particle on a line, $\langle q|e^{-\beta H}|q\rangle = 1/\sqrt{2\pi\beta}$.

The particle on a circle entered through a Lagrangian, however — hardly a clean break from the old paradigm. So let us fill the same inventory from the opposite end of physics: a statistical lattice with no Lagrangian, no continuum, and at first sight no quantum mechanics at all.

Example I.1.3 (The Ising chain, read as a field theory)

Put classical spins $s_i = \pm 1$ on N sites of a periodic chain, with energy favoring alignment, and define

$$Z_N = \sum_{\{s_i = \pm 1\}} \prod_{i=1}^N e^{J s_i s_{i+1}}, \quad (\text{I.1.14})$$

$$s_{N+1} \equiv s_1,$$

where $J > 0$ absorbs the temperature. This is a Euclidean QFT presentation in which spacetime itself is discrete: the chain is a one-dimensional “spacetime,” (I.1.14) its partition function on a (discrete) circle.

The structure of the inventory emerges from a single move. Define the 2×2 **transfer matrix** with entries $T_{ss'} = e^{J s s'}$,

$$T = \begin{pmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{pmatrix} = e^J I + e^{-J} \sigma_x, \quad (\text{I.1.15})$$

$$\lambda_{\pm} = 2 \cosh J, 2 \sinh J.$$

Multiplying along the chain and summing over spins,

$$Z_N = \text{tr } T^N = \lambda_+^N + \lambda_-^N. \quad (\text{I.1.16})$$

Compare item by item with the previous example. The *Hilbert space* assigned to a slice of this spacetime — a single site — is $\mathbb{C}^2$, the span of $|\uparrow\rangle, |\downarrow\rangle$. The *Euclidean time evolution* over one lattice step is T , the discrete stand-in for e^{-aH} . The *partition function on a circle* is a trace, exactly as $\text{tr } e^{-\beta H}$ was. And the *correlators*: inserting σ_z at sites 0 and r ,

$$\langle s_0 s_r \rangle = \frac{\text{tr} (\sigma_z T^r \sigma_z T^{N-r})}{\text{tr } T^N} \xrightarrow{N \rightarrow \infty} \left(\frac{\lambda_-}{\lambda_+} \right)^r = (\tanh J)^r = e^{-r/\xi}, \quad (\text{I.1.17})$$

with **correlation length** $\xi = 1/\ln \coth J$ (Exercise I.1.9). Two general morals hide in this formula. The correlation length is the inverse of the *gap*: $1/\xi = \ln(\lambda_+/\lambda_-)$ is the spacing of the transfer matrix’s spectrum, the discrete avatar of $m = E_1 - E_0$. And **cluster decomposition** — correlations factorize at long distance, $\langle s_0 s_r \rangle \rightarrow \langle s \rangle^2$ — is on explicit display: it is nothing but the statement that the largest eigenvalue dominates.

Now take the *continuum limit*, the move that will occupy all of Part VIII; here it can be done exactly. Give the lattice a physical spacing a and send $a \rightarrow 0$, $N \rightarrow \infty$. For the limit

to retain any physics, the correlation length *in lattice units* must diverge, so $J \rightarrow \infty$ too; the combination to hold fixed is the physical gap. Writing $T = e^J (I + e^{-2J} \sigma_x + \dots)$ and matching to $T \simeq e^{-aH + \text{const}}$,

$$\begin{aligned} H &= -\Gamma \sigma_x, \\ \Gamma &= \frac{e^{-2J}}{a} \text{ held fixed as } a \rightarrow 0. \end{aligned} \tag{I.1.18}$$

The continuum limit of the classical Ising chain is the quantum mechanics of a single spin in a transverse field — Hilbert space $\mathbb{C}^2$, gap 2Γ , physical correlation length $\xi_{\text{phys}} = a\xi \rightarrow 1/2\Gamma$. Every microscopic detail (the lattice, the specific nearest-neighbor form, corrections to the exponentiation) has washed out into one parameter. This is **universality** in its simplest instance, and it is the mechanism that makes “QFT from a lattice” meaningful at all: the continuum theory remembers only the data that survives the limit.

Two systems could hardly look more different than a quantum particle on a circle and a classical chain of spins, yet they filled the same inventory in the same pattern: Hilbert space on a slice, (Euclidean) evolution operator, trace on a circle, correlators with definite decay. That pattern — not any Lagrangian — is what Definition I.1.1 axiomatizes. The chain also displays, in miniature, the claim from the chapter opening that the standard paradigm gets *relocated* rather than discarded: “quantize a Lagrangian” lives inside the inventory picture as one family of presentations (start from classical configurations, build T or e^{iS} , read off the data), valid whenever it converges on something — and the chain shows there are presentations, perfectly good ones, that never mention a classical action in the continuum.

One disclaimer, before the lattice picture settles in too comfortably as “the” nonperturbative definition. In $d = 1$ the continuum limit (I.1.18) is a theorem of elementary analysis. In higher dimensions, “the continuum limit exists and is universal” is a theorem only in special cases, a well-supported conjecture in most, and — this is the surprise of the 2020s — *false* in instructive ones, where lattice models produce continuum descriptions that violate cherished axioms (fractons, Part VIII.5) or where the limit exists but is free despite all appearances (triviality, crack 4). The lattice is a presentation too, with its own dictionary problems; Part VIII is the audit.

Exercise I.1.8 (easy). Add a magnetic field: $Z_N = \sum_{\{s\}} e^{\sum_i (J s_i s_{i+1} + h s_i)}$. Find the symmetric transfer matrix, its eigenvalues, and the magnetization $m = \langle s_i \rangle$ in the thermodynamic limit.

Hint: Use $T_{ss'} = e^{J s s' + \frac{h}{2}(s+s')}$ to keep T symmetric. Then $m = \partial \ln \lambda_+ / \partial h$ at fixed J .

Answer: $\lambda_{\pm} = e^J \cosh h \pm \sqrt{e^{2J} \sinh^2 h + e^{-4J}}$, giving $m = \sinh h / \sqrt{\sinh^2 h + e^{-4J}}$: smooth for every $J < \infty$, with no spontaneous magnetization at $h = 0$.

Exercise I.1.9 (classic). Derive (I.1.17): insert σ_z operators into the trace, evaluate in the eigenbasis of T , and keep track of the finite- N corrections. Verify cluster decomposition and extract ξ .

Hint: σ_z is off-diagonal in the eigenbasis $|\pm\rangle = (|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$: $\sigma_z |\pm\rangle = |\mp\rangle$. The exact answer is

$$\frac{\lambda_+^{N-r} \lambda_-^r + \lambda_-^{N-r} \lambda_+^r}{\lambda_+^N + \lambda_-^N}.$$

Exercise I.1.10 (medium). Make the continuum limit (I.1.18) precise. Show that the corrections to $T \approx e^J e^{-aH}$ are of relative order $e^{-4J} = (a\Gamma)^2$ and hence vanish in the limit; show that the limit of N -site quantities with $Na = \beta_{\text{phys}}$ fixed reproduces the spin- $\frac{1}{2}$ thermal traces; and verify $\xi_{\text{phys}} \rightarrow 1/2\Gamma$ from (I.1.17).

Exercise I.1.11 (thinker). The chain never orders: Exercise I.1.8 shows the free energy is analytic in (J, h) for all finite J . Give two proofs that no phase transition can occur in the 1d Ising model with short-range interactions: (a) algebraic — the transfer matrix has strictly positive entries; invoke the Perron–Frobenius theorem to show $\lambda_+ > \lambda_-$ always, so the gap never closes; (b) physical — estimate the free energy of a single domain wall in a chain of length L and show entropy always wins (Peierls’ argument, run in reverse). What do these arguments use about $d = 1$, and which step fails in $d = 2$?

Hint: For (b): a wall costs fixed energy $2J$ but can sit at any of $\sim L$ positions, so $\Delta F = 2J - T \ln L \rightarrow -\infty$. In $d = 2$ a wall is a line whose energy *and* entropy both scale with its length — the competition becomes a genuine fight, and Peierls showed order wins at low temperature.

I.1.4 Locality, the one non-negotiable

The inventory of Definition I.1.1, as stated, is satisfied by systems nobody would call a field theory — nothing yet distinguishes the Ising chain from a quantum computer’s register or from a single harmonic oscillator, which also assigns Hilbert spaces and thermal traces. What makes a theory a theory *of fields* is not the size of its Hilbert space but the way its observables are organized by *geometry*: degrees of freedom are distributed over space, and they interact only with their neighbors. Every word of “quantum field theory” except “locality” is negotiable — we have already negotiated away the fields — and the modern attitude is to treat locality as the actual subject. QFT is what you get when many degrees of freedom are organized locally; everything else is presentation.

This one idea wears different uniforms in different formulations, and learning to recognize it in all of them is part of the training:

- *In the operator content*: operators are labeled by the points (and submanifolds) where they live, and the product of two nearby local operators is again a local operator — the **operator product expansion**, the subject of chapter I.2.
- *In Lorentzian signature*: measurements at spacelike separation are independent,

$$[\mathcal{O}_1(x), \mathcal{O}_2(y)] = 0 \text{ for } (x - y)^2 > 0,$$

the statement of **microcausality** (Exercise I.1.12).

- *In Euclidean correlators*: **cluster decomposition**, already met in (I.1.17) — distant regions decorrelate.
- *In the dynamics*: the Hamiltonian is an integral of a local density. Better, because it is presentation-independent: the theory possesses a local **stress tensor** $T_{\mu\nu}(x)$ — the operator that measures energy and momentum density, and the response of the theory to deforming the geometry it lives on. The stress tensor is the one operator every local QFT shares, and its existence is the sharpest single criterion for “is this a local field theory?” (Chapter I.2 develops it; Part X makes it the protagonist.)
- *In the path integral and on the lattice*: the action/energy is a sum of terms each involving only nearby points — nearest-neighbor couplings in (I.1.14) — which is what makes *cutting* spacetime along a slice meaningful and gives the gluing rules of item 3 of the inventory their content (chapter I.4).

These are not independent axioms but facets of one property, and their precise interrelation (which implies which, under what hypotheses) is genuinely subtle — it is the content of the reconstruction theorems discussed in chapter I.2. For orientation, the slogan suffices:

Idea I.1.2 (Locality is the subject)

A quantum field theory is a quantum system whose observables are organized by the geometry of spacetime, with independence at large separation and a local stress tensor as the operator witness. Locality is the one ingredient of the textbook definition that survives unweakened into every modern formulation — and the boundary of QFT as a subject traces exactly where locality starts to fail.

The final clause matters because the failure modes of locality are themselves instructive. A quantum system with all-to-all couplings — mean-field models, or the SYK model beloved of holographers — has no notion of spatial distance and is quantum mechanics, not field theory (often a useful caricature of one, which is its point). More exotically, the *little string theories* arising on NS5-branes have local energy *density* in a loose sense but provably no stress tensor and no local operators creating their excitations; they sit just outside the boundary, and Part XII.3 visits them as instructive aliens. And systems with power-law interactions, like the long-range Ising model $J_{ij} \sim |i - j|^{-1-s}$, interpolate: for s large enough they flow to ordinary local field theories, for small s to genuinely nonlocal continuum theories, with a phase boundary in between — locality of the IR theory is itself a dynamical question.

Note I.1.1. Locality should not be confused with *Lorentz invariance*, its frequent companion. Nonrelativistic, anisotropic, and Lifshitz-type field theories are perfectly local, organized by geometry, and within our subject (condensed matter supplies them in bulk); they simply weight time and space asymmetrically. The axiom systems of Section I.1.6, built around the Poincaré group, mostly exclude them — one of several signs that the existing axioms draw the boundary in the wrong place. Fractons (Part VIII.5) push on exactly this seam.

Why promote locality, rather than, say, unitarity or symmetry, to the definition's one fixed point? Because locality is what gives quantum field theory its unreasonable universality — the empirical fact, which the next section turns into a picture, that the same continuum theories appear as descriptions of magnets, fluids, metals, the QCD vacuum, and the early universe. The mechanism is the renormalization group: whatever the microscopic degrees of freedom, so long as they are many and locally coupled, viewing them at longer and longer wavelengths erases microscopic detail in a controlled way, and the possible long-distance destinations are heavily constrained precisely *because* locality survives the coarse-graining. Wilson's insight, restated in the vocabulary of this chapter: QFT is not fundamentally a theory of elementary particles; it is the universal effective description of locally-organized many-body systems, of which elementary particle physics is one (spectacular) instance.¹²

Exercise I.1.12 (classic). (Microcausality for the free scalar.) In the free massive scalar theory in $d = 4$ with mostly-plus signature, the commutator of fields is a c-number, $[\phi(x), \phi(y)] = \Delta(x - y)I$. Compute Δ from the mode expansion and prove it vanishes whenever $x - y$ is spacelike.

¹²Seiberg's *Thoughts About QFT* presses this point: “quantum field theory is the modern calculus — the natural language for describing diverse phenomena.” The high-energy and condensed-matter views of what a QFT is are compared, as two charts on one space of theories, in Section I.1.5.

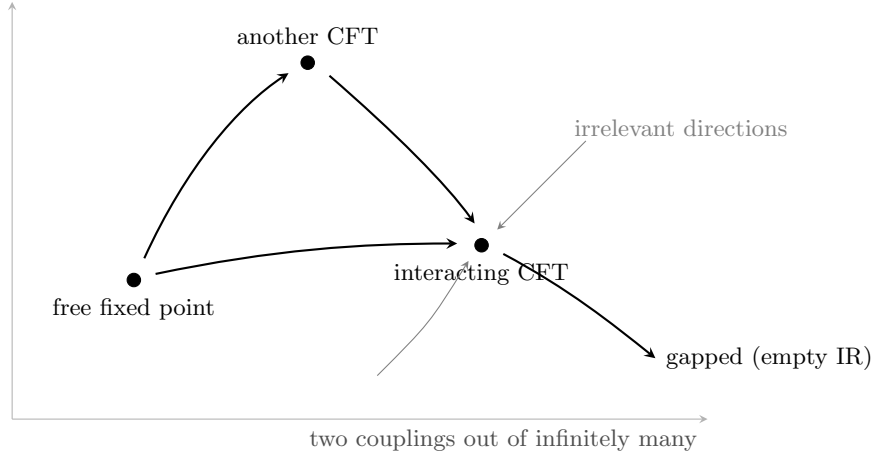


Figure 1: Theory space, schematically. Points are theories; the renormalization group moves every non-fixed theory toward longer distances along the arrows; fixed points (dots) are scale-invariant theories — CFTs — and serve as the landmarks from which the rest of the space is charted. Flows out of a fixed point follow its relevant directions (finitely many); the gray arrows are irrelevant directions, along which flows converge *into* a fixed point, producing universality.

Hint: $\Delta(x-y) = \int \frac{d^3 p}{(2\pi)^3 2E_p} \left(e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)} \right)$ with $p \cdot x = -Et + \mathbf{p} \cdot \mathbf{x}$ in mostly-plus. For spacelike separation there is a Lorentz frame with $x^0 = y^0$; show the integrand is then odd under $\mathbf{p} \rightarrow -\mathbf{p}$. Why does this argument fail (as it must) for timelike separation?

I.1.5 Theory space: the picture to keep

We can now assemble the chapter’s deliverable: the picture within which all twelve parts of these notes take place. It needs three ingredients, all already on the table — the inventory (a theory is a consistent assignment of observables), locality (the assignment is organized by geometry), and universality (microscopically different systems share long-distance descriptions).

Imagine, then, the set of all quantum field theories in d dimensions: every consistent inventory, whether or not anyone has presented it. Call it **theory space**. A point of theory space is a theory — the 3d Ising CFT is a point, the free photon is a point, QCD with physical quark masses is a point. A Lagrangian is not a point: it is a *name* for a point (and dualities are the statement that points have several names). This space is the arena. Its geography is given by the **renormalization group**: inspecting a theory at successively longer distances — integrating out, coarse-graining, “zooming out” — produces a one-parameter motion through theory space,

$$\frac{d}{d \ln(\text{scale})}(\text{theory}) = \beta(\text{theory}), \quad (\text{I.1.19})$$

a flow, of which chapter I.6 gives the honest construction. Here we need only its fixed-point structure, and the picture in Figure 1.

A **fixed point** of the flow is a theory invariant under zooming out: a scale-invariant theory. In all known unitary local examples scale invariance comes enhanced to conformal invariance,¹³ so the

¹³Proven in $d = 2$ (Zamolodchikov–Polchinski), established under mild assumptions in $d = 4$, and true in every known unitary example; see Nakayama’s review 1302.0884. We will use “fixed point” and “CFT” interchangeably, flagging the (conjectural) gap when it matters.

landmarks of theory space are **conformal field theories** — theories with no intrinsic length scale, looking the same at every magnification. The 3d Ising CFT of crack 3 is precisely such a landmark, which is why it could be studied with no reference to any flow that reaches it. It is worth pausing on how recent the consensus behind this picture is: when Polyakov suggested at a 1970 conference that critical points possess scale (indeed conformal) invariance, both C. N. Yang and T. T. Wu rose to object that scale invariance in statistical mechanics was neither mathematically nor physically justified.¹⁴

Near a fixed point, the flow linearizes, and the eigendirections sort themselves by their behavior under zooming out: **relevant** perturbations grow (they carry you away from the fixed point), **irrelevant** ones shrink (flows converge back), **marginal** ones sit at the threshold. This tangent-space structure is computable the moment you know the scaling dimensions of the fixed point's operators. At the free (Gaussian) fixed point of a single scalar, demanding that the kinetic action $\int d^d x (\partial\phi)^2$ be dimensionless fixes

$$\begin{aligned} [\phi] &= \frac{d-2}{2}, \\ \left[\int d^d x g_n \phi^n\right] &= 0, \\ [g_n] &= d - n \frac{d-2}{2}. \end{aligned} \tag{I.1.20}$$

The perturbation $g_n \phi^n$ is relevant when $[g_n] > 0$, i.e. $n < 2d/(d-2)$: in $d = 4$ the relevant interactions stop at ϕ^3 with ϕ^4 marginal; in $d = 3$, at ϕ^5 with ϕ^6 marginal; in $d = 6$, already ϕ^3 is the marginal frontier; and in $d = 2$, where $[\phi] = 0$, *every* potential is relevant — the first hint that two dimensions is a world of its own (Part III.4). Operators with more derivatives, like $(\partial\phi)^4$, are irrelevant in all $d \geq 2$. The pattern generalizes beyond free fixed points — at an interacting CFT the same counting runs on the true scaling dimensions Δ , with “relevant” meaning $\Delta < d$ — and the crucial finiteness persists: *a fixed point has finitely many relevant directions*.

That finiteness demystifies, in two sentences, the step of the standard recipe that was always presented as a mystery. “Renormalizable interactions” are exactly the relevant and marginal directions at the Gaussian fixed point: the textbook instruction *write only renormalizable terms* is the geometric statement *parametrize the flows leaving this fixed point*, and the miracle that infinitely many possible counterterms reduce to a few measurable couplings is no miracle — the irrelevant directions die under the flow, so the IR physics could never have depended on them. Conversely, every QFT used in physics is an **effective field theory**: a local chart around a fixed point, accurate up to corrections suppressed by powers of the distance scale, with the irrelevant operators recording — in a decaying, organized hierarchy — the memory of whatever lies upstream. The expanded version of these claims, with the Wilson–Fisher fixed point in $d = 4 - \epsilon$ constructed as the first nontrivial landmark, is chapter I.6’s job.

The same picture finally lets us say sharply what was right about the two traditional answers to “what is a QFT?”, which can now be seen as two *charts* on theory space.¹⁵ The high-energy chart starts at a fixed point and deforms by its finitely many relevant and marginal couplings (plus, implicitly, infinitesimal irrelevant ones): a small, rigid parametrization, anchored on a landmark — this is the Lagrangian paradigm, properly understood. The condensed-matter chart starts from a lattice model at the cutoff and allows *arbitrary* local couplings and arbitrary added degrees of freedom: an enormous, floppy parametrization, in which the natural questions are about the possible

¹⁴The exchange is transcribed in Rychkov [2509.02779](#), §2 — worth reading in full as a calibration of how non-obvious the now-obvious can be.

¹⁵This comparison is drawn from Seiberg’s *Thoughts About QFT*, whose slides phrase it as “QFT in high-energy theory” vs. “QFT in condensed-matter physics.”

long-distance *phases* and the walls between them. The two charts overlap where both converge, and translating between them — which lattice models flow to which continuum landmarks, and what data survives the trip — is a running concern of these notes, culminating in Part VIII.

Remark I.1.2 (How much structure does theory space have?). It is tempting to ask global questions: is theory space connected? What are its components? A sharp version uses the notion of a **deformation class**: two theories are in the same class if one can interpolate between them by adding degrees of freedom at high energy and moving along couplings, preserving the symmetries one cares about. Seiberg has conjectured that the deformation class is determined by precisely two pieces of data — the global symmetry and its 't Hooft anomalies — both of which are exactly preserved along any flow. If true, this gives theory space a remarkably simple coarse topology, with the anomalies of Part II as its complete invariants. The conjecture motivates much current work; we will be in a position to appreciate its content (and its supersymmetric stress tests) after Part II. For now it is a preview of how seriously the picture of this section can be taken: theory space is not a metaphor but an object with topology, invariants, and conjectures of its own.

The rest of these notes is organized as questions about Figure 1. What structures are transported *exactly* along the flow lines, surviving from any presentation down to the deep infrared? (Symmetries and their anomalies: Part II.) What are the fixed points, studied as intrinsic objects? (Conformal field theory: Part III.) Which flows between landmarks are possible, and is the flow irreversible? (Monotonicity theorems and infrared constraints: Part IV.) How do gauge theories, the duality webs, and the solvable supersymmetric corners populate the map? (Parts V–VII.) When does a lattice presentation actually converge to a point of this space? (Part VIII.) The picture is simple; what the last twenty years discovered is how much of the subject it organizes.

Exercise I.1.13 (easy). Tabulate $[\phi]$ and the relevant/marginal/irrelevant verdicts for ϕ^n ($n \leq 6$) and $(\partial\phi)^4$ in $d = 2, 3, 4, 6$. For a Dirac fermion (kinetic term $\int d^d x i\bar{\psi}\not{\partial}\psi$), find $[\psi]$ and classify the four-fermion interaction $(\bar{\psi}\psi)^2$ in $d = 2, 3, 4$. Which famous theories live at the marginal frontier in each d ?

Answer: $[\psi] = (d-1)/2$, so $[(\bar{\psi}\psi)^2] = 2(d-1)$ and the coupling has dimension $2-d$: marginal in $d = 2$ (Thirring, Gross–Neveu), irrelevant in $d > 2$ — which is why the 4d Fermi theory of weak interactions demanded a UV completion.

Exercise I.1.14 (medium). Check (I.1.20) against an explicit correlator: compute the Euclidean two-point function of a free massless scalar, $\langle\phi(x)\phi(0)\rangle = \frac{C_d}{|x|^{d-2}}$, from $\int d^d p \frac{e^{ip\cdot x}}{p^2}$, and confirm that the power of $|x|$ matches $2[\phi]$ with $[\phi] = (d-2)/2$. What replaces the power law in $d = 2$, and how does this connect to the “every potential is relevant” verdict above?

Hint: In $d = 2$ the integral is IR-divergent and the two-point function grows logarithmically: $\langle\phi\phi\rangle \sim -\frac{1}{2\pi} \ln|x|$. A dimensionless field is on the edge — whence both the richness (vertex operators $e^{i\alpha\phi}$ of continuously variable dimension) and the delicacy (no SSB of continuous symmetries) of $d = 2$.

Exercise I.1.15 (thinker). Nothing in this section’s picture forbids closed RG trajectories: flows that leave a fixed point and return, or limit cycles with no fixed point at all. Reflect on what a limit cycle would mean physically (physics periodic in the *logarithm* of the scale — discrete scale invariance) and why one might expect coarse-graining, which discards information, to be gradient-like instead. This intuition is correct for unitary theories in even dimensions ≤ 4 , where it is a theorem (*c*- and *a*-theorems, Part IV.1) — but the proofs need real input beyond “information is lost,” and non-unitary counterexamples exist.

I.1.6 What would count as an answer?

The question in this chapter’s title has, today, no complete answer — but the partial answers are substantial, and a reader should know the shape of the existing attempts before chapters I.2–I.4 borrow structure from all of them. Each of the four main axiomatic traditions formalizes a different row of the inventory of Definition I.1.1.¹⁶

The **Wightman axioms** (1950s) and their Euclidean counterpart, the **Osterwalder–Schrader axioms** (1970s), axiomatize item 2: fields as operator-valued distributions, correlators with specified analyticity, positivity, and causality properties, and reconstruction theorems that rebuild the Hilbert space from the correlators. Within their domain they are a complete success — the spin-statistics and CPT theorems are their fruits, and the constructive program built genuine interacting examples in $d = 2, 3$ satisfying them. Their domain, however, is local operators in flat space: gauge theories fit awkwardly (the gauge-variant fields are not Wightman fields, and the physical content includes the extended operators of item 4, about which the axioms are silent), and the $(2, 0)$ theory — with no fields at all in any classical sense — fits not at all.

Algebraic QFT (Haag–Kastler, 1960s onward) drops fields in favor of assigning to each spacetime *region* the algebra of observables measurable inside it, with compatibility under inclusion and commutativity at spacelike separation. It is the formulation in which the deep structural results about quantum information in field theory live — the type classification of local algebras, Tomita–Takesaki modular theory, the Reeh–Schlieder theorem — and Part X of these notes is, in effect, its payoff. Its classic limitation is mirror to Wightman’s: theories are easy to axiomatize but hard to construct, and topological sectors (instantons, again the extended operators) strain the axioms’ locality assumptions in ways that current research is still sorting out.

Functorial QFT (Atiyah–Segal, late 1980s; Kontsevich–Segal for the non-topological case) axiomatizes items 1 and 3 directly: a QFT is an assignment of Hilbert spaces to space-slices and of evolution maps to spacetime cobordisms between them, compatible with gluing — the path integral’s cutting-and-sewing behavior elevated to the definition. For *topological* theories this definition is not merely rigorous but spectacularly classifiable, and it is the natural home of every “assign data to manifolds” instinct in this chapter (chapter I.4 is built on it). For theories with metric dependence it faces real open problems: which complex metrics are allowable (the Kontsevich–Segal criterion [2105.10161](#), chapter I.4’s modern answer to “what is Wick rotation?”), and how to accommodate the scheme ambiguities of genuine QFTs, where the partition function on a curved background is not a number but a number-up-to-local-counterterms.

And the **lattice** tradition — physics’ working definition — defines a theory as the continuum limit of a regularized system, with constructive field theory as its rigorous wing. Its strengths and its audit findings (triviality; exotic limits) have already appeared in this chapter, and Part VIII is devoted to it.

Four traditions, four fragments — and the honest summary is that none captures class $\mathcal{S}$, large- N limits, *and* fracton models simultaneously, while the developments of Parts II–XI (symmetry categories as part of a theory’s identity; defects on equal footing with local operators) are only beginning to be reflected in any of them. Dedushenko’s verdict stands: *formulating a complete definition of QFT is an outstanding open problem*, of equal interest to physics and mathematics. These notes take the working position already staked out in this chapter — the inventory, locality,

¹⁶This entire section compresses Dedushenko’s survey [2203.08053](#), which should be consulted for precise statements and for the hundred-some references the one-line summaries below conceal. The comparison returns, with the benefit of the whole book’s examples, in Part XII.1.

and consistency, used opportunistically through whatever presentation converges — and treat the missing definition not as an embarrassment but as the subject’s standing research frontier, revisited with better equipment in Part XII.1.

Note I.1.2 (What these notes assume). Operationally, the chapters ahead assume of every theory: unitarity (reflection positivity in Euclidean signature), a local stress tensor, cluster decomposition, and — in flat space — Poincaré invariance, flagging explicitly each time one of these is relaxed (non-unitary fixed points, little strings, fractons, theories without stress tensors all make scheduled appearances). When an argument depends on an unproven structural assumption — existence of a continuum limit, convergence of an OPE beyond CFT, enhancement of scale to conformal invariance — the dependence is stated where it occurs.

Chapter exercises

Exercise I.1.16 (classic). (Two Lagrangians, one theory, in $d = 0$.) In the integral (I.1.2), change variables $\phi = \psi + c\psi^3$ for small constant c . Show that the resulting “theory of ψ ” has a different-looking action — new interaction vertices $\psi^4, \psi^6, \dots$ with c -dependent couplings, plus a vertex from the Jacobian — yet *identical* partition function, and (after matching sources) identical correlators of the original operator. Conclude that coupling constants of presentations are not, in general, observables. Which combinations of the new couplings are invariant?

Hint: The Jacobian $d\phi = (1 + 3c\psi^2)d\psi$ exponentiates into the action as $-\ln(1 + 3c\psi^2)$. This freedom of field redefinition is the $d = 0$ shadow of the equivalence theorem; “the theory” is the equivalence class.

Exercise I.1.17 (medium). (Large orders from the complex saddle.) Represent the coefficients of Example I.1.1 as $a_n = \oint \frac{d\lambda}{2\pi i} \frac{Z(\lambda)}{\lambda^{n+1}}$ and evaluate the large- n behavior by saddle point: the contour is pushed onto the negative- λ cut, where the discontinuity of Z is computed by the instanton saddle $\phi_*^2 = -6/\lambda$ with action $-3/2\lambda$. Recover the growth $|a_n| \sim n!(2/3)^n$ of (I.1.5), including the power of n in the prefactor if you are feeling thorough.

Hint: $\text{Disc}Z(-|\lambda|) \propto e^{-3/2|\lambda|}/\sqrt{|\lambda|}$ from the Gaussian integral over the unstable direction at the saddle. Then $a_n \approx \frac{1}{\pi} \int_0^\infty \frac{d|\lambda|}{|\lambda|^{n+1}} \text{Im}Z(-|\lambda|)$ is dominated by $|\lambda| \sim 3/2n$. This dispersive route from instantons to large orders is the $d = 0$ version of the Lipatov/Bender–Wu analysis.

Exercise I.1.18 (medium). (A correlator on the circle.) For the particle on a circle at general θ , compute the Euclidean two-point function $\langle e^{iq(\tau)} e^{-iq(0)} \rangle_\beta$ by inserting energy eigenstates, and examine its $\beta \rightarrow \infty$ limit. Show that the answer decays exponentially in τ with rate set by the gap, in parallel with (I.1.17) — except at $\theta = \pi$, where the degeneracy makes part of the correlator *persist* at infinite separation. Interpret the persistent piece as spontaneous breaking of the C symmetry between the two ground states, in the only way quantum mechanics permits: exact degeneracy.

Hint: $e^{iq}|n\rangle = |n+1\rangle$, so the correlator is $\frac{1}{Z} \sum_n e^{-(\beta-\tau)E_n - \tau E_{n+1}}$. At $\theta = \pi$, the $n = 0$ term contributes $e^{-\beta E_0}$ with no τ -dependence in the exponent’s leading piece, since $E_0 = E_1$.

Exercise I.1.19 (thinker). Each of the four cracks of Section I.1.2 embarrasses at least one axiom system of Section I.1.6. Match them: which axioms could in principle accommodate the $(2,0)$ theory but cannot construct it? Which formalize a duality like (I.1.7) as an exact statement, and which cannot even state it? Which tradition does the triviality theorem vindicate, and which does it humble? Use Dedushenko 2203.08053 §3 as a referee for your answers.

Exercise I.1.20 (thinker). Idea I.1.1 says a theory is its observables. Here is a stress test you can hold in mind through chapter I.3: could two *distinct* theories have identical correlation functions of all *local* operators on $\mathbb{R}^d$, differing only in their extended operators or in their partition functions on topologically nontrivial spacetimes? (No proof expected — but formulate precisely what data would have to differ, and what experiment, in a finite-size sample, would tell the two theories apart.) The answer is yes, the canonical examples being gauge theories with the same algebra and different global structure, e.g. $SU(2)$ vs. $SO(3)$ Yang–Mills; this is the content of “reading between the lines” [1305.0318](#), and the reason item 4 sits in Definition I.1.1.

Reading

Dedushenko, *The Quest to Define QFT*, [2203.08053](#). The survey behind Section I.1.6: a short overview of every axiom system, with its successes and limitations, and a bibliography that leads to the original literature. Read §§1, 3 now and §2 as reference.

Rychkov [2509.02779](#). *Conformal bootstrap: from Polyakov to our times*. The history that makes the modern viewpoint feel inevitable rather than fashionable: the 1970 Kyiv exchange, Polyakov’s 1974 manifesto, and the road to the Ising island. Sections 2–5 complement this chapter; the rest seeds Part III.

Seiberg, *Thoughts About Quantum Field Theory* (IAS). The reorganization manifesto in slide form: theories without Lagrangians, deformation classes, symmetries-and-anomalies as the invariants of theory space. Twenty minutes that frame the whole book.

Gaiotto–Kapustin–Komargodski–Seiberg, [1703.00501](#), **Appendix D**. The particle on a circle, done as a warm-up for Yang–Mills at $\theta = \pi$ — the source for Example I.1.2 and the bridge to Parts II.7 and V.3.

Seiberg–Senthil–Wang–Witten, [1606.01989](#), §1. What duality means, by example; the duality web behind crack 2.

Aizenman–Duminil-Copin, [1912.07973](#), §1. The triviality theorem. The introduction is readable by physicists and states exactly what is and is not proven.

Poland–Rychkov–Vichi, [1805.04405](#), §1. The data-centric view of CFT from the practitioners; the source of the numbers in (I.1.8).

Books (for the shelf, not the week): Glimm–Jaffe, *Quantum Physics: A Functional Integral Point of View* — what “constructing a QFT” means when done in earnest; Weinberg, *The Quantum Theory of Fields I*, ch. 1 — the historical case that QFT was forced on us, the perfect foil for this chapter’s question.

I.2 Observables I: local operators

Chapter I.1 dismantled a definition and left an inventory in its place (Definition I.1.1). This chapter makes the book’s first affirmative claim: the oldest entry of that inventory — the local operators and their correlation functions — is *primary data*. Local operators are not records kept about a theory that truly lives in some Lagrangian; they are the theory’s observables in the fullest sense of Idea I.1.1: data from which the rest of the theory’s life on $\mathbb{R}^d$ — states, dynamics, symmetries, the response to geometry — can be rebuilt, and whose precise limits, charted at the chapter’s end, are what the rest of Part I exists to cross. A claim like that has content only if three questions can be answered. One must say what a correlation function actually *is* — in particular how the Euclidean correlators that lattices and path integrals produce are related to the Lorentzian amplitudes that quantum mechanics demands. One must show that correlators *contain* the theory — that the Hilbert space, the Hamiltonian, and the operators themselves can be recovered from $\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle$, so that whoever specifies consistent correlators has specified everything. And one must find structure in the list: the set of all n -point functions of all operators is enormously redundant, and the claim that it is primary data is useful only if the redundancy is organized by some definite structure. This chapter settles these three questions for pointlike operators. It will turn out that the three answers are closely related: all three rest on a single analytic and Hilbert-space structure that positivity of the energy builds into the correlation functions.

Question I.2.1 — Are the local operators of a quantum field theory in fact its *observables* — primary data that determines the theory — rather than bookkeeping attached to a presentation? Concretely: in what sense do correlators *contain* states, operators, and a Hamiltonian; what organizes the infinite list of correlation functions into usable data; and which operators must every local theory contain?

The chapter’s centerpiece is the operator product expansion. In the quantum field theory the reader already knows, the OPE is an asymptotic statement about short distances, one further entry in the perturbative toolbox. In a conformal field theory it is something stronger: a *convergent* expansion, valid at finite separation, with an explicitly known radius of convergence — and therefore an exact multiplication law on the space of local operators. The modern subject depends on this fact; the bootstrap of Part III is built on it. We will prove it by a quantum-mechanical argument, radial quantization, whose central point can be stated in advance: the reconstruction theorems, which build a Hilbert space out of Euclidean correlators by cutting spacetime along planes, can be run just as well cutting along *spheres* — and the OPE then becomes an ordinary expansion of a state in a complete basis, convergent for the same reason Fourier series are. The chapter closes with the two operators that locality forces on every QFT — the stress tensor and the conserved currents — and with a worked counterexample, the generalized free field, showing exactly what goes missing when they are absent.

Throughout, “the theory” is presented however convenient — lattice, path integral, canonical — in the spirit of chapter I.1: presentations are scaffolding, and everything we establish is a statement about the correlators themselves. There is also a question prior to the three above, one that textbooks answer so implicitly that the reader may never notice it being answered: what, exactly, *is* a local operator? Since the chapter is about to build theories out of these objects, we begin there.

I.2.1 First: what is a local operator?

The inventory of chapter I.1 refers to “labels $\mathcal{O}_i$ drawn from a set of local operators,” and we should begin by asking what this set actually contains. In a Lagrangian presentation the natural guess is: the elementary field ϕ , its derivatives, and local polynomials built from them — ϕ^2 , ϕ^4 , $\phi\partial^2\phi$, and so on. This guess fails in two instructive ways, once in the continuum and once on the lattice, and examining the two failures will lead us to the correct definition.

The first failure is that products of fields at coincident points do not exist. Consider the free massless scalar in $d = 4$ and try to define $\phi(x)^2$. Its expectation value would be $\langle\phi(x)\phi(x)\rangle = \lim_{\varepsilon \rightarrow 0} \frac{1}{4\pi^2\varepsilon^2} = \infty$: the would-be operator has a divergent overlap with the identity, and its correlators with everything else are contaminated in the same way. The standard repair, familiar from canonical quantization, is **normal ordering**: split the free field into its annihilation and creation parts, $\phi = \phi^+ + \phi^-$ with $\phi^+|0\rangle = 0$, and define

$$:\phi^2:(x) \equiv \phi^-\phi^-(x) + 2\phi^-\phi^+(x) + \phi^+\phi^+(x), \quad (\text{I.2.1})$$

so that every annihilation operator stands to the right of every creation operator and $\langle:\phi^2:\rangle = 0$. This prescription works, but as stated it is tied to the mode expansion of a free field, and mode expansions are scaffolding: nothing like $\phi^\pm$ exists on a lattice or at an interacting fixed point. To extract the presentation-independent content of normal ordering, we rewrite it without reference to modes. Since the commutator $[\phi^+(x), \phi^-(y)] = \langle\phi(x)\phi(y)\rangle$ is a c-number, reordering the modes in a product of two fields produces exactly one contraction,

$$\phi(x+\varepsilon)\phi(x) = :\phi(x+\varepsilon)\phi(x): + \langle\phi(x+\varepsilon)\phi(x)\rangle, \quad (\text{I.2.2})$$

which is Wick’s theorem for two fields. The normal-ordered product on the right-hand side is regular as $\varepsilon \rightarrow 0$ — the singularity of the product lives entirely in the c-number term — so (I.2.2) can be solved for it and the limit taken:

$$:\phi^2:(x) = \lim_{\varepsilon \rightarrow 0} [\phi(x+\varepsilon)\phi(x) - \langle\phi(x+\varepsilon)\phi(x)\rangle]. \quad (\text{I.2.3})$$

This **point-splitting** form of normal ordering defines the same operator as (I.2.1) (Exercise I.2.1 checks this), but it is the better definition, because it refers only to a product of operators and to a correlation function: it makes sense in any theory and in any presentation. Note the structure of the repair: the singular part of the product of two local operators was itself proportional to a local operator — here the identity, with coefficient $\langle\phi\phi\rangle$ — and subtracting it left a finite local operator. In an interacting theory the same procedure requires more terms. The product $\phi(x+\varepsilon)\phi(x)$ then contains, besides the identity, singular multiples of other local operators, all of which must be subtracted; and the finite operators that survive mix among themselves as the splitting scale ε is varied, which is one way anomalous dimensions enter. The reader should recognize the pattern: this is the operator product expansion, encountered backwards, in its historical role as a renormalization procedure. (Section I.2.5 runs the same structure forwards, as an exact expansion.) The “set of local operators” of an interacting theory is therefore not the classical polynomial algebra. It is what survives the organization of coincident-point singularities — smaller in some respects, because equations of motion such as $\partial^2\phi = 0$ in the free theory hold as operator identities and remove combinations from the list; larger in others; and with dimensions shifted away from naive counting.

The second failure appears on the lattice, where there is no field ϕ from which to build polynomials in the first place. Consider the two-dimensional Ising model at its critical temperature — the spins

of chapter I.1's Example I.1.3, now placed on a square lattice of spacing a , where (unlike in $d = 1$) the model has a critical point. The candidate local operators are functions of the spins in a bounded neighborhood of a point: the spin s_x itself, the product $s_x s_{x+\hat{e}}$ of two adjacent spins, three-spin clusters, and so on. The problem with these candidates is that none of them, taken individually, has a good continuum limit: their correlation functions are homogeneous under rescaling only asymptotically, and different candidates mix with one another under coarse-graining.

The renormalization group converts this difficulty into a definition. First make the coarse-graining concrete: partition the lattice into 3×3 blocks and assign to each block the majority value of its nine spins. This produces an Ising-like system on a lattice of spacing $3a$, and at the critical point the statistics of the blocked spins reproduce the statistics of the original spins — this self-reproduction is what “fixed point” means, as in chapter I.1's theory-space picture. The same blocking acts on operators: summing over the fine-grained spins at fixed block spins converts any operator $\mathcal{O}$ supported near a point x into an operator $R[\mathcal{O}]$ supported near x on the coarser lattice, and the map R is linear, because expectation values are linear in the inserted operator. A linear map can be diagonalized. Its eigenvectors are called **scaling operators**,

$$\begin{aligned} R[\mathcal{O}_k] &= b^{-\Delta_k} \mathcal{O}_k, \\ b &= 3 \text{ (the blocking factor)}, \end{aligned} \tag{I.2.4}$$

and the eigenvalue defines the **scaling dimension** Δ_k . The reason this is the right notion is that eigenoperators, and only they, have homogeneous correlation functions. Applying the blocking n times inside a two-point function multiplies each insertion by $b^{-n\Delta_k}$ and reduces the separation, measured in lattice units, by b^n :

$$\begin{aligned} \langle \mathcal{O}_k(0) \mathcal{O}_k(r) \rangle &= b^{-2n\Delta_k} \langle \mathcal{O}_k(0) \mathcal{O}_k(r/b^n) \rangle, \\ \langle \mathcal{O}_k(0) \mathcal{O}_k(r) \rangle &\propto \frac{1}{r^{2\Delta_k}}, \end{aligned} \tag{I.2.5}$$

where the implication follows by choosing $b^n \sim r$: a pure power law, with the exponent given by the RG eigenvalue.

A generic lattice operator is not an eigenvector, but it can be expanded in eigenvectors, and the expansion is the working dictionary between the lattice and the continuum. The spin s_x is odd under the global flip $s \rightarrow -s$, so only $\mathbb{Z}_2$ -odd scaling operators can appear in its expansion:

$$s_x = c_\sigma a^{\Delta_\sigma} \sigma(x) + c_{\sigma'} a^{\Delta_{\sigma'}} \sigma'(x) + \dots, \tag{I.2.6}$$

where σ is the lowest-dimension odd operator ($\Delta_\sigma = \frac{1}{8}$ for the 2d Ising model), σ' is the next-lowest, and the powers of the lattice spacing are fixed by dimensional analysis: s_x is a dimensionless number, while the normalization $\langle \sigma(x) \sigma(0) \rangle = |x|^{-2\Delta_\sigma}$ assigns σ the dimension (length) $^{-\Delta_\sigma}$. The constants $c_\sigma, c_{\sigma'}, \dots$ are non-universal — they depend on the microscopic model — but the operators and their dimensions are properties of the fixed point alone. Taking the expectation value of two copies of (I.2.6) makes the dictionary quantitative:

$$\langle s_0 s_r \rangle = c_\sigma^2 \left(\frac{a}{r} \right)^{2\Delta_\sigma} [1 + O((a/r)^{\Delta_{\sigma'} - \Delta_\sigma})], \tag{I.2.7}$$

a power law with subleading **corrections to scaling** controlled by the higher operators in the expansion. This, corrections included, is precisely the form against which Monte Carlo data for critical systems is fit in practice. Read in the opposite direction, the scaling operator σ expressed

in lattice variables is an infinite sum, $s_x + (\text{three-spin terms}) + \dots$, with contributions of every size; Wilson constructed such eigenvectors numerically for the 2d Ising model already in 1975.¹⁷

The two constructions — point-splitting in the continuum, RG eigenvectors on the lattice — are different scaffoldings around the same object, and the object itself can now be defined with no scaffolding at all. What both constructions produced were operators whose correlation functions scale homogeneously, labeled by a scaling dimension. At a fixed point, where dilatations are a symmetry, this property can be taken as the definition. Because everything in these notes is built on this notion, we state it formally and then explain why it is the right choice.

Definition I.2.1 (Local operators of a scale-invariant theory)

In a scale-invariant theory, a **local operator** of scaling dimension Δ is an object $\mathcal{O}(x)$, labeled by a point of space(time), whose correlation functions with other such operators are defined at non-coincident points and transform homogeneously under dilatations,

$$\langle \mathcal{O}_1(\lambda x_1) \mathcal{O}_2(\lambda x_2) \cdots \mathcal{O}_n(\lambda x_n) \rangle = \lambda^{-(\Delta_1 + \Delta_2 + \cdots + \Delta_n)} \langle \mathcal{O}_1(x_1) \mathcal{O}_2(x_2) \cdots \mathcal{O}_n(x_n) \rangle, \quad (\text{I.2.8})$$

and covariantly under rotations, according to the operator's spin. Equivalently, it is an eigenvector of the dilatation generator with eigenvalue Δ . The **operator content** of the theory is the list of all pairs (Δ, spin) that occur, with their multiplicities.

Three remarks explain why this definition, rather than some other, is the correct one. First, it is exhaustive: as the lattice discussion showed, every reasonable local observable of every presentation is a linear combination of scaling operators, so nothing is lost by working in this basis. It is the basis that diagonalizes the renormalization group, in the same sense in which energy eigenstates diagonalize time evolution. Second, it is presentation-independent: (I.2.8) mentions only correlation functions, which are the data all presentations share. Third, it is intrinsic exactly where it needs to be: at a fixed point, dilatation is a symmetry, so the eigenvector condition refers to nothing outside the theory. For a theory that is not scale invariant the definition is inherited rather than intrinsic: by chapter I.1's picture, a generic theory is an RG flow leaving a fixed point, and its operators match onto the fixed point's operator content, with corrections analytic in the couplings along the flow. (Section I.2.5 will sharpen the definition once more — in a conformal theory, a local operator is the same thing as a state in the radial Hilbert space — but the present form is what makes that sharpening possible.)

Idea I.2.1 (The operator content is data, not bookkeeping)

The set of local operators of a theory — which dimensions occur, with which spins and multiplicities — is intrinsic, presentation-independent data, on the same footing as the correlators themselves. It is constructed, not read off: from products of fields by organizing their coincident-point singularities, from lattice variables by diagonalizing the renormalization group. It can

¹⁷Wilson's calculation, and the eigenvector picture of operators sketched here, are described in Rychkov [1601.05000](#), §1.2. Two caveats. The blocking map R and its eigenvectors depend on the chosen blocking scheme; what is scheme-independent is the spectrum $\{\Delta_k\}$ and the continuum correlators — one more instance of the distinction between presentation and physics. And (I.2.6) is an asymptotic statement about the $a \rightarrow 0$ limit, of the kind Part VIII scrutinizes; in some models, symmetries of the continuum limit descend from lattice *translations* rather than from any internal lattice symmetry (the emanant symmetries of chapter VIII.2).

be smaller than a presentation suggests, because equations of motion remove operators from the list; and its gaps carry physics, as the chapter's closing example — a candidate “theory” disqualified by a single missing operator — will show.

Exercise I.2.1 (classic). (The simplest composite operator.) For the free massless scalar in $d = 4$ with $\langle \phi(x)\phi(0) \rangle = \frac{1}{4\pi^2 x^2}$: (a) verify that (I.2.3) defines an operator with finite correlators by computing, via Wick's theorem,

$$\langle :\phi^2:(x) :\phi^2:(0) \rangle = \frac{2}{(4\pi^2)^2 x^4},$$

and conclude $\Delta_{:\phi^2:} = 2 = 2\Delta_\phi$: free-field composites have additive dimensions. (b) Verify that the mode definition (I.2.1) and the point-split definition (I.2.3) give the same operator, by checking that their correlators with $\phi(y_1)\phi(y_2)$ agree. (c) At the interacting 3d Wilson–Fisher fixed point, the operator occupying this slot is ϵ , with $\Delta_\epsilon = 1.412625(10)$, while $2\Delta_\phi \simeq 1.036$: the operator survives the interaction, but additivity of dimensions does not. Convince yourself that nothing in the point-splitting construction required additivity — the dimension of a composite operator is independent dynamical data, fixed by the theory and not by arithmetic.

Hint: For (a) and (b): Wick's theorem; the only correlators needed are products of two-point functions. For (b), note that both definitions differ from $\phi(x + \varepsilon)\phi(x)$ by the same c -number.

1.2.2 Correlators and their analytic structure

We turn to the correlation functions themselves. Two inequivalent-looking definitions of a correlation function are in circulation, corresponding to the two kinds of presentation met in chapter I.1, and the first task is to state both precisely. The main result of this section is the exact relation between them.

In a Euclidean presentation, a correlator is an average. On the lattice, $\langle s_{i_1} \cdots s_{i_n} \rangle$ is a weighted sum over configurations, as in the Ising chain of Example I.1.3; in a Euclidean path integral,

$$\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle = \frac{1}{Z} \int \mathcal{D}\phi \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) e^{-S_E[\phi]}, \quad (\text{I.2.9})$$

the insertions are ordinary commuting functionals and the correlator is a moment of a probability-like measure. Nothing here is quantum: no operator ordering, no Hilbert space in sight, just statistics. These are called **Schwinger functions** in the axiomatic literature, and they are what every nonperturbative presentation — lattice Monte Carlo, the ϵ -expansion's Euclidean diagrams, the bootstrap's crossing equations — actually computes.

In a Lorentzian, Hamiltonian presentation, a correlator is a transition amplitude. Operators act on a Hilbert space, $\mathcal{O}(t, \mathbf{x}) = e^{iHt} \mathcal{O}(0, \mathbf{x}) e^{-iHt}$, and the basic objects are the vacuum expectation values of *ordered* products,

$$W(x_1, \dots, x_n) = \langle 0 | \mathcal{O}_1(x_1) \mathcal{O}_2(x_2) \cdots \mathcal{O}_n(x_n) | 0 \rangle, \quad (\text{I.2.10})$$

the **Wightman functions**. The ordering is physical: $\mathcal{O}_1$ acts last. Because the operators need not commute, different orderings define different functions. For two operators, the difference of the two orderings is the expectation value of a commutator,

$$W(x_1, x_2) - W(x_2, x_1) = \langle 0 | [\mathcal{O}_1(x_1), \mathcal{O}_2(x_2)] | 0 \rangle, \quad (\text{I.2.11})$$

which microcausality requires to vanish at spacelike separation but not elsewhere. The time-ordered correlators of perturbation theory are yet another object, assembled from the Wightman functions by step functions in the times:

$$\langle 0|T\{\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\}|0\rangle = \theta(t_1 - t_2)W(x_1, x_2) + \theta(t_2 - t_1)W(x_2, x_1), \quad (\text{I.2.12})$$

with $n!$ such terms for n points. The Wightman functions are thus the primitive objects: every ordering — time-ordered, retarded, or otherwise — is a linear combination of them, while the step functions cannot be undone to recover Wightman functions from time-ordered ones alone.

We therefore have two definitions in hand. In Euclidean presentations a correlator is a moment of a statistical measure, symmetric under permutations of its arguments; in Hamiltonian presentations it is a matrix element of ordered, noncommuting operators. The remainder of this section establishes the exact relation between them: the Wightman functions of a theory with energy bounded below are boundary values of a single analytic function of complexified times, and the Schwinger functions are the values of that same function at purely imaginary times. The analyticity follows from positivity of the energy alone; no conformal symmetry, weak coupling, or Lagrangian is required.

Positive energy means analyticity

Take the two-point Wightman function of a Hermitian scalar operator in any Poincaré-invariant theory with $H \geq 0$, and insert a complete set of energy eigenstates:

$$W(t, \mathbf{x}) = \langle 0|\mathcal{O}(t, \mathbf{x})\mathcal{O}(0)|0\rangle = \sum_n |\langle 0|\mathcal{O}(0)|n\rangle|^2 e^{-iE_n t + i\mathbf{p}_n \cdot \mathbf{x}}. \quad (\text{I.2.13})$$

Each term oscillates in t ; the sum, a priori, is a distribution and nothing more. But now let t wander into the complex plane, $t \rightarrow t - i\sigma$ with $\sigma > 0$. Every term acquires a factor $e^{-E_n \sigma} \leq 1$: the positivity of the spectrum converts the lower half of the complex time plane into a *damping* region, where the sum converges better than on the real axis and defines a holomorphic function of t . The real-time Wightman function is the boundary value of this function as $\sigma \rightarrow 0^+$:

$$\langle 0|\mathcal{O}(t, \mathbf{x})\mathcal{O}(0)|0\rangle = \lim_{\sigma \rightarrow 0^+} W(t - i\sigma, \mathbf{x}). \quad (\text{I.2.14})$$

The opposite ordering, $\langle 0|\mathcal{O}(0)\mathcal{O}(t, \mathbf{x})|0\rangle$, is by the identical argument the boundary value from the *upper* half-plane. The two orderings are therefore boundary values of one analytic function, approached from opposite sides of the real axis; their difference, the commutator (I.2.11), is the discontinuity across the axis.

The Euclidean correlator can now be located inside this structure. Put $t = -i\tau$ with $\tau > 0$: this is a point deep inside the lower half-plane, where the sum (I.2.13) converges absolutely,

$$W(-i\tau, \mathbf{x}) = \sum_n |\langle 0|\mathcal{O}|n\rangle|^2 e^{-E_n \tau + i\mathbf{p}_n \cdot \mathbf{x}} = \langle 0|\mathcal{O}(0, \mathbf{x})e^{-H\tau}\mathcal{O}(0)|0\rangle, \quad (\text{I.2.15})$$

which is precisely the quantity a Euclidean path integral computes with two insertions separated by Euclidean time τ . The Schwinger function is therefore the Wightman function evaluated at imaginary time: equal to it at interior points of a common domain of analyticity, not merely related to it by a formal substitution. Notice also what has happened to the question of operator ordering. Continuing the *opposite* ordering $\langle 0|\mathcal{O}(0)\mathcal{O}(t, \mathbf{x})|0\rangle$ to imaginary time requires $\tau < 0$ for convergence, since its damping region is the upper half-plane. A given Euclidean configuration

therefore corresponds to exactly one operator ordering — the one in which Euclidean times decrease from left to right — and this is why the Euclidean correlator (I.2.9) can be symmetric in its arguments without contradiction: the orderings are not lost, but each is recovered from a different boundary of the same function. The n -point version of the argument runs the same way — positivity of H gives analyticity in each successive time difference, in the tube $\text{Im}(t_{k+1} - t_k) < 0$ — and a classic theorem of the axiomatic tradition, spelled out with the n -point bookkeeping in Section I.2.2 below, upgrades this to the following general statement.¹⁸

Idea I.2.2 (One function, many correlators)

In a theory with energy bounded below, all orderings of Lorentzian correlators — Wightman functions, time-ordered products, retarded commutators — and the Euclidean Schwinger functions are boundary values and special sections of a *single* analytic function of complexified times. “Wick rotation” is not a substitution $t \rightarrow i\tau$ performed on formulas; it is analytic continuation through a domain whose existence is equivalent to the positivity of the energy.

This is the modern division of labor: *compute where the function is nice* (the Euclidean section, where correlators are permutation-symmetric, single-valued, and often given by convergent statistical sums), *then continue to where the physics asks the question* (the Lorentzian boundary, where causality, scattering, and unitarity live). The entire Euclidean strategy of modern QFT — and of these notes — rests on the continuation being lossless, and the reconstruction theorems of Section I.2.3 are the precise statement that it is.

Before the example, let us fix the dictionary in the form we will use constantly (the general conventions are in Appendix A):

Definition I.2.2 (Wightman and Schwinger functions)

Let $\mathcal{O}_i$ be local operators of a Poincaré-invariant theory with $H \geq 0$.

1. The **Wightman functions** are the vacuum expectation values of operator products in a fixed order, $W_n(x_1, \dots, x_n) = \langle 0 | \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) | 0 \rangle$, defined as distributions on real Minkowski space. They are boundary values of functions holomorphic for ordered imaginary parts, $\text{Im}(t_{k+1} - t_k) < 0$.
2. The **Schwinger functions** are the restrictions of those holomorphic functions to Euclidean points $t_k = -i\tau_k$ (with $\tau_1 > \tau_2 > \cdots$ enforcing convergence, and then extended by symmetry): $S_n(\tau_k, \mathbf{x}_k) = W_n$ at imaginary times. They are real-analytic away from coincident points and symmetric under permutations.
3. Conversely, Wightman functions are recovered from Schwinger functions by continuing $\tau \rightarrow it$ and approaching the real-time axis from the side prescribed by the desired operator ordering: $\tau_k \rightarrow it_k + \epsilon_k$ with $\epsilon_1 > \epsilon_2 > \cdots$ reproducing the ordering $\mathcal{O}_1 \cdots \mathcal{O}_n$.

The third clause is the famous “ $i\epsilon$ -prescription,” demystified: the small imaginary parts that textbooks sprinkle on propagators are the residue of the Euclidean ordering $\tau_1 > \tau_2 > \cdots$, remembering

¹⁸Kontsevich–Segal 2105.10161, §2, opens with precisely this boundary-value picture before generalizing it; we meet their generalization in Section I.2.4.

from which side of the cut each ordering approaches the real axis. Let us now watch every clause work in the one theory where all the integrals can be done.

Example I.2.1 (The free scalar two-point function, across signatures)

Take a free massless scalar in $d = 4$, mostly-plus signature. The mode expansion gives, for the Wightman function at separation $x = (t, \mathbf{x})$, $r = |\mathbf{x}|$,

$$W(t, \mathbf{x}) = \int \frac{d^3 p}{(2\pi)^3 2|\mathbf{p}|} e^{-i|\mathbf{p}|t + i\mathbf{p}\cdot\mathbf{x}} = \frac{1}{4\pi^2 r} \int_0^\infty dp \sin(pr) e^{-ipt}. \quad (\text{I.2.16})$$

As it stands the last integral does not converge — the spectral sum on the real-time axis never does — but for $t \rightarrow t - i\sigma$, $\sigma > 0$, it converges absolutely, exactly as the general argument promised, and gives

$$W(t, \mathbf{x}) = \frac{1}{4\pi^2} \frac{1}{r^2 - (t - i\sigma)^2} \Big|_{\sigma \rightarrow 0^+}. \quad (\text{I.2.17})$$

We now examine the analytic structure of this function and verify each clause of the definition above.

The Euclidean section. Set $t = -i\tau$: $W = \frac{1}{4\pi^2} (r^2 + \tau^2)^{-1} = \frac{1}{4\pi^2 x_E^2}$, the Euclidean propagator — single-valued, symmetric, with its only singularity at coincident points. This is the function the Euclidean path integral hands you, and we have just exhibited it as the midplane section of (I.2.17).

The cut. As a function of complex t at fixed r , (I.2.17) has poles at $t = \pm r$ — on the light cone — and, once smeared into the massive or interacting case, branch cuts along the real axis for $|t| > r$: the *timelike* region. For $|t| < r$ (spacelike separation) the real axis is free of singularities, so the boundary values from above and below agree there: the two operator orderings coincide, which is microcausality, $[\mathcal{O}(x), \mathcal{O}(0)] = 0$ at spacelike separation — derived this time not from a mode-expansion accident (Exercise I.1.12 of chapter I.1) but from the location of the singularities, the form in which it survives interactions.

The commutator as a discontinuity. At timelike separation the orderings differ by the jump across the cut:

$$\begin{aligned} \langle 0 | [\mathcal{O}(x), \mathcal{O}(0)] | 0 \rangle &= \frac{1}{4\pi^2} \left[\frac{1}{r^2 - (t - i\sigma)^2} - \frac{1}{r^2 - (t + i\sigma)^2} \right]_{\sigma \rightarrow 0^+} \\ &= -\frac{i}{2\pi} \text{sgn}(t) \delta(r^2 - t^2), \end{aligned} \quad (\text{I.2.18})$$

by the Sokhotski formula applied to $X = r^2 - t^2$. The commutator of the massless field is supported *on* the light cone — it vanishes at spacelike separation, as causality requires, and also at timelike separation, a $d = 4$ massless peculiarity (Huygens' principle) that interactions destroy.

The orderings as $i\epsilon$'s. Finally, the time-ordered two-point function used in perturbation theory is $\theta(t)W(t) + \theta(-t)W(-t)$, which the same algebra assembles into

$$\langle 0 | T \{ \mathcal{O}(x) \mathcal{O}(0) \} | 0 \rangle = \frac{1}{4\pi^2} \frac{1}{r^2 - t^2 + i\epsilon} = \frac{1}{4\pi^2} \frac{1}{x^2 + i\epsilon}, \quad (\text{I.2.19})$$

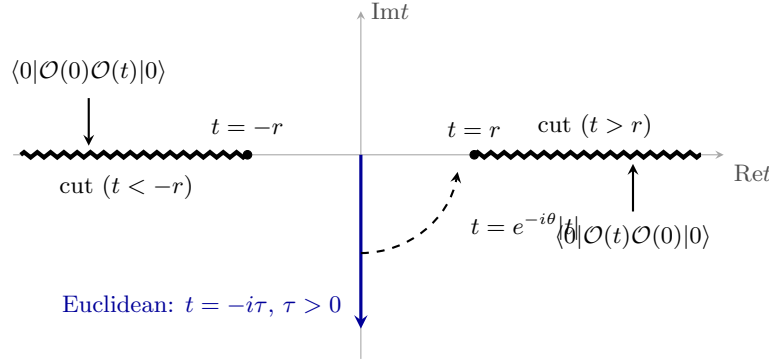


Figure 2: The complex time plane of the two-point function at fixed spatial separation r . Positivity of the energy makes the function holomorphic off the real axis, and its singularities reach the real axis only at timelike separation $|t| \geq r$. The two operator orderings are the boundary values from below and from above (one arrow of each is drawn; the same applies on both cuts), and their mismatch across the cuts is the commutator (I.2.18). The Euclidean correlator is the section along the negative imaginary axis; continuing it back to real time along the dashed path $t = e^{-i\theta}|t|$, $\theta: \frac{\pi}{2} \rightarrow 0$, recovers the Lorentzian boundary values, which is the inverse of the Wick rotation.

the Feynman propagator in mostly-plus signature: a *single* $i\epsilon$ attached to the invariant x^2 , approaching the cut from one fixed side regardless of the sign of t . Each of the standard two-point functions — Wightman, time-ordered, retarded, Euclidean — is thus a boundary value or section of the same analytic function, distinguished only by the direction from which the real axis is approached.

Exercise I.2.2 (easy). Repeat Example I.2.1 for the massive free scalar in $d = 4$ at purely spatial separation, and at purely timelike separation. Show that the Euclidean propagator decays as e^{-mr} at large r (the Yukawa/OZ form, as anticipated by the correlation-length discussion around (I.1.17)), while at large timelike separation the Wightman function *oscillates* as e^{-imt} with a power-law envelope. One analytic function: exponential decay in one direction of the complex plane is oscillation in the orthogonal one.

Hint: Both behaviors follow from the Källén–Lehmann representation below with $\rho(s) = \delta(s - m^2)$; no integral needs to be done exactly. For the envelope, saddle-point the mode integral.

Exercise I.2.3 (medium). Define the retarded function $G_R(x) = -i\theta(t)\langle 0|[\mathcal{O}(x), \mathcal{O}(0)]|0\rangle$. Show from the representation (I.2.14) and its conjugate that G_R is the boundary value of a function holomorphic when both $\text{Im}t$ and the energy variable of its Fourier transform are continued appropriately: in momentum space, $\tilde{G}_R(\omega, \mathbf{p})$ is holomorphic for $\text{Im}\omega > 0$. This upper-half-plane analyticity is the QFT ancestor of the Kramers–Kronig dispersion relations, and the germ of the dispersive bootstrap methods of Part IX.

Hint: $\theta(t) = \frac{1}{2\pi i} \int d\omega' \frac{e^{i\omega' t}}{\omega' - i\epsilon}$; convolve. Causality (support of G_R in $t \geq 0$) and analyticity in ω are Fourier duals of one another.

The n -point dictionary, and how to use it

The two-point function fit in one complex plane; the general entry of the dictionary must be spelled out once, carefully, because every “Wick rotate the correlator” step in the rest of these notes silently

invokes it. The structure is conveniently organized in four steps, each resting on one physical input.¹⁹

Step 1 (positive energy): the tube. Repeating the damping argument of (I.2.13) between each adjacent pair of operators, the n -point Wightman function $W_n(x_1, \dots, x_n)$ — one fixed ordering — extends holomorphically to complex times with *ordered* imaginary parts, $\text{Im}(t_{k+1} - t_k) < 0$: each successive factor $e^{-iH(t_{k+1}-t_k)}$ becomes a damping $e^{-H|\text{Im}\Delta t|}$. (Translation invariance makes this a statement about the $n - 1$ difference variables; with Lorentz invariance the domain widens in the spatial directions too — the “extended tube” — but the time story carries all the intuition.)

Step 2 (microcausality): the orderings agree where they can. Distinct orderings of the same operators are boundary values of *a priori* different analytic functions. But at spacelike separations the operators commute, so the functions agree on a real open set — and holomorphic functions agreeing on a real open set of their common boundary are continuations of one another (the *edge-of-the-wedge* theorem, the one piece of hard analysis this story genuinely needs). All $n!$ orderings thus assemble into a *single* analytic function on the union of the permuted domains (the “permuted extended tube”), and “the correlator” becomes a well-defined single object with many real-time faces — the n -point upgrade of Figure 2.

Step 3 (the theorem): Euclidean points are inside. The classic result of the axiomatic era is that the purely Euclidean configurations $t_k = -i\tau_k$, at non-coincident points, lie in the interior of that common domain. The restriction there — the Schwinger function S_n — is real-analytic and, because the domain is permutation-glued, *symmetric*: Euclidean correlators have no memory of operator order, which is why a statistical-mechanics presentation, which never knew about order, can produce them.

Step 4 (the recipe): returning to Lorentzian signature. To recover any Lorentzian correlator from Euclidean data, give each operator a small positive Euclidean time offset encoding its place in the desired order and continue:

$$\langle 0 | \mathcal{O}_1(t_1) \mathcal{O}_2(t_2) \cdots \mathcal{O}_n(t_n) | 0 \rangle = \lim_{\epsilon_1 > \epsilon_2 > \cdots > \epsilon_n \rightarrow 0^+} S_n(\tau_k = \epsilon_k + it_k) : \quad (\text{I.2.20})$$

operator order = order of the ϵ 's. Time-ordered correlators correspond to choosing the ϵ 's in the same order as the real times themselves — equivalently, to the uniform rotation $t_k \rightarrow (1 - i\delta)t_k$ — which for two points collapses to the single “ $x^2 + i\epsilon$ ” of Example I.2.1. This recipe is worth committing to memory: it is how every Lorentzian statement in these notes (commutators, response functions, the chaos correlators of Part XI) will be extracted from Euclidean computations.

The recipe is not hostage to free fields or to perturbation theory, and it pays to see it act on an interacting correlator at least once.

Example I.2.2 (A Lorentzian correlator of the 2d Ising CFT)

The critical 2d Ising model has $\langle \sigma(x) \sigma(0) \rangle_E = |x|^{-1/4}$ (with $\Delta_\sigma = \frac{1}{8}$ — a lattice-derivable fact we import; Part III.4 computes it three ways). What does the *Lorentzian* critical Ising magnet

¹⁹This subsection states what is proved in Streater–Wightman, *PCT, Spin and Statistics, and All That*, chs. 2–3 (the tube theorems and the Bargmann–Hall–Wightman analysis). The proofs are essentially exercises in several-variable complex analysis, and we cite them rather than reproduce them.

do? Continue by the recipe: with $x_E^2 = r^2 + \tau^2 \rightarrow r^2 - (t - i\epsilon)^2$,

$$\langle 0 | \sigma(t, r) \sigma(0) | 0 \rangle = \frac{1}{[r^2 - (t - i\epsilon)^2]^{1/8}} = \begin{cases} (r^2 - t^2)^{-1/8} & \text{spacelike,} \\ (t^2 - r^2)^{-1/8} e^{-i\pi/8} & \text{timelike, } t > 0, \end{cases} \quad (\text{I.2.21})$$

since for $t > r$ the argument $r^2 - (t - i\epsilon)^2$ crosses to the negative reals from the upper side, $(|X|e^{i\pi})^{-1/8} = |X|^{-1/8}e^{-i\pi/8}$. Three comments. (i) At spacelike separation the correlator is real and independent of the operator ordering, as causality requires. (ii) At timelike separation it acquires the universal phase $e^{-i\pi\Delta_\sigma}$ — the operator ordering now matters, the reversed order giving $e^{+i\pi\Delta_\sigma}$ — and the imaginary part

$$\langle 0 | [\sigma(t, r), \sigma(0)] | 0 \rangle = -2i \sin\left(\frac{\pi}{8}\right) \frac{\text{sgn}(t)}{(t^2 - r^2)^{1/8}} \quad (\text{I.2.22})$$

is a nonvanishing commutator filling the entire interior of the forward lightcone. Compare the free massless field (I.2.18), whose commutator was supported on the lightcone alone: in an interacting CFT, a local disturbance is felt at every timelike-separated point, not only on the wavefront. (iii) The branch-point structure at the lightcone $t = r$, with its exponent set by an anomalous dimension, is the starting point of Lorentzian CFT technology — lightcone limits, where the analytic bootstrap of Part III.3 operates.

Exercise I.2.4 (medium). Check (I.2.21) against unitarity: Fourier transform the 2d Wightman function $W(t, r)$ at r -momentum zero and verify the spectral weight is supported at positive energy and is positive. Then verify consistency with the Källén–Lehmann formula (I.2.28) of the next subsection at $d = 2$, $\Delta = \frac{1}{8}$.

Hint: It is easiest to trust the general theorem and check the phase: the positivity computation is the $d = 2$ case of Exercise I.2.5.

The spectral representation, and theories without particles

The two-point function earns one more pass, because in an interacting theory it carries a complete, presentation-independent record of the spectrum, in a form — the **Källén–Lehmann representation** — that we will use several times in this chapter. We derive it now.

Return to the spectral sum (I.2.13) and organize the intermediate states by their total momentum. In a Poincaré-invariant theory the Hilbert space is spanned by states $|\lambda, \mathbf{p}\rangle$ of definite spatial momentum $\mathbf{p}$ and energy $E_\lambda(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m_\lambda^2}$, where λ runs over all remaining labels — single- and multi-particle content, internal quantum numbers, relative momenta — organized into Lorentz multiplets of invariant mass m_λ . With covariant normalization, completeness reads

$$\mathbf{1} = |0\rangle\langle 0| + \sum_\lambda \int \frac{d^{d-1}p}{(2\pi)^{d-1}2E_\lambda(\mathbf{p})} |\lambda, \mathbf{p}\rangle\langle \lambda, \mathbf{p}|. \quad (\text{I.2.23})$$

Insert this between the two operators in $W(x) = \langle 0 | \mathcal{O}(x) \mathcal{O}(0) | 0 \rangle$, taking $\langle \mathcal{O} \rangle = 0$ so that the vacuum term drops out, and use translation covariance to extract the position dependence, $\langle 0 | \mathcal{O}(x) | \lambda, \mathbf{p} \rangle = e^{ip \cdot x} \langle 0 | \mathcal{O}(0) | \lambda, \mathbf{p} \rangle$ with p^0 on shell:

$$W(x) = \sum_\lambda \int \frac{d^{d-1}p}{(2\pi)^{d-1}2E_\lambda(\mathbf{p})} |\langle 0 | \mathcal{O}(0) | \lambda, \mathbf{p} \rangle|^2 e^{ip \cdot x}. \quad (\text{I.2.24})$$

Now use Lorentz invariance. For a scalar operator the matrix element $\langle 0|\mathcal{O}(0)|\lambda, \mathbf{p}\rangle$ is unchanged by boosts, so its modulus can depend on $\mathbf{p}$ only through the invariant mass m_λ ; the momentum integral in (I.2.24) is then exactly the integral that produced the free-field Wightman function in Example I.2.1, evaluated at mass m_λ . All dependence on the dynamics therefore collapses into a single nonnegative function of one variable, the **spectral density**

$$\rho(s) \equiv \sum_{\lambda} |\langle 0|\mathcal{O}(0)|\lambda\rangle|^2 \delta(s - m_\lambda^2) \geq 0, \quad (\text{I.2.25})$$

and the two-point function becomes a weighted superposition of free two-point functions of every mass:

$$W(x) = \int_0^\infty ds \rho(s) W_{\text{free}}(x; s). \quad (\text{I.2.26})$$

This is the Källén–Lehmann representation. The derivation used only completeness, translation and Lorentz covariance, and unitarity ($\rho \geq 0$, being a sum of squared amplitudes for $\mathcal{O}$ to create states from the vacuum); it therefore holds in any unitary Poincaré-invariant theory, with or without a Lagrangian. Since (I.2.26) holds throughout the common domain of analyticity, it continues directly to Euclidean points, where the free propagator of mass² = s is $1/(p^2 + s)$ in momentum space:

$$\begin{aligned} \langle \mathcal{O}(x)\mathcal{O}(0) \rangle_{\text{E}} &= \int_0^\infty ds \rho(s) G_{\text{free}}(x; s), \\ \tilde{G}(p) &= \int_0^\infty ds \frac{\rho(s)}{p^2 + s}. \end{aligned} \quad (\text{I.2.27})$$

Analytically, (I.2.27) says the momentum-space two-point function is a Stieltjes transform: holomorphic in the cut p^2 plane, with all its singularities on the negative real axis $p^2 = -s$ (the Lorentzian on-shell region, $p^2 = -m^2$ in mostly-plus), and with ρ recoverable as the discontinuity across the cut. Everything we said about complex time in the previous subsection has become, after Fourier transformation, a statement about complex momentum.

The representation turns qualitative questions about the spectrum into questions about the *shape* of ρ , answerable by inspecting correlators alone:

- *A gapped theory with a stable particle:* $\rho(s) = Z\delta(s - m^2) + (\text{continuum starting at } s \geq (2m)^2)$. The isolated δ -function is the particle; the gap between it and the multiparticle continuum is what makes asymptotic states, and hence an S-matrix, exist.
- *The free massless field:* $\rho(s) = \delta(s)$, one massless particle and nothing else.
- *A conformal field theory:* power-law correlators force a power-law density. For $\langle \mathcal{O}\mathcal{O} \rangle_{\text{E}} = 1/x^{2\Delta}$ in d dimensions, Fourier transformation gives $\tilde{G}(p) \propto (p^2)^{\Delta-d/2}$ and

$$\rho(s) = \frac{\pi^{d/2} 2^{d-2\Delta}}{\Gamma(\Delta)\Gamma(\Delta - \frac{d}{2} + 1)} s^{\Delta-d/2}, \quad (\text{I.2.28})$$

a featureless continuum from $s = 0$ to infinity: no δ -function, no gap, no threshold (Exercise I.2.5). This makes precise the claim of chapter I.1 that a CFT has no particles. There is no state for an asymptotic wavepacket to converge to, hence no S-matrix; the correlators are not a stepping stone to the observables — they *are* the observables.²⁰

²⁰This point, with the same computation, opens Rychkov’s EPFL lectures [1601.05000](#), §1.3.1. A dimension close to a free value, $\Delta = \frac{d-2}{2} + \gamma$ with γ small, gives $\rho \sim s^{\gamma-1}$: a free particle’s δ -function “melted” into a sharp but integrable peak — the right mental image for an almost-free fixed point like Banks–Zaks.

Positivity of ρ now constrains which power laws can occur as two-point functions. The prefactor in (I.2.28) changes sign when $\Gamma(\Delta - \frac{d}{2} + 1)$ does: for the power law $1/x^{2\Delta}$ to be the two-point function of *any* operator in a unitary theory requires

$$\Delta \geq \frac{d-2}{2}, \quad (\text{I.2.29})$$

with equality forcing $\rho \rightarrow \delta(s)$ — the free field. This is the scalar **unitarity bound**, here derived purely from the Lorentzian spectral representation, before conformal symmetry has even entered the chapter. (Part III.1 re-derives it, together with its spin- ℓ siblings, from radial quantization, where it generalizes more easily.) We will reuse both the bound and the computation behind it when we build the generalized free field in Example I.2.6.

Exercise I.2.5 (classic). Derive (I.2.28). First establish the Fourier transform $\int d^d x e^{-ip \cdot x} (x^2)^{-\Delta} = \pi^{d/2} 2^{d-2\Delta} \frac{\Gamma(d/2-\Delta)}{\Gamma(\Delta)} (p^2)^{\Delta-d/2}$ (valid by analytic continuation in Δ), then extract the spectral density as the discontinuity of the Stieltjes transform (I.2.27), $\rho(s) = \frac{1}{\pi} \text{Im} \tilde{G}(p^2 = -s - i0)$, and use the reflection formula $\Gamma(z)\Gamma(1-z) = \pi / \sin \pi z$ to reach the quoted form. Verify: (a) $\rho \geq 0$ iff $\Delta \geq \frac{d-2}{2}$ (for Δ below the free value the would-be density oscillates in sign); (b) as $\Delta \rightarrow \frac{d-2}{2}$, $\rho(s) \rightarrow \delta(s)$ in the distributional sense.

Hint: For the Fourier transform, use the Schwinger parametrization $(x^2)^{-\Delta} = \frac{1}{\Gamma(\Delta)} \int_0^\infty du u^{\Delta-1} e^{-ux^2}$ and do the Gaussian integral. For (b), note $s^{\gamma-1}/\Gamma(\gamma) \rightarrow \delta(s)$ as $\gamma \rightarrow 0^+$ on test functions, after checking the overall normalization.

Answer: The sign of ρ is the sign of $1/\Gamma(\Delta - \frac{d}{2} + 1)$, positive for $\Delta > \frac{d}{2} - 1$, and the limit (b) is the statement that the free massless particle is the boundary case of the conformal continuum.

Remark I.2.1 (Correlators are distributions). At coincident points correlators blow up — $1/x^{2\Delta}$ is not a function at $x = 0$ — and the axiomatic tradition correspondingly treats fields as operator-valued *distributions*, to be smeared with test functions before they act on the Hilbert space. The Wightman axioms (chapter I.1, §6) make temperedness an axiom; theories with very fast-growing spectra can violate it and require finer test-function classes. In these notes we work at non-coincident points, where everything is real-analytic, and treat the short-distance behavior not as an analytic nuisance but as the *subject*: the operator product expansion of Section I.2.5 is exactly the statement that the coincident-point singularity is structured, computable, and — in a CFT — the generating structure of the whole theory.

I.2.3 Reflection positivity: how correlators become a theory

The previous section moved in one direction: given a quantum theory — Hilbert space, positive Hamiltonian, operators — we produced Euclidean correlators and located them inside an analytic structure. But the working direction of modern physics is the reverse. A lattice model, a Euclidean path integral, a set of bootstrap data: each hands you Schwinger functions and nothing else. Nobody hands you a Hilbert space. So the central question of this section is: *which* sets of Euclidean correlation functions are the Euclidean section of a quantum theory? A generic statistical ensemble has no reason to describe a quantum theory: unitary evolution and the positivity of probabilities are additional structure, and something in the Euclidean data must encode that structure — in a form that can be checked without ever leaving Euclidean signature.

The required condition is called reflection positivity, and it can be motivated by asking how a state of the theory is described in Euclidean language. In a Hamiltonian presentation one builds

states by acting on the vacuum with operators in the past. Under the continuation of the previous section, “the past” becomes the lower half-space $\tau < 0$, so a state on the $\tau = 0$ slice is specified by a *collection of operator insertions in the lower half-space*. Call such a collection F — finitely many operators at points $x_1, \dots, x_k$ with $\tau_i < 0$, or any linear combination of such. Its conjugate state, by the continuation rule (I.2.14) read backwards, is the *time-reflected, complex-conjugated* collection ΘF , living in the upper half-space: Euclidean time reflection $\theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x})$ is what Hermitian conjugation looks like after Wick rotation.²¹ The norm of the state is then a correlator with mirror-symmetric insertions,

$$\|F\|^2 = \langle (\Theta F) F \rangle, \quad (\text{I.2.30})$$

and quantum mechanics demands it be nonnegative. That is the condition.

Definition I.2.3 (Reflection positivity)

A collection of Schwinger functions is **reflection positive** with respect to a hyperplane (say $\tau = 0$) if for every finite linear combination $F = \sum_{\alpha} c_{\alpha} \mathcal{O}_1(x_1^{\alpha}) \cdots \mathcal{O}_{k_{\alpha}}(x_{k_{\alpha}}^{\alpha})$ of insertions supported at $\tau < 0$,

$$\sum_{\alpha, \beta} \bar{c}_{\alpha} c_{\beta} \left\langle \theta [\mathcal{O}_1(x_1^{\alpha}) \cdots]^* \mathcal{O}_1(x_1^{\beta}) \cdots \right\rangle \geq 0, \quad (\text{I.2.31})$$

where θ reflects all insertion points through the hyperplane and conjugates coefficients. Reflection positivity is the Euclidean avatar of unitarity, and we will use the two words interchangeably once the dictionary is established.

Two remarks before we test the definition on a lattice. First, the condition involves a *choice* of hyperplane, but Euclidean invariance of the correlators promotes positivity across one plane to positivity across all of them — a freedom that looks innocent and is not: making radically different choices of “cut” is precisely how radial quantization will fall out of this machinery in Section I.2.5. Second, the definition never mentions a Hilbert space; it is a property of a list of functions, checkable configuration by configuration on a lattice. That is its entire value.

Example I.2.3 (Reflection positivity on the lattice, by transfer matrix)

Return to the Ising chain of Example I.1.3: spins $s_i = \pm 1$, weight $\prod_i e^{J s_i s_{i+1}}$, and now take the chain infinite (or long, with free ends — the boundary will not matter). Reflect through the *bond* between sites 0 and 1: $\theta(i) = 1 - i$ maps the right half $\{i \geq 1\}$ onto the left half $\{i \leq 0\}$. Let F be any real function of the spins on the right half. Reflection positivity asks for $\langle (\Theta F) F \rangle \geq 0$.

Sum first over the spins strictly inside each half, holding the boundary spins fixed. All the dependence collapses into a single function of one spin,

$$v_F(s_1) = \sum_{\{s_i\}_{i \geq 2}} F(s_1, s_2, \dots) \prod_{i \geq 1} e^{J s_i s_{i+1}}, \quad (\text{I.2.32})$$

²¹One line of algebra: Euclidean Heisenberg evolution $\mathcal{O}(\tau, \mathbf{x}) = e^{H\tau} \mathcal{O}(0, \mathbf{x}) e^{-H\tau}$ is *not* unitary, so for a scalar Hermitian at $\tau = 0$, $\mathcal{O}(\tau, \mathbf{x})^{\dagger} = e^{-H\tau} \mathcal{O}(0, \mathbf{x}) e^{H\tau} = \mathcal{O}(-\tau, \mathbf{x})$: the conjugate lives at the mirror point. Operators with spin pick up a reflection acting on their indices; see Simmons-Duffin [1602.07982](#), §7.1, whose treatment we follow, and Appendix A.

and, by the mirror symmetry of the construction, the left half produces the *same* function of s_0 . The full average is then a 2×2 quadratic form threaded by the one bond that crosses the mirror:

$$\begin{aligned} \langle (\theta F) F \rangle &\propto \sum_{s_0, s_1} v_F(s_0) e^{J s_0 s_1} v_F(s_1) \\ &= v_F^T T v_F, \\ T &= \begin{pmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{pmatrix}. \end{aligned} \tag{I.2.33}$$

This is nonnegative for every F if and only if the bond matrix T — the transfer matrix of Example I.1.3 — is *positive definite*. Its eigenvalues are $2 \cosh J$ and $2 \sinh J$: both positive for the ferromagnet $J > 0$. The Ising chain is reflection positive, and the proof is two lines once the transfer matrix exists.

Three features of this short computation illustrate general phenomena. (i) What reflection positivity actually delivered is the statement that T is a positive operator, $T = e^{-aH}$ for a Hermitian H bounded below (here $H = -\frac{1}{a} \log T$, finite because both eigenvalues are positive). In chapter I.1 we *matched* $T \approx e^{-aH}$ in the continuum limit as a computational device; reflection positivity is the structural reason a Hamiltonian had to be there. A statistical weight whose mirror-symmetric averages are positive contains a quantum theory, and the Hamiltonian is the logarithm of the weight. (ii) The positivity was not automatic. For the antiferromagnet $J < 0$ the bond matrix has a negative eigenvalue and bond-reflection positivity *fails* — though reflection through a *site* ($\theta(i) = -i$, which fixes site 0) still works: summing out everything except the mirror spin gives $\langle (\theta F) F \rangle \propto \sum_{s_0} u_F(s_0)^2 \geq 0$ identically. Which reflections are positive is a real, useful invariant of a lattice model (site- vs. bond-centered reflections are the germ of the chessboard estimates of rigorous statistical mechanics), and on the bipartite lattice the antiferromagnet is rescued by a sublattice flip — a baby example of positivity holding only after a dressing. (iii) Complex couplings violate the hypotheses: for the chain in an imaginary magnetic field, $e^{h s_i}$ with h imaginary, the weight is not positive, reflection positivity fails, and yet the model has a perfectly sharp critical point — the Lee–Yang edge singularity, governed by a *non-unitary* conformal field theory (Exercise I.2.7). Reflection positivity is a restrictive axiom, not a formality: nature’s statistical-mechanics corner is populated by theories that violate it and are interesting for that very reason.

Exercise I.2.6 (easy). Extend the bond-reflection argument to the two-dimensional Ising model on a rectangular lattice (reflect through a line of bonds; the boundary data is now a whole column of spins $\mathbf{a} = (s_{1,j})_j$). Show $\langle (\theta F) F \rangle = \sum_{\mathbf{a}, \mathbf{b}} v_F(\mathbf{a}) K(\mathbf{a}, \mathbf{b}) v_F(\mathbf{b})$ with $K(\mathbf{a}, \mathbf{b}) = \prod_j e^{J a_j b_j}$, and prove K is a positive matrix.

Hint: $e^{Jab} = \cosh J + ab \sinh J$ for $a, b = \pm 1$, so K is a tensor product of 2×2 positive matrices when $J > 0$.

Exercise I.2.7 (thinker). (Lee–Yang, or life without positivity.) Consider the Ising chain in a purely imaginary field, weight $\exp[\sum_i (J s_i s_{i+1} + i h_I s_i)]$ with h_I real. (a) Show the transfer matrix is no longer Hermitian, find where its eigenvalues collide as h_I grows at fixed J , and conclude the free energy has a singularity at real h_I — the Lee–Yang edge, which in $d \geq 2$ becomes a true critical point. (b) Exactly which step of Example I.2.3 fails? (c) The corresponding CFT (Fisher; in 2d the minimal model $\mathcal{M}(2, 5)$, cf. Part III.4) has an operator with $\Delta < 0$, flatly violating (I.2.29). Reconcile: which physical statements of this chapter survive without unitarity (analyticity? OPE?) and which do not (norms? bounds?).

Hint: For (a): eigenvalue collision of a non-Hermitian 2×2 matrix means a square-root branch point — the edge exponent in $d = 1$. For (c): analytic continuation needed only H bounded below in some sector; positivity of norms is what fails.

The reconstruction theorem

Reflection positivity is necessary for a quantum interpretation. The remarkable fact — the central structural theorem of axiomatic QFT, and the license for everything these notes do in Euclidean signature — is that together with a short list of statistical-mechanics properties it is *sufficient*: the entire quantum theory can be excavated from the Schwinger functions. The hard analysis is due to Osterwalder and Schrader and we will not reproduce it, but the *construction* at its heart is short, beautiful, and worth doing for real — it is the construction we will reuse, with spheres in place of planes, to prove OPE convergence. We follow the blueprint already visible in the lattice example.

Suppose, then, we are given Schwinger functions S_n for some set of local operators, satisfying: Euclidean covariance; symmetry under permutations; reflection positivity (I.2.31); cluster decomposition; and a temperedness/growth condition on their short-distance singularities (this last is the fine print — see the Note below — and we suppress it until it matters). Construct:

Step 1: the pre-Hilbert space. Let V be the complex vector space of formal finite linear combinations of insertion-collections in the lower half space, $F = \sum_{\alpha} c_{\alpha} \mathcal{O}(x_1^{\alpha}) \cdots \mathcal{O}(x_{k_{\alpha}}^{\alpha})$, all τ 's negative. Define a sesquilinear form on V using only the given correlators:

$$\langle F, G \rangle \equiv \langle (\Theta F) G \rangle = \sum_{\alpha\beta} \bar{c}_{\alpha} c_{\beta} S_{k_{\alpha}+k_{\beta}} \left(\theta x_1^{\alpha}, \dots; x_1^{\beta}, \dots \right). \quad (\text{I.2.34})$$

Reflection positivity says precisely that this form is positive semidefinite.

Step 2: quotient and complete. The null vectors $N = \{F : \langle F, F \rangle = 0\}$ form a subspace (Cauchy–Schwarz, which positive semidefinite forms obey), and on the quotient V/N the form is an inner product. Complete it:

$$\mathcal{H} \equiv \overline{V/N}. \quad (\text{I.2.35})$$

This is *the* Hilbert space of the theory — not a Hilbert space postulated alongside the correlators, but one constructed from them. The vacuum is the class of the empty insertion, $|0\rangle = [\mathbf{1}]$, with $\langle 0|0\rangle = S_0 = 1$.

The quotient by null vectors requires explanation, because it is physics rather than hygiene. A **null configuration** is a linear combination F of insertions whose correlators with *every* configuration in the upper half-space vanish: by Cauchy–Schwarz, $\langle F, F \rangle = 0$ implies $\langle G, F \rangle = 0$ for all G , so a null F is indistinguishable from zero by any conceivable future measurement, and the quotient simply identifies it with zero. Two examples make this concrete. In the free massless scalar theory, the insertion of $\partial^2 \phi(x)$ is null — its correlators with everything vanish because ∂^2 annihilates the propagator at separated points — so the quotient is how the equation of motion, an operator identity of Section I.2.1, is implemented in the reconstructed Hilbert space. On the Ising chain, any two right-half configurations with the same boundary function v_F in (I.2.33) differ by a null configuration, and the quotient collapses the enormous space of insertion patterns to the two-dimensional boundary Hilbert space $\mathbb{C}^2$, as Exercise I.2.9 makes explicit. (In gauge theories, the same quotient enforces constraints such as Gauss's law; we will see this in Part V.)

Step 3: the Hamiltonian. Euclidean time translation acts on insertions: $(T_{\sigma} F)$ shifts every insertion by $\tau \rightarrow \tau - \sigma$, $\sigma > 0$, staying in the lower half-space. The form (I.2.34) is reflection-symmetric, so T_{σ} is a symmetric operator on V/N , and a two-line Cauchy–Schwarz induction

(Exercise I.2.8) shows it is a *contraction*: $\|T_\sigma F\| \leq \|F\|$. A symmetric contraction semigroup has a self-adjoint, nonnegative generator:

$$\begin{aligned} T_\sigma &= e^{-\sigma H}, \\ H &= H^\dagger \geq 0. \end{aligned} \tag{I.2.36}$$

Note the direction of the logic here. In Section I.2.2, positivity of the energy was the *input* and Euclidean analyticity the output. Here the direction is reversed: reflection positivity of statistical averages *produced* a positive Hamiltonian. The two positivities are the same fact seen from the two sides of the Wick rotation — this is the precise sense in which Euclidean field theory is not a trick but an equivalent formulation.

Step 4: operators, and back to Wightman. A local operator at $\tau = 0$ is the limit of insertions approaching the mirror from below; smeared suitably, these become densely defined operators on $\mathcal{H}$. The remaining Euclidean symmetries come along: spatial translations and rotations act on insertions and descend to unitaries on $\mathcal{H}$; the rotations that mix Euclidean time with space do *not* descend to unitaries — they become, upon the analytic continuation now justified by (I.2.36) and the spectral theorem, the *boosts* of a unitary representation of the Poincaré group, and the analytically continued correlators of the reconstructed operators are Wightman functions satisfying the full set of Wightman axioms. This last step — controlling the continuation of n -point functions, not just of the semigroup — is where the real work and the growth condition live, and is what we cite rather than derive.²²

Theorem I.2.1 (Osterwalder–Schrader reconstruction)

Schwinger functions satisfying Euclidean covariance, permutation symmetry, reflection positivity, cluster decomposition, and a linear-growth temperedness condition are the Euclidean continuation of a unique unitary quantum field theory satisfying the Wightman axioms — with Hilbert space, vacuum, positive Hamiltonian, Poincaré representation, and field operators all constructed from the Schwinger functions as in Steps 1–4. Cluster decomposition of S_n becomes uniqueness of the vacuum.

Note I.2.1. The growth condition is a substantive hypothesis, not a technical formality. Osterwalder and Schrader’s first paper claimed reconstruction without it; the proof contained a gap, and the linear-growth hypothesis of their second paper (each S_n bounded by factorially many derivatives, in a controlled way) is what makes the continuation of all n -point functions converge. There exist sets of Schwinger functions satisfying every other axiom whose continuation fails: positivity across planes by itself does not control how fast the correlators grow with n . For the theories considered in these notes the condition is verified wherever rigorous constructions exist (the $d = 2, 3$ scalar theories of chapter I.1’s constructive inventory), and for CFTs the OPE of the next section provides much stronger control over coincident-point behavior. We will assume the condition from here on without further comment.

²²Osterwalder–Schrader, *Axioms for Euclidean Green’s functions* I, Comm. Math. Phys. **31** (1973) 83, and II, **42** (1975) 281. Neighboring axiom systems with their own reconstruction theorems (Glimm–Jaffe’s, built on the generating functional; Nelson’s, built on the Markov property) are treated in Glimm–Jaffe, *Quantum Physics*, chs. 6 and 19.

Idea I.2.3 (Correlators are the theory)

With reflection positivity, the slogan of chapter I.1 becomes a construction: the Hilbert space is built from Euclidean correlators (insertions modulo null configurations), the Hamiltonian is the generator of Euclidean time shifts acting on them, and the field operators are insertions approaching the cut. Nothing else needs to be given. Whoever specifies consistent correlators — a lattice at criticality, a bootstrap dataset $\{\Delta_i, c_{ijk}\}$, a mythical path integral — has specified a complete quantum theory, whether or not any Lagrangian is anywhere in sight.

One aspect of the construction should be isolated as a principle, because we will reuse it. Examine what was actually used: not the plane as such, but a *reflection*. The construction needed three ingredients — an involution θ exchanging two halves of Euclidean space, positivity of the correlators on mirror-symmetric configurations, and a family of translates of the mirror along which to evolve — and the plane $\tau = 0$ entered only as the fixed locus of the mirror. Stated invariantly: *a quantization of Euclidean data is a choice of reflection-with-positivity together with a compatible foliation*. Parallel planes are one such choice; and since Euclidean correlators have no preferred time direction, every hyperplane provides one, all of them unitarily equivalent. This raises a natural question: are there reflections of a different shape, which quantize the *same* correlators in different and useful coordinates? For conformal theories the answer is yes, and it is the subject of Section I.2.5. Before that, we address a question raised by the reconstruction theorem itself: on which backgrounds should the Euclidean correlators be defined in the first place?

Exercise I.2.8 (medium). Prove the contraction property in Step 3: for any F in the lower half-space and T_σ the shift, show

$$\|T_\sigma F\|^2 = \langle F, T_{2\sigma} F \rangle \leq \|F\| \|T_{2\sigma} F\|,$$

iterate to

$$\|T_\sigma F\| \leq \|F\|^{1-2^{-n}} \|T_{2^n \sigma} F\|^{2^{-n}},$$

and use the temperedness of correlators (no blow-up of $\|T_\sigma F\|$ as $\sigma \rightarrow \infty$, by clustering) to conclude $\|T_\sigma F\| \leq \|F\|$. Where did reflection symmetry of the inner product enter?

Hint: $\langle T_\sigma F, T_\sigma F \rangle = \langle F, T_{2\sigma} F \rangle$ uses $\Theta T_\sigma = T_{-\sigma} \Theta$: shifting both halves away from the mirror by σ is one shift by 2σ across it.

Exercise I.2.9 (classic). (Reconstruction, executed completely.) Run Steps 1–3 on the Ising chain. Take as input *only* the correlators $\langle s_{i_1} \cdots s_{i_n} \rangle$ computed in Example I.1.3. Show: (a) the space of right-half insertion collections, modulo null vectors, is exactly two-dimensional (every collection of insertions at sites ≥ 1 is equivalent, as a state, to a combination of “nothing” and “one spin at site 1”); (b) the inner product reproduces $\mathbb{C}^2$ with the transfer-matrix inner product; (c) the reconstructed e^{-aH} is T/λ_+ (normalizing the ground-state energy to zero), with H matching the continuum limit (I.1.18) as $a \rightarrow 0$. This is the smallest nontrivial instance of Theorem I.2.1, with every analytic subtlety absent.

Hint: For (a): on the lattice, v_F in (I.2.33) is the state; two functions of one Ising spin span a two-dimensional space. Null vectors are collections with $v_F = 0$.

1.2.4 Interlude: which Euclidean backgrounds are allowed?

The reconstruction theorem connects flat Euclidean space to flat Minkowski space, and for this chapter that is nearly all we need. But the inventory of chapter I.1 assigns partition functions and correlators to *curved* backgrounds too, and there the phrase “Wick rotation” quietly loses its meaning: a generic curved spacetime has no preferred time coordinate to rotate, and a generic smooth manifold has no canonical complexification for t to wander into. Since these notes will spend their lives on spheres, tori, and cylinders — starting two sections from now, when a sphere becomes a Hilbert space — we will need the modern answer, due to Kontsevich and Segal, to the question: *on which backgrounds should a QFT’s Euclidean data be expected to exist at all?* Their move is characteristically simple: do not complexify time — complexify the *metric*.²³

The setup is the following. At each point of the manifold, allow the metric to be a *complex-valued* symmetric bilinear form on the (real) tangent space. Any such form can be diagonalized in a real basis,

$$g = \sum_{i=1}^d \lambda_i dy_i^2, \quad (I.2.37)$$

$$\lambda_i \in \mathbb{C} \setminus \{0\},$$

so the question “which complex metrics are allowed?” becomes a question about the complex eigenvalues λ_i . The criterion for admitting a metric must be that the Euclidean weight e^{-S} *damps* rather than oscillates or grows: the real part of the action should be positive for every field configuration, since this is what makes the functional integral converge. We therefore compute the action of the simplest matter on the background (I.2.37) and demand positivity of its real part.

For a free scalar, the curved-space action $S = \frac{1}{2} \int \sqrt{\det g} [g^{ij} \partial_i \phi \partial_j \phi + m^2 \phi^2]$ involves the metric in two places: the volume element, which for the diagonal metric (I.2.37) is $\sqrt{\det g} = \prod_i \lambda_i^{1/2}$ (taking for each square root the branch with positive real part, the one continuously connected to the Euclidean case), and the inverse metric in the kinetic term, which contributes $1/\lambda_i$ for the gradient in the i -th direction. Explicitly, for a constant diagonal metric,

$$S[\phi] = \frac{1}{2} \int d^d y \left(\prod_k \lambda_k^{1/2} \right) \left[\sum_i \frac{1}{\lambda_i} (\partial_i \phi)^2 + m^2 \phi^2 \right]. \quad (I.2.38)$$

The integrand is a sum of squares of real quantities ($\partial_i \phi$ and ϕ) with complex coefficients, so $\text{Re} S > 0$ for all configurations exactly when every coefficient has positive real part. The mass term requires $|\arg \prod_k \lambda_k^{1/2}| < \frac{\pi}{2}$, and the gradient terms require

$$\left| \arg \frac{\prod_k \lambda_k^{1/2}}{\lambda_i} \right| < \frac{\pi}{2} \text{ for each } i, \quad (I.2.39)$$

$$\left| \frac{1}{2} \sum_{k \neq i} \arg \lambda_k - \frac{1}{2} \arg \lambda_i \right| < \frac{\pi}{2}.$$

The scalar alone, however, does not determine the final answer; we should ask the same question for the other standard free fields, and it is here that the criterion takes its final form. The natural

²³Kontsevich–Segal 2105.10161, §2, which this section follows including its notation. The idea: the archetype of every path integral is the oscillatory Gaussian $\int e^{\frac{i}{2} x^T A x}$, which one *defines* as a boundary value from the domain of complex symmetric A with positive-definite imaginary part, where the integral converges. The space of complex metrics below is the field-theory version of that domain; Lorentzian signature will sit on its boundary exactly as the real A ’s sit on the boundary of the Siegel half-plane.

family to consider consists of the abelian gauge fields of all ranks: the Maxwell field A_μ with field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and action $\frac{1}{2} \int F \wedge \star F$, and its higher-rank generalizations, p -form gauge fields $A_{\mu_1 \dots \mu_p}$ with $(p+1)$ -form field strengths.²⁴ The pattern of metric factors generalizes directly: the kinetic term of a $(p+1)$ -form field strength carries one inverse metric per index, so the quadratic form for the components of F along the directions $I = \{i_1 < \dots < i_{p+1}\}$ has coefficient $(\prod_k \lambda_k^{1/2}) / (\lambda_{i_1} \dots \lambda_{i_{p+1}})$. Demanding positive real part for all ranks p and all index sets I gives a family of conditions of the form (I.2.39), with one $\arg \lambda$ subtracted for each index in I ; the strongest of these conditions is the one in which I collects all the eigenvalues whose arguments share a sign, and imposing that worst case is equivalent to a single inequality (Exercise I.2.10 carries out the maximization):

Theorem I.2.2 (Allowable complex metrics (Kontsevich–Segal))

A complex metric $g = \sum_i \lambda_i dy_i^2$ admits convergent quadratic actions for p -form fields of every degree if and only if

$$\sum_{i=1}^d |\arg \lambda_i| < \pi. \quad (\text{I.2.40})$$

A QFT's partition functions and correlators are expected to extend holomorphically over the space of metrics satisfying (I.2.40) pointwise — and no further.

Consider now what the inequality (I.2.40) says about *signature*. For a real metric, each eigenvalue λ_i is real, so each $\arg \lambda_i$ is 0 (positive eigenvalue) or π (negative eigenvalue, understood as a limit), and the sum in (I.2.40) is π times the number of negative eigenvalues. Three cases result. Euclidean metrics, with no negative eigenvalues, satisfy the inequality strictly: they lie in the interior of the allowable domain, where path integrals converge. Lorentzian metrics, with exactly one negative eigenvalue, give $\sum_i |\arg \lambda_i| = \pi$: they saturate the inequality, and so lie on the *boundary* of the domain — not themselves allowable, but reachable as limits of allowable metrics. And real metrics with two or more time directions give $\sum_i |\arg \lambda_i| \geq 2\pi$: they are not even on the closure of the domain, and cannot be reached from Euclidean signature by any limit of allowable metrics. The old question of why the time direction is special — nothing in the formalism of QFT appears to single out one eigenvalue of the metric — thus receives a sharp answer internal to quantum field theory: *Lorentzian spacetimes are precisely the boundary points of the domain of convergent Euclidean field theory*, the locus where damping degenerates into pure oscillation, in the same way that e^{iHt} is the boundary behavior of $e^{-H\tau}$.²⁵

Our flat-space Wick rotation now embeds as the simplest possible path through the domain. Rotate the time coordinate by phase, $t = e^{-i\theta} \tau$: the metric component along it is $\lambda_t = e^{-2i\theta}$, all other λ 's real positive, so $\sum_i |\arg \lambda_i| = 2\theta$, which sweeps from 0 (Euclidean) to π (Lorentzian) as

²⁴Why this family? Free p -form fields exist as sensible quantum field theories in any dimension, so a criterion for “backgrounds on which QFT makes sense” must accommodate them; and they turn out to be exactly the family that sharpens the scalar condition into (I.2.40) below. Kontsevich–Segal also note an argument of Witten that the same conditions are natural from the standpoint of the energy-momentum tensor. The scalar is the $p = 0$ member of the family.

²⁵Kontsevich–Segal sharpen “boundary” to *Shilov boundary* — the minimal set controlling maxima of holomorphic functions, on which the time-oriented Lorentzian metrics sit — and prove the criterion guarantees convergence not just of actions but of the p -form partition functions themselves on tori. We take up that refinement, the application to curved backgrounds, and the Segal-style gluing axioms it serves, in chapter I.4.

$\theta: 0 \rightarrow \pi/2$ — always allowable until the endpoint. The $i\epsilon$ prescriptions of Section I.2.2, and the dashed quarter-circle in Figure 2, are this path seen by the correlators: the small imaginary part is the last trace of the damping domain, kept alive until the boundary value is taken.

Preview I.2.1 (Where this goes)

Chapter I.4 uses allowability to answer “on which complex backgrounds does the inventory make sense?” — including the cylinder-with-complexified-time geometries that interpolate between partition functions and return amplitudes, and the curved-space version of the state-operator map. Beyond these notes’ Part I: the criterion has become a working tool in quantum gravity and cosmology (which complex saddles of the gravitational path integral are admissible), precisely because it is a statement about QFT on a background, needing no theory of the metric’s own dynamics.

Exercise I.2.10 (medium). Prove Theorem I.2.2 from the p -form conditions. The coefficient for the $(p+1)$ -form field strength along directions $I = \{i_1 < \dots < i_{p+1}\}$ has argument $\frac{1}{2} \sum_j \arg \lambda_j - \sum_{i \in I} \arg \lambda_i$. Show that as I ranges over all subsets (all p), the supremum of the absolute value of this quantity is exactly $\frac{1}{2} \sum_j |\arg \lambda_j|$, attained by taking I to be the set of all indices with $\arg \lambda_i > 0$ (or its complement). Conclude that all coefficients have positive real part iff (I.2.40) holds.

Hint: Adding to I an index with $\arg \lambda_i > 0$ decreases the running total by $\arg \lambda_i$; starting from $\frac{1}{2} \sum_j \arg \lambda_j$, the extremes over subsets are $\pm \frac{1}{2} \sum_j |\arg \lambda_j|$.

Exercise I.2.11 (easy). (Allowability in $d = 1$ is something you already know.) The partition function of the particle on a circle, Example I.1.2, was $Z(\beta) = \text{tr } e^{-\beta H}$. For which complex β does the trace converge? Interpret the answer via Theorem I.2.2 for the metric $g = \beta^2 dt^2$ on the circle of unit coordinate length, and identify the boundary points $\beta = \pm iT$ as the Lorentzian return amplitudes $\text{tr } e^{\mp iHT}$ — the $d = 1$ shadow of “Lorentzian signature lives on the boundary of the allowable domain.”

Answer: $\text{Re } \beta > 0$, i.e. $|\arg \beta| < \pi/2$, i.e. $|\arg \lambda| = |2 \arg \beta| < \pi$: the KS criterion verbatim.

1.2.5 The operator product expansion

We come to the third of the opening’s three questions: what organizes the set of correlators. The set of all n -point functions of all local operators is an infinite tower of functions of ever more variables, subject to consistency conditions; for the local data to serve as a definition of the theory, this tower must be generated by some finitely describable structure. The structure that accomplishes this is the **operator product expansion**: the statement that the product of two nearby local operators is itself expandable in local operators,

$$\mathcal{O}_i(x) \mathcal{O}_j(0) = \sum_k C_{ijk}(x) \mathcal{O}_k(0), \quad (\text{I.2.41})$$

with c-number coefficient functions C_{ijk} , valid as an operator equation — that is, inside arbitrary correlation functions, provided all other operators in the correlator are sufficiently far away (the precise condition appears below). Used iteratively, (I.2.41) reduces any n -point function to a sum of lower-point ones: the tower of correlators collapses onto knowledge of the OPE coefficients and the two-point functions. Wilson proposed exactly this in 1969 — under the name *operator*

algebra, and as a candidate *definition* of a theory, an ambition the textbooks quietly downgraded²⁶ — and the central fact of this section is that in a conformal field theory the original ambition was right: the expansion (I.2.41) *converges* at finite separation, with a computable, geometric radius of convergence and an exponential rate. Not an asymptotic series that happens to work: a convergent expansion in the same sense as a Fourier series, and for the same reason — it *is* an expansion of a Hilbert-space vector in a complete orthogonal basis. But first we should understand why *any* expansion of the form (I.2.41) should exist, in any local theory, conformal or not — because the argument for existence already contains, in embryo, the argument for convergence.

Why an OPE at all: integrating out a ball

Here is the path-integral argument, which works in any presentation with a notion of locality and makes no mention of conformal symmetry. Consider a correlator with two nearby insertions $\mathcal{O}_i(x_1)\mathcal{O}_j(x_2)$ and arbitrary spectators far away. Draw a sphere S of radius R around the close pair, small enough to exclude every spectator, and compute the path integral in two stages: first over the field configurations *inside* S with boundary values ϕ_∂ held fixed, then over everything outside. Locality of the action is what permits the split — the weight e^{-S_E} factorizes across the sphere up to the shared boundary data. The inside integral,

$$\Psi_{ij}[\phi_\partial] = \int_{\phi|_S=\phi_\partial} \mathcal{D}\phi \mathcal{O}_i(x_1)\mathcal{O}_j(x_2) e^{-S_E[\phi]_{\text{inside}}}, \quad (\text{I.2.42})$$

is a functional of the boundary data alone, and it is the *only* place x_1 and x_2 appear: as far as the outside world is concerned, the pair of operators is equivalent to a single functional of the boundary data on S . Now expand this functional in any complete set of boundary functionals, and choose for that set the functionals generated by *single* insertions at the center: $\Psi_k[\phi_\partial] = \int \mathcal{D}\phi \mathcal{O}_k(0) e^{-S_E}$. The expansion coefficients are c-numbers carrying all the x_1, x_2 dependence,

$$\begin{aligned} \Psi_{ij} &= \sum_k C_{ijk}(x_1, x_2) \Psi_k, \\ \mathcal{O}_i(x_1)\mathcal{O}_j(x_2) &= \sum_k C_{ijk}(x_1, x_2) \mathcal{O}_k(0) \end{aligned} \quad (\text{I.2.43})$$

inside any correlator whose other insertions stay outside S . That is the OPE, with its domain of validity — a separating sphere — already attached by the derivation. Two operators close together *are* a superposition of single operators, in precisely the sense that two charges close together are a multipole expansion: seen from outside, only the total “moments” survive, and the moments of a local theory are its local operators.

What the argument does not deliver is control. “Expand in a complete set of boundary functionals” raises three questions — complete by what criterion, in which topology, with what convergence — and in a generic massive theory none has a useful answer: the expansion (I.2.43) holds only *asymptotically* in the limit $x_1 \rightarrow x_2$, with coefficient functions C_{ijk} that depend on all the scales and must themselves be computed. (This asymptotic version, with logarithms resummed by the renormalization group, is the OPE of the QCD textbooks: the organizing principle of deep inelastic

²⁶K. Wilson, Phys. Rev. **179** (1969) 1499; Kadanoff’s contemporaneous operator algebra of the 2d Ising model, PRL **23** (1969) 1430, was the statistical-mechanics twin. Polyakov’s 1974 bootstrap manifesto (chapter I.1, crack 3) is this proposal plus the demand of crossing. The downgraded, asymptotic version became a precision instrument in QCD — deep inelastic scattering is organized by it — which is part of why its original ambition was forgotten.

scattering, where it earns its keep order by order without ever being summed.) The missing ingredient is a *Hilbert space* statement: were the boundary functionals on S organized into an orthogonal basis adapted to the geometry, the expansion would be exact and convergent by the elementary theory of Hilbert spaces. Building exactly that structure is what reflection positivity, run on spheres, does for us in a CFT — and it is the only ingredient of the construction that needs conformal symmetry.

Radial quantization: reconstruction with spheres

Section I.2.3 ended with a principle: a quantization of Euclidean correlators is a reflection-with-positivity plus a compatible foliation, the constant- τ planes being merely the first instance. Once the theory is *conformal*, a second instance exists, and it is strictly superior for studying local operators: take for the mirror the **inversion** $x^\mu \rightarrow x^\mu/x^2$, whose fixed locus is the unit sphere, and for the foliation the concentric spheres S^{d-1} that it preserves — inversion exchanges their interiors and exteriors exactly as θ exchanged the half-spaces.

Run the reconstruction verbatim. A *state* is now a collection of insertions *inside* the unit sphere, modulo null collections; the Hilbert space $\mathcal{H}_{S^{d-1}}$ is their completion — this is item 3 of the inventory (Definition I.1.1), realized concretely for the $(d-1)$ -sphere. Evolution from sphere to sphere is generated by the **dilatation** operator D , which plays the role of the Hamiltonian: in a scale-invariant theory, rescaling $x \rightarrow \lambda x$ maps each leaf of the foliation to another, exactly as time translation slid the planes. Writing $r = e^\tau$, radial evolution is $e^{-\tau D}$, and “radial time” $\tau = \log r$ runs from $-\infty$ (the origin) to $+\infty$ (the point at infinity).²⁷ The role of time reflection θ — Hermitian conjugation’s Euclidean shadow — passes to the inversion, and reflection positivity across planes becomes *inversion positivity*; this is not a new assumption, because a conformal transformation conjugates the planar mirror into the spherical one and carries the one positivity rigorously onto the other.²⁸

What makes the spherical cut *better* than the planar one for local operators is the location of its degenerate leaf. The planes’ foliation degenerates nowhere; the spheres’ foliation degenerates at the origin — a *point*, exactly the habitat of a local operator. The sphere of radius $\rightarrow 0$ wraps a single point, and therefore:

- Inserting *nothing* inside a small sphere prepares the vacuum $|0\rangle$ — which is consequently D -invariant, with radial energy zero.
- Inserting one operator $\mathcal{O}(0)$ at the origin prepares a state

$$\begin{aligned} |\mathcal{O}\rangle &\equiv \mathcal{O}(0)|0\rangle, \\ D|\mathcal{O}\rangle &= \Delta|\mathcal{O}\rangle. \end{aligned} \tag{I.2.44}$$

²⁷Radial quantization conventions follow Simmons-Duffin [1602.07982](#), §§6–7 (our default for CFT material; see Appendix A, which collects the conformal algebra and the inversion conjugation we use). The one fact this chapter leans on: P_μ (translations) and K_μ (special conformal transformations) raise and lower radial energy, $[D, P_\mu] = P_\mu$ and $[D, K_\mu] = -K_\mu$. We use the algebra as bookkeeping; its derivation from conformal invariance, and the representation theory built on it, is chapter III.1’s business.

²⁸The plane-to-sphere dictionary is a conformal conjugation: the generator $K_1 + P_1$, whose flow foliates $\mathbb{R}^d$ by spheres through two fixed points, is conjugate to D inside the conformal group, and the corresponding “N–S quantization” interpolates between planar and radial pictures — see Rychkov [1601.05000](#), §3.1.5, and the original treatment of Lüscher–Mack. Conjugation in radial quantization is $\mathcal{O}(x)^\dagger = x^{-2\Delta}\mathcal{O}(x^\mu/x^2)$ for a Hermitian scalar, whence $P_\mu^\dagger = K_\mu$ and $D^\dagger = D$.

an *eigenstate* of the radial Hamiltonian, with eigenvalue the scaling dimension. (Insert the operator anywhere else and the state is no eigenstate — dilatations move the insertion — but it is a convergent superposition of eigenstates, as we are about to exploit.)

- Conversely — and this is the nontrivial direction — *every* eigenstate of D on the sphere arises this way: given $|\psi\rangle$ with $D|\psi\rangle = \Delta_\psi|\psi\rangle$, cutting a hole around the origin in any correlator’s path integral and gluing $|\psi\rangle$ into the boundary defines a local operator $\mathcal{O}_\psi(0)$ whose correlators are the overlaps $\langle \cdots |\psi\rangle$, and which reproduces $|\psi\rangle$ when re-inserted.

This two-way traffic is the **state–operator correspondence**: in a CFT, the Hilbert space on S^{d-1} and the space of local operators are the same space, with radial energy equal to scaling dimension. We will treat the gluing construction at the level just stated — it is the cut-and-glue axiomatics of chapter I.4 that makes “cutting a hole and gluing a state” fully precise, and chapter III.1 that mines the correspondence for representation theory.²⁹ What this chapter needs from it is one structural consequence, which organizes the entire operator content and which we now state as a definition. The conformal algebra sorts the D -eigenstates into ladders: acting with P_μ raises the radial energy by one unit (this is the commutator $[D, P_\mu] = P_\mu$), while acting with K_μ lowers it by one unit. In a unitary theory dimensions are bounded below ((I.2.29) was a first instance of this), so descending any ladder must terminate: every eigenstate belongs to a tower built on a lowest rung.

Definition I.2.4 (Primary and descendant operators)

A **primary** operator of a CFT is a local operator annihilated by all the special conformal generators at the origin,

$$[K_\mu, \mathcal{O}(0)] = 0 \iff K_\mu|\mathcal{O}\rangle = 0. \quad (\text{I.2.45})$$

the bottom of a ladder. Its **descendants** are the operators obtained by applying momenta, that is, derivatives,

$$\begin{aligned} &|\mathcal{O}\rangle, \\ &P_\mu|\mathcal{O}\rangle = |\partial_\mu\mathcal{O}\rangle, \\ &P_\mu P_\nu|\mathcal{O}\rangle = |\partial_\mu\partial_\nu\mathcal{O}\rangle, \dots \end{aligned} \quad (\text{I.2.46})$$

with dimensions $\Delta, \Delta + 1, \Delta + 2, \dots$; a primary together with its descendants is a **conformal multiplet**. The operator content of a CFT is a list of primaries with their dimensions and spins; every other local operator is a derivative of these.

The reader who learned conformal field theory from a textbook may know a different-looking definition: a primary is an operator that transforms covariantly under finite conformal transformations $x \rightarrow x'$,

$$\mathcal{O}(x) \longrightarrow b(x)^{-\Delta} R[M(x)] \mathcal{O}(x'), \quad (\text{I.2.47})$$

where $b(x)$ is the local scale factor of the map and $R[M(x)]$ the rotation matrix acting on the operator’s spin indices, with *no* inhomogeneous terms — whereas a descendant such as $\partial_\mu\mathcal{O}$ picks up extra terms involving derivatives of $b(x)$. The two definitions are equivalent: K_μ generates the special conformal transformations, which are exactly the conformal maps whose scale factor varies to first order at the origin, so (I.2.45) is the infinitesimal statement of (I.2.47) at the point where

²⁹For the careful flat-space construction, including why no separate convergence assumption is needed, see Simmons-Duffin [1602.07982](#), §6.2, or Polchinski’s *String Theory*, Vol. 1, ch. 2, where the 2d case is done with loving care.

the operator sits. (Chapter III.1 develops the transformation-law side systematically; in this chapter the algebraic form (I.2.45) is the useful one.)

Intuition I.2.1 (Radial quantization is reconstruction, pointed at a point)

Nothing in radial quantization is new machinery. It is the Osterwalder–Schrader construction of Section I.2.3 run with spheres instead of planes: states from insertions inside the cut, inner product from inversion (the spherical mirror), Hamiltonian from the scale flow. The payoff of the spherical cut is that its limiting leaf is a point, so the basis of the reconstructed Hilbert space is labeled by *local operators* — the state–operator correspondence — and the completeness of a Hilbert-space basis becomes available as a statement about products of local operators at a point. Conformal invariance is what lets the sphere’s radius be changed at will, making the foliation as good as the planes’; in a massive theory radial evolution is still defined but D is not a symmetry, and the spheres’ Hilbert spaces, while they exist, admit no useful energy basis.

Derivation of the OPE

Now let two scalar primaries approach each other. Insert $\mathcal{O}_i$ at x and $\mathcal{O}_j$ at the origin, both inside the unit sphere, and read the configuration as a state:

$$|\Psi\rangle = \mathcal{O}_i(x)\mathcal{O}_j(0)|0\rangle \in \mathcal{H}_{S^{d-1}}. \quad (\text{I.2.48})$$

Expand $|\Psi\rangle$ in the energy eigenbasis. By the state–operator correspondence that basis is exhaustively labeled by primaries and their descendants, so the expansion collects into conformal multiplets:

$$\mathcal{O}_i(x)\mathcal{O}_j(0)|0\rangle = \sum_{\mathcal{O} \text{ primary}} \lambda_{ij\mathcal{O}} C_{\mathcal{O}}(x, \partial_y) \mathcal{O}(y)|_{y=0}|0\rangle, \quad (\text{I.2.49})$$

where $C_{\mathcal{O}}(x, \partial_y)$ is a power series in ∂_y packaging the contribution of the whole multiplet of $\mathcal{O}$. That — modulo removing the vacuum vector from both sides, which the convergence analysis below justifies — is the OPE, *derived*: the product of two local operators, viewed through any sphere enclosing both, is a convergent superposition of single local operators at the center. Three structural facts follow immediately, each by pure symmetry:

(i) *The powers are fixed by dimensional bookkeeping.* Acting with D on both sides of (I.2.49) and matching eigenvalues term by term forces the leading coefficient of each multiplet to scale as

$$C_{\mathcal{O}}(x, \partial_y) = \frac{1}{|x|^{\Delta_i + \Delta_j - \Delta_{\mathcal{O}}}} (1 + \#x \cdot \partial_y + \cdots), \quad (\text{I.2.50})$$

(Exercise I.2.12): operators of low dimension dominate at short distance, and the identity operator — present in the $\mathcal{O} \times \mathcal{O}$ OPE of any operator with itself — carries the leading singularity $|x|^{-\Delta_i - \Delta_j}$, which is nothing but the two-point function factoring out.

(ii) *Each multiplet costs one number.* The descendant terms hidden in the ellipsis of (I.2.50) are fixed recursively by acting with K_μ on both sides — conformal symmetry relates the coefficient of $\partial^n \mathcal{O}$ to that of $\mathcal{O}$ (Exercise I.2.13 computes the first two). What symmetry cannot fix is one overall constant $\lambda_{ij\mathcal{O}}$ per primary: the **OPE coefficients**, the dynamical content. The same constants govern the three-point functions, $\langle \mathcal{O}_i \mathcal{O}_j \mathcal{O} \rangle \propto \lambda_{ij\mathcal{O}}$, whose coordinate dependence is again completely fixed by conformal symmetry.

(iii) *Correlators reduce recursively.* Using (I.2.49) inside an n -point function strips it to $(n-1)$ -point functions, then $(n-2)$, down to the two-point functions that normalize the operators. A conformal field theory is therefore completely specified by its **CFT data**:

$$\{\Delta_i(\text{and spins}); \lambda_{ijk}\} \implies \text{every correlator of local operators.} \quad (\text{I.2.51})$$

This is the finitely-describable generating structure the chapter opening demanded, and the precise content of “abstract CFT data” as a *presentation* of a theory in the sense of chapters I.1 and I.5. The consistency condition this data must satisfy — the same correlator reduced in two different pairings must agree, **crossing symmetry** — is the single equation from which Part III will extract the 3d Ising critical exponents. We record the definition in its final form, and then turn to the question of convergence.

Definition I.2.5 (Operator product expansion)

In any local QFT, the product of two local operators admits an asymptotic short-distance expansion (I.2.41) (Wilson). In a CFT, the expansion organizes into conformal multiplets,

$$\mathcal{O}_i(x)\mathcal{O}_j(0) = \sum_{\mathcal{O} \text{ primary}} \frac{\lambda_{ij\mathcal{O}}}{|x|^{\Delta_i+\Delta_j-\Delta_{\mathcal{O}}}} (1 + \dots) \mathcal{O}(0), \quad (\text{I.2.52})$$

with all descendant terms $(\dots)$ fixed by conformal symmetry and one constant $\lambda_{ij\mathcal{O}}$ per primary; and it is an *exact, convergent* operator equation, valid inside any correlator whose remaining insertions all lie outside a sphere separating x and 0 from them.

Convergence: the centerpiece

Here is the claim to be established, in its sharpest useful form. Insert the OPE (I.2.49) into a four-point function of identical scalars ϕ of dimension Δ_ϕ — four points being the first place convergence is at stake and the configuration on which the bootstrap runs. Use conformal transformations to standardize: put one operator at the origin, one at ∞ (i.e. consider $\lim_{x_4 \rightarrow \infty} x_4^{2\Delta_\phi} \langle \phi(x_4) \dots \rangle$), which is the radial-quantization out-state $\langle \phi |$), and the remaining two at radii $r_2 < r_3$:

$$\langle \phi | \phi(x_3) \phi(x_2) | \phi \rangle. \quad (\text{I.2.53})$$

The OPE in the $(x_2, 0)$ pairing is the expansion of the ket $\phi(x_2)|\phi\rangle$ in radial energy eigenstates; conjugating, the bra $\langle \phi | \phi(x_3)$ expands in the dual basis. The four-point function becomes a sum over the exchanged eigenstates,

$$\langle \phi | \phi(x_3) \phi(x_2) | \phi \rangle = \sum_E \langle \phi | \phi(x_3) | E \rangle \langle E | \phi(x_2) | \phi \rangle = \sum_E \rho_E e^{-E\beta}, \quad (\text{I.2.54})$$

where $E = \Delta_{\mathcal{O}} + n$ runs over all primaries and descendants in the $\phi \times \phi$ OPE, the angular dependence is absorbed into the coefficients, and

$$e^{-\beta} = \frac{r_2}{r_3} \quad (\text{I.2.55})$$

identifies the expansion parameter as the *radial time* separating the two insertions: the OPE is a Laplace series in cylinder time, a thermal-looking sum over the spectrum. Convergence of the OPE

is convergence of (I.2.54) for all $\beta > 0$, i.e. whenever the radii are strictly ordered — whenever some sphere centered at the origin separates $\{0, x_2\}$ from $\{x_3, \infty\}$.

The proof needs no information about the spectrum or the size of the OPE coefficients — only unitarity, through the fact that (I.2.53) is an inner product of two Hilbert-space vectors:

$$\begin{aligned} |\Phi\rangle &= \phi(x_2)|\phi\rangle, \\ \langle\Psi| &= \langle\phi|\phi(x_3). \end{aligned} \tag{I.2.56}$$

Both have finite norm: $\|\Phi\|^2$ is the same four-point function evaluated at an inversion-symmetric configuration, with insertions at radii r_2 and $1/r_2$ — this is where reflection positivity, in its inversion form, enters the proof — and likewise $\|\Psi\|^2$. Now recall the elementary Hilbert-space fact: the scalar product of two finite-norm vectors, expanded in any orthogonal eigenbasis, converges *absolutely*. Concretely, the tail of the series (I.2.54) above energy E_* is controlled by Cauchy–Schwarz,

$$\left| \sum_{E>E_*} \langle\Psi|E\rangle\langle E|\Phi\rangle \right| \leq \left(\sum_{E>E_*} |\langle E|\Psi\rangle|^2 \right)^{1/2} \left(\sum_{E>E_*} |\langle E|\Phi\rangle|^2 \right)^{1/2} \xrightarrow{E_*\rightarrow\infty} 0, \tag{I.2.57}$$

because each factor is the tail of a convergent sum of positive terms (the norms). That is the whole proof. Its hypotheses are worth reading back: a Hilbert space built on spheres came from the reconstruction theorem, the energy basis labeled by local operators from the state–operator correspondence, and the finiteness of the norms from reflection positivity — each of the preceding sections supplied exactly one hypothesis.³⁰

How fast does it converge? Here the same physical reading — the OPE as a thermal sum — gives the rate almost for free, and the answer is the second half of the centerpiece. Define the weighted spectral density of the exchanged states, $f(E) = \sum_k \rho_k \delta(E - E_k)$ with all $\rho_k \geq 0$ at the reflection-symmetric configuration, so that the four-point function is its Laplace transform $L(\beta) = \int_0^\infty dE f(E) e^{-E\beta}$. Two observations pin down everything:

First: the $\beta \rightarrow 0$ behavior of L is known exactly. Colliding $x_2 \rightarrow x_3$ is governed by the leading term of the OPE in the *crossed* pairing (x_2, x_3) — the identity operator — which gives the free-field-like singularity $L(\beta) \sim (\text{const})\beta^{-2\Delta_\phi}$ as $\beta \rightarrow 0$. The high-energy tail of f is thus tied, through a Laplace transform, to a short-distance singularity fixed by dimensional analysis. This crossed-channel trick — the UV of one expansion is the leading term of another — is the embryonic form of the crossing equations of Part III, and it already computes things: by the Hardy–Littlewood Tauberian theorem, the *integrated* density must grow as a pure power,

$$\begin{aligned} F(E) \equiv \int_0^E dE' f(E') &\sim \frac{E^{2\Delta_\phi}}{\Gamma(2\Delta_\phi + 1)}, \\ \text{as } E \rightarrow \infty : \end{aligned} \tag{I.2.58}$$

no assumption on the spectrum — the growth of the OPE data is universal, fixed by the dimension of the colliding operators.

³⁰The argument is Pappadopulo–Rychkov–Espín–Rattazzi [1208.6449](#), §§2 and 4.1, following the radial-quantization logic of Polchinski’s book; the rigorous statement within the Wightman framework — including the recursive use of the OPE to all orders — is a 1976/77 theorem of Mack (Comm. Math. Phys. **53** (1977) 155), building on Lüscher. The reader allergic to “insert a complete set of states” may take Mack’s theorem as the licensed version of everything in this subsection.

Second: a power-law density makes Laplace tails exponentially small. With (I.2.58) in hand, the part of the four-point function carried by states above E_* is

$$\begin{aligned} L(\beta; E_*) &= \int_{E_*}^{\infty} dE f(E) e^{-E\beta} \\ &\lesssim \frac{E_*^{2\Delta_\phi}}{\Gamma(2\Delta_\phi + 1)} e^{-E_*\beta}, \\ &\text{for } E_* \gg \Delta_\phi/\beta, \end{aligned} \tag{I.2.59}$$

by integrating by parts against (I.2.58). Truncating the OPE at dimension E_* costs an error $e^{-E_*\beta}$ times a power: the expansion converges *exponentially* in the dimension of the discarded operators, at a rate β that is literally a geometric quantity — the logarithm of the ratio of radii, the cylinder-time gap between the colliding pair and the spectator pair. Numerical bootstrap computations depend on this estimate: it is what makes truncations of the OPE controlled approximations with quantifiable errors.³¹

Theorem I.2.3 (OPE convergence)

In a unitary CFT, the operator product expansion of $\mathcal{O}_i(x_1)\mathcal{O}_j(x_2)$, applied inside any correlation function, converges absolutely whenever some sphere encloses x_1 and x_2 while excluding all other insertion points. The error from truncating the expansion at dimension Δ_* falls off exponentially, $\sim \Delta_*^{\Delta_i+\Delta_j} e^{-\Delta_*\beta}$, where $e^{-\beta}$ measures the separation of scales (the ratio of inner to outer radii in the frame where the sphere is centered at the origin). [Mack 1977; Pappadopulo–Rychkov–Espin–Rattazzi 1208.6449.]

It remains to see the machinery actually produce numbers — and to see the convergence radius emerge by hand, which the following example does in the simplest possible arena.

Example I.2.4 (The two-point function, computed by radial quantization)

Forget for a moment that conformal kinematics fixes the scalar two-point function, and *compute* $\langle \mathcal{O}(y)\mathcal{O}(x) \rangle$, $|x| < |y|$, by the radial rules, as a test that the Hilbert-space picture reproduces correlators with the advertised convergence.

The ket is $\mathcal{O}(x)|0\rangle = e^{x\cdot P}\mathcal{O}(0)e^{-x\cdot P}|0\rangle = e^{x\cdot P}|\mathcal{O}\rangle$: a coherent superposition of descendants, with the level- N term carrying $|x|^N$. The bra is the inversion conjugate, $\langle 0|\mathcal{O}(y) = y^{-2\Delta}\langle \mathcal{O}|e^{\tilde{y}\cdot K}$ with $\tilde{y} = y^\mu/y^2$. Therefore

$$\langle \mathcal{O}(y)\mathcal{O}(x) \rangle = \frac{1}{y^{2\Delta}} \langle \mathcal{O}|e^{\tilde{y}\cdot K}e^{x\cdot P}|\mathcal{O}\rangle = \frac{1}{y^{2\Delta}} \sum_{N=0}^{\infty} \frac{1}{(N!)^2} \langle \mathcal{O}|(\tilde{y}\cdot K)^N (x\cdot P)^N |\mathcal{O}\rangle, \tag{I.2.60}$$

the cross terms vanishing because states of different radial energy are orthogonal. Each matrix

³¹Rates and prefactor: 1208.6449, §§4.2–4.4, whose elegant two-line alternative derivation we reproduce: for any $0 < \beta' < \beta$, $L(\beta; E_*) \leq e^{-E_*(\beta-\beta')} L(\beta') \leq \text{const}(\beta')^{-2\Delta_\phi} e^{-E_*(\beta-\beta')}$; optimizing $\beta' = 2\Delta_\phi/E_*$ gives (I.2.59) up to the same constant. The paper also extends convergence from the OPE sum over *states* to the sum over conformal *blocks* (multiplets repackaged into functions of cross-ratios), the form Part III uses.

element is pure algebra: commute the K 's rightward until they annihilate $|\mathcal{O}\rangle$. At level one,

$$\tilde{y}^\mu x^\nu \langle \mathcal{O} | K_\mu P_\nu | \mathcal{O} \rangle = \tilde{y}^\mu x^\nu \langle \mathcal{O} | (2\delta_{\mu\nu} D - 2M_{\mu\nu}) | \mathcal{O} \rangle = 2\Delta(\tilde{y} \cdot x) = 2\Delta t c, \quad (\text{I.2.61})$$

where $t \equiv |x|/|y|$ and $c \equiv \hat{x} \cdot \hat{y}$, and we used $D|\mathcal{O}\rangle = \Delta|\mathcal{O}\rangle$, $M_{\mu\nu}|\mathcal{O}\rangle = 0$ for a scalar. The level- N elements work the same way, each a polynomial in c times t^N .

Now compare with the exact answer, organized as a power series in t :

$$\frac{1}{|x-y|^{2\Delta}} = \frac{1}{y^{2\Delta}} \frac{1}{(1-2tc+t^2)^\Delta} = \frac{1}{y^{2\Delta}} \sum_{N=0}^{\infty} C_N^{(\Delta)}(c) t^N, \quad (\text{I.2.62})$$

the generating function of the Gegenbauer polynomials. The series data matches term by term — $C_0^{(\Delta)} = 1$ is the primary, $C_1^{(\Delta)}(c) = 2\Delta c$ is the level-one element just computed, and so on up the multiplet (Exercise I.2.14) — and the comparison makes three abstract claims of this section concrete:

Convergence radius, on the nose. The generating function (I.2.62) has its singularities at $t = e^{\pm i \arccos c}$, on the unit circle $|t| = 1$: the series converges for $|x| < |y|$, *exactly* the criterion “a sphere centered at the origin separates x (and the origin) from y (and infinity).” Push y inside the sphere through x and the expansion dies at the moment the geometry says it must, with the divergence rate as $t \rightarrow 1$ controlled — as in (I.2.59) — by how nearly the separating sphere is pinched.

Positivity, visibly. At the reflection-symmetric configuration $c = 1$ (operators on the same ray) every coefficient $C_N^{(\Delta)}(1) = \binom{N+2\Delta-1}{N} > 0$: the terms are norms of descendant projections, as unitarity demands — and indeed positivity of *these* coefficients for all spins is precisely what fails when Δ dips below the unitarity bound, reconnecting with (I.2.29).

Exponential rate, numerically. At, say, $t = 1/2$, $c = 1$, the tail beyond level N_* of $\sum_N \binom{N+2\Delta-1}{N} 2^{-N}$ falls like $N_*^{2\Delta-1} 2^{-N_*}$: truncating the “OPE” of this two-point function at level 10 for $\Delta = 1$ already reproduces the exact answer to better than a part in 10^2 , at level 30 to a part in 10^8 — $e^{-N_*\beta}$ with $\beta = \log 2$, the promised cylinder-time gap.

One practical refinement is worth recording here, because every modern bootstrap computation uses it and because it answers a natural objection. The convergence rate β depends on the choice of radial frame — where we put the origin and infinity — while the four-point function itself depends only on conformal invariants. So one should *optimize*: for fixed physical configuration, choose the frame that maximizes the gap β . Conformal invariance reduces four points to a single complex cross-ratio z (defined so that the operators sit at $0, z, 1, \infty$); the optimal frame instead places the colliding pair symmetrically about the origin, at $\pm\rho$, and the spectator pair at ± 1 , with

$$\rho = \frac{z}{(1 + \sqrt{1-z})^2}, \quad (\text{I.2.63})$$

$$e^{-\beta_{\text{opt}}} = |\rho|.$$

The expansion in ρ converges for $|\rho| < 1$ — which covers *every* configuration except $z \in [1, \infty)$, the locus where the spectator operator 1 sits between the colliding pair 0 and z and no separating sphere exists at all. The OPE thus converges not merely “at short distance” but on essentially the entire configuration space, with a rate one can read off a ruler: at the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$, where the numerical bootstrap pitches its tent, $\rho = 3 - 2\sqrt{2} \approx 0.172$, so each unit of

dimension buys a factor $e^{-\beta} \approx 0.17$ — truncating at $\Delta_* \approx 10$ is already a 10^{-7} -level approximation. The 3d Ising computation of chapter I.1’s crack 3 is downstream of exactly this estimate.³²

Exercise I.2.12 (easy). Derive the power in (I.2.50): act with D on both sides of $\mathcal{O}_i(x)\mathcal{O}_j(0)|0\rangle \supset |x|^{-k}(\mathcal{O}(0) + \dots)|0\rangle$, using $[D, \mathcal{O}(x)] = (\Delta + x \cdot \partial)\mathcal{O}(x)$ on the left and $D|\mathcal{O}\rangle = \Delta_{\mathcal{O}}|\mathcal{O}\rangle$ on the right, and match: $k = \Delta_i + \Delta_j - \Delta_{\mathcal{O}}$. Conclude that in the $\phi \times \phi$ OPE the unit operator reproduces the two-point function, and that “which operators have low dimension” is the same question as “which OPE terms dominate at short distance.”

Exercise I.2.13 (medium). Fix the first descendants. For $\Delta_i = \Delta_j = \delta$ and a scalar primary Φ of dimension Δ exchanged, write $C_{\Phi}(x, \partial_y) = |x|^{\Delta-2\delta} (1 + ax \cdot \partial_y + \alpha x^{\mu} x^{\nu} \partial_{\mu} \partial_{\nu} + \beta x^2 \partial^2 + \dots)$ and determine a, α, β by matching the exact three-point function $\langle \mathcal{O}_i(x) \mathcal{O}_j(0) \Phi(z) \rangle$ (fixed by conformal symmetry) against $\lambda_{ij\Phi} C_{\Phi}(x, \partial_y) \langle \Phi(y) \Phi(z) \rangle|_{y=0}$, expanding in x .

Answer: $a = \frac{1}{2}$, $\alpha = \frac{\Delta+2}{8(\Delta+1)}$, $\beta = -\frac{\Delta}{16(\Delta-\frac{d}{2}+1)(\Delta+1)}$ (Rychkov [1601.05000](#), §3.3). Note β ’s pole at $\Delta = \frac{d}{2} - 1$: the free-field dimension, where the multiplet shortens because $\partial^2 \Phi = 0$ — the same boundary case as in (I.2.28) and Exercise I.2.5.

Exercise I.2.14 (classic). (a) Verify $C_2^{(\Delta)}(c) = 2\Delta(\Delta+1)c^2 - \Delta$ from the generating function, then reproduce it from the level-two matrix element $\frac{1}{4} \langle \mathcal{O} | (\tilde{y} \cdot K)^2 (x \cdot P)^2 | \mathcal{O} \rangle$ using the algebra (you will need $[K_{\mu}, P_{\nu}]$ twice; watch the $M_{\mu\nu}$ term act nontrivially in the middle of the computation even though it kills the end states). (b) Show that the level- N term decomposes into spins $N, N-2, \dots$ — the traceless-symmetric content of N derivatives — and that at $c=1$ all spin contributions add with positive weights.

Hint: For (b), expand $C_N^{(\Delta)}(c)$ in Gegenbauer polynomials $C_j^{(\nu)}(c)$ with $\nu = \frac{d-2}{2}$: these are the sphere harmonics’ generating polynomials, and the expansion coefficients are the squared norms of the spin- j projections of $(x \cdot P)^N | \mathcal{O} \rangle$.

Exercise I.2.15 (medium). (Null operators.) Using the radial-quantization algebra, show that in the free massless scalar theory the state $P^2|\phi\rangle = \partial^2\phi(0)|0\rangle$ has zero norm, so that $\partial^2\phi = 0$ holds as an operator equation: the classical equation of motion survives in the quantum theory as a statement about the operator content. Then explain why at the interacting Wilson–Fisher fixed point, where $\Delta_{\phi} \neq \frac{d-2}{2}$, the same state has positive norm, so that $\partial^2\phi$ is a nonzero operator there — related by the dynamics to ϕ^3 , with the consequence that the interacting theory has *fewer* primaries than free-field counting suggests (the multiplets of ϕ and ϕ^3 recombine into one). Null states and multiplet shortening return as the engine of the unitarity bounds in Part III.1.

Hint: Evaluate $\|P^2|\phi\rangle\|^2 = \langle \phi | K^2 P^2 | \phi \rangle$ with the commutators of Section I.2.5, as in Example I.2.4; the result is proportional to $\Delta_{\phi} - \frac{d-2}{2}$ times a positive factor. Compare Exercise I.2.13, where the same factor appeared as a pole.

Exercise I.2.16 (easy). (Wick’s theorem is an OPE.) For the free massless scalar in d dimensions, show directly from Wick contractions that

$$\phi(x)\phi(0) = \frac{C_d}{x^{d-2}} \cdot \mathbf{1} + :\phi^2:(0) + x^{\mu}:\phi\partial_{\mu}\phi:(0) + \dots,$$

identify the primaries among the first few terms, and explain why the expansion of any free-field correlator in this form converges with exactly the radius of Theorem I.2.3.

Hint: In a free theory the “sum over the spectrum” is the binomial expansion of products of propagators $1/(x_i - x_j)^{d-2}$; its convergence condition is the geometric series’.

³²The ρ -coordinate analysis is [1208.6449](#), §5; the systematic ρ -expansion of conformal blocks built on it is Hogervorst–Rychkov, [1303.1111](#). Exercise I.2.26 runs the numbers on the 2d Ising correlator, where the answer is exactly known and the bound can be graded.

Exercise I.2.17 (thinker). Why exactly does all this fail in a massive theory? Take a free scalar field of mass m , or the 2d Ising model slightly off its critical temperature (where the gap is $m \sim 1/\xi$). Radial foliation, sphere Hilbert spaces, and the OS construction all survive. Identify every step of Sections I.2.5–I.2.5 that breaks: which operator generates radial evolution when D is not a symmetry? what replaces the discrete dilatation eigenbasis? why does the expansion of $\phi(x)\phi(0)|0\rangle$ in *any* basis no longer organize into finitely many local-operator multiplets with power-law coefficients? Conclude that Wilson’s OPE in a massive theory is an expansion in $|x|$ *asymptotic* at $|x| \ll 1/m$, with coefficient functions $C_k(x; m)$ that are themselves nontrivial (they resum the would-be descendants), and reconcile this with the RG: the massive theory’s short-distance limit *is* a CFT, whose convergent OPE the massive expansion borrows, order by order in $m|x|$.

Hint: The cleanest statement: the massive theory is a point on an RG trajectory leaving a fixed point (chapter I.1’s picture); its operators match onto fixed-point operators with corrections analytic in the relevant coupling. OPE convergence is a property of the fixed point, not of the flow.

Associativity

Because the OPE is convergent, it can be applied inside a correlator in more than one way, and the results must agree; this consistency condition is the only one the CFT data has to satisfy, and it is the equation on which Part III runs. Take the four-point function $\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4) \rangle$ and reduce it in two different ways: apply the OPE to the pair (12), which is legitimate whenever a sphere separates x_1, x_2 from x_3, x_4 , or to the pair (14), which is legitimate whenever a sphere separates x_1, x_4 from x_2, x_3 (Figure 3). A generic configuration admits both spheres at once, and there both expansions converge to the same correlator:

$$\sum_{\mathcal{O}} \lambda_{\phi\phi\mathcal{O}}^2 g_{\mathcal{O}}^{(12)(34)}(x_i) = \sum_{\mathcal{O}} \lambda_{\phi\phi\mathcal{O}}^2 g_{\mathcal{O}}^{(14)(23)}(x_i), \quad (\text{I.2.64})$$

where $g_{\mathcal{O}}$ — a **conformal block** — packages the contribution of the full multiplet of the primary $\mathcal{O}$ in the indicated pairing (a function of the two cross-ratios, computable from kinematics alone; chapter III.1 constructs them). Equation (I.2.64) is **crossing symmetry**, or OPE **associativity**: the same words multiplied in a different order.

Two facts give this condition its power. First, it is *the complete list* of consistency conditions: imposing associativity on all four-point functions guarantees that every higher-point reduction is pairing-independent as well, so CFT data $\{\Delta_i, \lambda_{ijk}\}$ satisfying (I.2.64), together with unitarity, defines a consistent set of local correlators outright.³³ Second, it is highly restrictive, though one needs the convergence theorem to see why. Each side of (I.2.64) is an exponentially convergent sum dominated by a small number of low-dimension operators; demanding that two such sums, organized around *different* corners of configuration space, agree everywhere is a strong condition on the few numbers that dominate both. Combined with unitarity ($\lambda^2 \geq 0$ — positivity of norms once again), the condition can be implemented on a computer as an exclusion problem. That is the conformal bootstrap, the subject of Part III, and the method that produced the Ising dimensions of chapter I.1’s crack 3. Establishing that the equation handed to Part III is exact, rather than asymptotic, was this chapter’s task.

³³Rychkov 1601.05000, §4.3, for the five-point argument that four-point associativity suffices; the careful statement is again Mack’s. Note the connection with Idea I.2.3: consistent correlators *are* a theory, so (I.2.64) plus unitarity is as close as the subject currently comes to a usable finite *definition* of “CFT” — up to the locality caveat that Example I.2.6 will make precise.

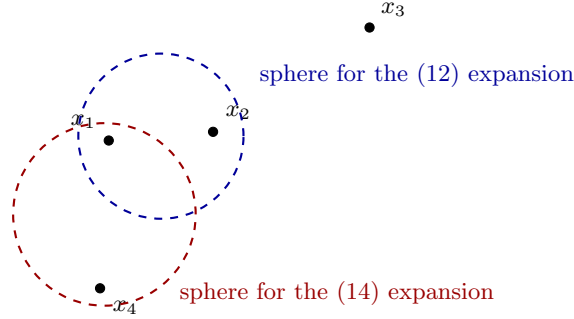


Figure 3: One four-point configuration, two ways to expand it. The blue sphere separates x_1, x_2 from x_3, x_4 , so the OPE may be applied to the pair (12); the red sphere separates x_1, x_4 from x_2, x_3 , so the OPE may equally be applied to (14). Both expansions converge, by Theorem I.2.3, and both compute the same correlator; their equality is the crossing equation (I.2.64).

I.2.6 The stress tensor

Stand back and audit what the chapter has assumed about its subject. Analytic continuation needed a positive Hamiltonian; reconstruction needed reflection positivity; the OPE needed conformal symmetry on top. Nowhere did we use — or even state — the property chapter I.1 declared the “one non-negotiable”: locality. And the omission is not cosmetic. Everything so far would be satisfied by systems nobody calls a field theory; the sharpest way to see this is that we will shortly construct, in Example I.2.6, a full set of conformal, reflection-positive, OPE-convergent correlators describing no local theory at all. What separates quantum field theory from “quantum mechanics with correlators” must therefore live somewhere specific in the operator content. Chapter I.1 told us where: in the existence of a local **stress tensor** — the operator that measures energy and momentum *density*, equivalently the response of the theory to local deformations of the geometry it lives on. This section constructs that operator, extracts the identities that make it useful, and then exhibits the counterexample that makes it necessary.

Definition and Ward identities

In any presentation with an action on a background metric, the definition is a functional derivative (we work in Euclidean signature; Lorentzian signs and the index dictionary are pinned in Appendix A):

$$T_{\mu\nu}(x) \equiv \frac{2}{\sqrt{g}} \frac{\delta S_E}{\delta g^{\mu\nu}(x)}, \quad (\text{I.2.65})$$

the source conjugate to the metric, exactly as a current is the source conjugate to a background gauge field.

This definition should be compared with the one most textbooks give first. In flat space, translation invariance of a Lagrangian $L(\phi, \partial\phi)$ yields, by Noether’s procedure, the **canonical** stress tensor

$$T_{\mu\nu}^{\text{can}} = \frac{\partial L}{\partial(\partial^\mu \phi)} \partial_\nu \phi - \delta_{\mu\nu} L, \quad (\text{I.2.66})$$

conserved on the equations of motion. For fields with spin the canonical tensor is generally not symmetric, and even for scalars it is generally not traceless; the metric definition (I.2.65), by contrast, is symmetric by construction. The two definitions agree up to identically conserved *improvement*

terms (for spinning fields this is the Belinfante construction), and in particular they generate the same charges P^μ ; we will meet the improvement freedom concretely in Example I.2.5 below. The metric definition is the one we take as primary, for a reason that goes beyond convenience: it characterizes $T_{\mu\nu}$ by what it *does* — it measures the response of the theory to a change of geometry — rather than by a formula in a particular presentation.

Both definitions, however, still mention an action. The structure that survives in any presentation is the set of **Ward identities** obeyed by correlators with a $T_{\mu\nu}$ insertion, and the modern attitude is to take that structure as the definition: a local QFT is a theory possessing a symmetric tensor operator $T_{\mu\nu}$, of dimension exactly d at a fixed point, satisfying the identities derived below. Before deriving them, we should say why these identities are the right objects to want. They are the form that symmetry takes at the level of correlation functions — the only level shared by all presentations — and they accomplish three things at once: they make “this theory has energy-momentum density” a sharp, checkable property of the correlators (a property the generalized free field below will fail); they constrain correlators quantitatively (fixing, for example, $\Delta_T = d$, and relating coefficients that would otherwise be independent); and they construct the charges that implement translations, rotations, and conformal transformations on the Hilbert space — including, as we will verify, the generators that radial quantization borrowed in Section I.2.5.

To derive them, insert operators $\mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n)$ into the flat-space path integral and perform an infinitesimal diffeomorphism $x \rightarrow x' = x + \epsilon(x)$, with ϵ vanishing at infinity, treated as a mere change of integration variables — which changes the value of the integral not at all. Track where the relabeling acts. The measure is invariant (we assume; in chiral theories this very step can fail, which is one address of the anomalies of Part II). The action is *not* invariant — flat space has no reason to be diffeomorphism-invariant by itself — but its variation is, by the definition (I.2.65), the stress tensor contracted with the induced metric deformation $\delta g_{\mu\nu} = \partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu$:

$$\delta S_E = \frac{1}{2} \int d^d x T^{\mu\nu} \delta g_{\mu\nu} = \int d^d x T^{\mu\nu} \partial_\mu \epsilon_\nu = - \int d^d x \epsilon_\nu \partial_\mu T^{\mu\nu}, \quad (\text{I.2.67})$$

using the symmetry of T and integrating by parts. And the insertions transport: a scalar shifts by $\delta_\epsilon \mathcal{O}(x_i) = \epsilon^\nu(x_i) \partial_\nu \mathcal{O}(x_i)$. Since the relabeled integral equals the original,

$$0 = - \langle \delta S_E \mathcal{O}_1 \cdots \mathcal{O}_n \rangle + \sum_i \langle \mathcal{O}_1 \cdots \delta_\epsilon \mathcal{O}_i \cdots \mathcal{O}_n \rangle, \quad (\text{I.2.68})$$

and stripping the arbitrary function $\epsilon^\nu(x)$ gives the local identity

$$\partial_\mu \langle T^{\mu\nu}(x) \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle = - \sum_{i=1}^n \delta^d(x - x_i) \frac{\partial}{\partial x_i^\nu} \langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle. \quad (\text{I.2.69})$$

(Operators with spin contribute an extra rotation piece, $\propto \partial_\mu [\delta^d(x - x_i) \Sigma^{\mu\nu}]$, which the reader may restore when needed.) The identity (I.2.69) contains two distinct statements, which we separate.

Away from insertions, $T^{\mu\nu}$ is conserved as an operator equation. The right-hand side is a sum of contact terms, so at separated points $\partial_\mu T^{\mu\nu} = 0$ inside any correlator. This is Noether’s theorem in correlator form, with no Lagrangian on stage.

At insertions, the contact terms make charges act. Recall the relevant geometric vocabulary. A **Killing vector** of flat space is a vector field ξ generating an isometry — a map that preserves the metric — which infinitesimally means $\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 0$; the solutions are the constant vectors

(translations) and $\xi_\mu = \omega_{\mu\nu}x^\nu$ with ω antisymmetric (rotations). A **conformal Killing vector** preserves the metric up to a local rescaling,

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \frac{2}{d}(\partial \cdot \xi)\delta_{\mu\nu}, \quad (\text{I.2.70})$$

which admits in addition the dilatation $\xi^\mu = x^\mu$ and the special conformal vector fields $\xi^\mu = 2(b \cdot x)x^\mu - x^2 b^\mu$. Now integrate (I.2.69) against such a vector field ξ^ν over a region R bounded by a closed surface Σ . When ξ is Killing — or conformal Killing, provided T is symmetric and traceless — the bulk term $\propto \int_R T^{\mu\nu} \partial_\mu \xi_\nu$ vanishes, the integral collapses to the boundary, and one finds

$$\oint_\Sigma dS_\mu \xi_\nu \langle T^{\mu\nu}(x) \mathcal{O}_1(x_1) \cdots \rangle = - \sum_{x_i \in R} \langle \cdots \delta_\xi \mathcal{O}_i(x_i) \cdots \rangle, \quad (\text{I.2.71})$$

where $\delta_\xi \mathcal{O}$ is the operator's transformation. The object

$$Q_\xi[\Sigma] = \oint_\Sigma dS_\mu \xi_\nu T^{\mu\nu} \quad (\text{I.2.72})$$

is the conserved charge, and (I.2.71) says something stronger than conservation: the surface Σ can be *deformed at will* without changing any correlator, as long as no insertion point is crossed — the difference of two surface integrals is a bulk integral of (I.2.69)'s left side, which vanishes away from insertions. The charges of a local theory are **topological surface operators**: their value depends only on which operators they enclose, not on the surface's shape, size, or location. Crossing an insertion changes the correlator by exactly the enclosed operator's transformation. These are the first topological objects of these notes, and they are the thin end of the wedge that Part II drives through the whole subject — there, the logic is *reversed*, and a symmetry is *defined* as a topological operator, with (I.2.72) demoted to the special case in which the topological operator happens to be built from a local current.

In particular, the conformal generators that powered radial quantization are charges of exactly this kind — the loan taken out in Section I.2.5 can now be repaid. For each conformal Killing vector of flat space, evaluate (I.2.72) on a sphere:

$$\begin{aligned} P_\mu &= \oint dS_\rho T^\rho{}_\mu, & M_{\mu\nu} &= \oint dS_\rho (x_\mu T^\rho{}_\nu - x_\nu T^\rho{}_\mu), \\ D &= \oint dS_\rho x^\nu T^\rho{}_\nu, & K_\mu &= \oint dS_\rho (2x_\mu x^\nu - \delta_\mu{}^\nu x^2) T^\rho{}_\nu, \end{aligned} \quad (\text{I.2.73})$$

the last two conserved surface-independently only because T is traceless ($\xi_\nu T^{\mu\nu}$ has divergence $\frac{1}{d}(\partial \cdot \xi)T^\mu{}_\mu$ for a conformal Killing vector ξ). Shrinking the surface onto an insertion converts (I.2.71) into the commutator statement: in radial ordering, Q_ξ just inside minus Q_ξ just outside an insertion is

$$[Q_\xi, \mathcal{O}(x)] = \delta_\xi \mathcal{O}(x), \quad (\text{I.2.74})$$

which for the choices (I.2.73) reproduces exactly the algebra postulated in Section I.2.5: $[D, \mathcal{O}(0)] = \Delta \mathcal{O}(0)$, $[P_\mu, \mathcal{O}(x)] = \partial_\mu \mathcal{O}(x)$, and the rest. The bookkeeping device of radial quantization is thus generated, operator by operator, by the stress tensor — which is the precise sense in which conformal symmetry is a property *of a local theory* rather than of a coordinate change: no traceless conserved T , no charges (I.2.73), no radial Hamiltonian, no state-operator correspondence. (The generalized free field below is about to make this threat concrete: its conformal invariance is real but *global only* — inherited from correlator kinematics, with no local generator implementing it.)

Example I.2.5 (The stress tensor of the free scalar, canonical and improved)

For the free massless scalar, the definition (I.2.65) applied to $S_E = \frac{1}{2} \int \sqrt{g} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$ gives, in flat space,

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \delta_{\mu\nu} (\partial\phi)^2, \quad (\text{I.2.75})$$

which for this Lagrangian coincides with the canonical tensor (I.2.66): symmetric, conserved on-shell ($\partial^\mu T_{\mu\nu} = (\partial_\nu \phi) \partial^2 \phi$), and — a problem — not traceless:

$$T^\mu{}_\mu = \left(1 - \frac{d}{2}\right) (\partial\phi)^2 \neq 0 \text{ for } d > 2. \quad (\text{I.2.76})$$

A nonzero trace would, by the argument below, obstruct using (I.2.72) to build the dilatation and special conformal charges — yet the free massless scalar *is* conformal. The resolution is that (I.2.65) has a hidden ambiguity: actions that agree in flat space can disagree about the metric response. Add the non-minimal coupling $\frac{\xi}{2} \int \sqrt{g} \mathcal{R} \phi^2$ ($\mathcal{R}$ the Ricci scalar): it vanishes in flat space, changes no flat-space correlator, but shifts the stress tensor by an identically conserved **improvement term**,

$$T_{\mu\nu} \longrightarrow T_{\mu\nu} + \xi (\delta_{\mu\nu} \partial^2 - \partial_\mu \partial_\nu) \phi^2. \quad (\text{I.2.77})$$

The trace of the added piece is $\xi(d-1)\partial^2 \phi^2 = 2\xi(d-1) [(\partial\phi)^2 + \phi \partial^2 \phi]$, so on-shell

$$T^\mu{}_\mu|_{\text{improved}} = \left[\left(1 - \frac{d}{2}\right) + 2\xi(d-1) \right] (\partial\phi)^2 = 0 \text{ for } \xi_c = \frac{d-2}{4(d-1)}, \quad (\text{I.2.78})$$

the **conformal coupling** ($\xi_c = \frac{1}{6}$ in $d = 4$; $\xi_c = 0$ in $d = 2$, where (I.2.75) was traceless on-shell all along — another early hint that $d = 2$ is special). Three general lessons follow. (i) “The” stress tensor is an equivalence class: improvements (I.2.77) change the operator without changing the charges P^μ or the physics, and statements about T should be improvement-aware. (ii) Conformal invariance is the statement that a *traceless* representative exists; with it, $\xi_\nu T^{\mu\nu}$ is conserved for every conformal Killing vector ξ (the trace multiplies $\partial \cdot \xi$), and all the charges of radial quantization flow from (I.2.72). (iii) The improved T is the fixed-point operator: it is the spin-2 primary of dimension d in the free CFT’s spectrum, the operator that the OPE machinery sees (cf. Exercise I.2.16: the level-(0,2) composite in $\phi \times \phi$ is exactly (I.2.77)-improved).

The trace earns its own paragraph, because it is the operator that knows about scale. Repeating the derivation of (I.2.69) for a *Weyl* variation $\delta g_{\mu\nu} = 2\omega(x)g_{\mu\nu}$ instead of a diffeomorphism shows that $T^\mu{}_\mu(x)$ measures the response to local rescalings; in a flowing theory the schematic operator equation

$$T^\mu{}_\mu = \sum_i \beta^i(g) \mathcal{O}_i \quad (\text{I.2.79})$$

identifies the trace with the beta functions — the precise statement, with its subtleties (operator mixing, “virial current” ambiguities), is part of chapter I.6’s careful construction of the RG, and the question of when $T^\mu{}_\mu = 0$ exactly (scale versus conformal invariance) was flagged already in chapter I.1’s theory-space section. Here we need only the fixed-point moral: at a fixed point a traceless stress tensor exists in all known unitary local examples, and these notes assume it wherever “CFT” is written.

Exercise I.2.18 (classic). Test the Ward identity (I.2.69) on the free scalar. Using Wick’s theorem with (I.2.75) improved at ξ_c , compute $\langle T_{\mu\nu}(x)\phi(x_1)\phi(x_2) \rangle$ in $d = 4$ and verify: (a) it is conserved away from x_1, x_2 ; (b) integrating the $\nu = 0$ component of (I.2.69) over a ball around x_1 reproduces ∂_{x_1} acting on $\langle \phi\phi \rangle$; (c) the same correlator with the *unimproved* T fails tracelessness by contact terms plus the on-shell piece, and identify how the dilatation Ward identity then misreports Δ_ϕ .

Hint: Work in momentum space if the contact terms get confusing; or use the position-space identity $\partial^2(1/x^2) = -4\pi^2\delta^4(x)$ repeatedly.

Exercise I.2.19 (medium). The two-point function of stress tensors in a CFT takes the form $\langle T_{\mu\nu}(x)T_{\rho\sigma}(0) \rangle = C_T \frac{\mathcal{I}_{\mu\nu,\rho\sigma}(x)}{x^{2d}}$ with $\mathcal{I}$ a fixed tensor structure built from inversions, and $C_T > 0$ in a unitary theory — by now you can see why (it is a norm in radial quantization). Compute C_T for the free conformal scalar in $d = 4$ by Wick contraction of two improved stress tensors, in any normalization you state clearly, and check positivity. (C_T will return as a yardstick: it counts degrees of freedom well enough to order theories, and appears in the monotonicity discussion of Part IV.)

Hint: Repeated Wick contraction; the real content of the exercise is the check that the ξ_c -dependent pieces assemble into the conserved, traceless structure $\mathcal{I}$ — the unimproved tensor’s correlator is not even traceless. Compare your value against the general- d free-scalar C_T tabulated in Osborn–Petkou, [hep-th/9307010](#), §3, translating normalizations carefully (everyone’s C_T differs by factors of the sphere volume).

A consistent set of correlators with no theory underneath

We now construct the promised counterexample: a complete and consistent set of conformal correlators with no stress tensor, and hence no local theory underneath. The construction is very simple, and the simplicity is itself part of the lesson — nothing exotic is required to produce “correlators without a theory.”

Example I.2.6 (The generalized free field)

Pick a number $\Delta \geq \frac{d-2}{2}$ and *declare* a scalar operator $\mathcal{O}$ with two-point function

$$\langle \mathcal{O}(x)\mathcal{O}(0) \rangle = \frac{1}{x^{2\Delta}}, \quad (\text{I.2.80})$$

and all higher correlators given by Wick’s theorem — sums over pairings of products of two-point functions, exactly as for a Gaussian field:

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4) \rangle = \frac{1}{x_{12}^{2\Delta}x_{34}^{2\Delta}} + \frac{1}{x_{13}^{2\Delta}x_{24}^{2\Delta}} + \frac{1}{x_{14}^{2\Delta}x_{23}^{2\Delta}}, \text{ etc.} \quad (\text{I.2.81})$$

This is the **generalized free field** (GFF; “mean field theory” in the bootstrap literature). We now check it against every axiom introduced in this chapter, and every axiom will pass:

Euclidean covariance, permutation symmetry, clustering: immediate from the form of (I.2.81). In fact the correlators are covariant under the full *conformal* group, since the two-point function is and Wick products inherit it.

Reflection positivity: a Gaussian theory is reflection positive iff its covariance is, and Section I.2.2 already did the computation: the Källén–Lehmann density of $1/x^{2\Delta}$ is the positive function (I.2.28) precisely when $\Delta \geq \frac{d-2}{2}$. So the OS reconstruction of Section I.2.3 runs

without complaint and produces an honest Hilbert space — a Fock space generated by $\mathcal{O}$, with a positive Hamiltonian, a unitary conformal group, the lot.

OPE and its convergence: by Wick algebra,

$$\mathcal{O}(x)\mathcal{O}(0) = \frac{1}{x^{2\Delta}} \cdot \mathbf{1} + :\mathcal{O}^2:(0) + \cdots, \quad (\text{I.2.82})$$

and the full single- $\mathcal{O}$ -pair spectrum consists of the **double-twist** composites

$$\begin{aligned} [\mathcal{O}\mathcal{O}]_{n,\ell} &\sim :\mathcal{O}\partial^{2n}\partial_{\mu_1}\cdots\partial_{\mu_\ell}\mathcal{O}:, \\ \Delta_{n,\ell} &= 2\Delta + 2n + \ell, \\ &\text{with } \ell \text{ even,} \end{aligned} \quad (\text{I.2.83})$$

an integer-spaced tower starting at 2Δ . The four-point function (I.2.81) decomposes into these multiplets with computable, positive squared OPE coefficients (Exercise I.2.20), the expansion converging exactly as Theorem I.2.3 dictates. Crossing symmetry — the bootstrap’s one dynamical equation — holds by inspection: (I.2.81) is manifestly symmetric under permuting the points.

And yet there is no stress tensor. The complete list of primaries is $\mathbf{1}$, $\mathcal{O}$, and the towers $[\mathcal{O}\cdots\mathcal{O}]_{n,\ell}$ built on (I.2.83) and its multi- $\mathcal{O}$ generalizations. A stress tensor must be a spin-2 primary of dimension exactly d . The spin-2 candidates available have dimensions $2\Delta + 2n + 2$ and up: for generic Δ , *no operator of dimension d exists at all*, and the lone exception

$$2\Delta + 2 = d \iff \Delta = \frac{d-2}{2} \quad (\text{I.2.84})$$

is the free scalar, where $[\mathcal{O}\mathcal{O}]_{0,2}$ is precisely the improved tensor of Example I.2.5. The same counting kills conserved currents: a spin-1 double-twist would need $2\Delta + 1 = d - 1$, free field again. (For a single real $\mathcal{O}$ the odd-spin double-twists are absent as primaries — $[\mathcal{O}\mathcal{O}]_{0,1} \sim :\mathcal{O}\partial\mathcal{O}: = \frac{1}{2}\partial:\mathcal{O}^2:$ is a descendant — so strictly this is the *complex* GFF’s verdict, where the $\mathcal{O}^\dagger\mathcal{O}$ tower supplies the spin-1 candidate; either way no current of dimension $d - 1$ survives off the free point.) The moment Δ moves off the free value, the theory keeps every correlator-level virtue — unitarity, conformal symmetry, convergent operator algebra, crossing — and loses energy density, momentum density, and local charge density. There is no Hamiltonian *density*; only the global charges survive (they act fine on Fock space). By the standard of Idea I.1.2, this is not a quantum field theory. It is the quantum mechanics of a conformally organized continuum of oscillators.

Two comments on this example, one structural and one physical. Structurally, notice *where* the failure lives: not in any correlator one could falsify locally, but in the *absence* of an operator — a hole in the spectrum at spin 2, dimension d . No finite list of correlator measurements detects it; one must know the whole operator content to know the theory is nonlocal. This sharpens chapter I.1’s Idea I.1.1: “the theory is its observables” means the *complete* assignment, including which operators exist, not any finite sample of correlators — and it forewarns of chapter I.3, where theories agreeing on *all* local correlators will differ in their extended operators. It also matters practically: the numerical bootstrap must *input* the existence of a stress tensor (or not) as a hypothesis, exactly because crossing alone is satisfied by GFF-like solutions — the bootstrap’s exclusion plots are maps of where local theories can live inside the larger continent of crossing-symmetric data.

Physically, the GFF is anything but pathological — it is ubiquitous, appearing whenever a

local theory is viewed through a nonlocal window. It is the boundary description of a *free field in $(d+1)$ -dimensional anti-de Sitter space* (the extra dimension is where the missing locality went — and where the stress tensor lives: AdS gravity off), the kinematic skeleton on which holography is built in Part XI. It is the strict large- N limit of correlators of single-trace operators (factorization = Wick’s theorem, cf. Part XI.1). And it is the actual critical theory of the long-range Ising model in part of its phase diagram — the system chapter I.1 used to illustrate locality’s failure modes, now revealed as a GFF in the flesh.³⁴

Note I.2.2. “Generalized *free field*” refers to Wick factorization, not to a Lagrangian: away from the free value of Δ no local action in d dimensions produces these correlators — the would-be kinetic operator is a fractional power of the Laplacian, meaningful only as an integral kernel. The GFF is the cleanest standing illustration that the space of consistent “correlator systems” is strictly larger than the space of local QFTs, with the existence of a stress tensor as the criterion that separates the two.

Exercise I.2.20 (classic). Decompose the GFF four-point function. With u, v the standard cross-ratios, show from (I.2.81) that the (12)(34)-channel expansion contains the identity plus double-twist towers, and compute the squared OPE coefficient of the leading scalar $[\mathcal{OO}]_{0,0} = \frac{1}{\sqrt{2}} \mathcal{O}^2$: (answer: $\lambda^2 = 2$ in unit-normalized conventions). Verify positivity of the first few coefficients, and check that the spin-2, $n = 0$ coefficient stays finite and nonzero as $\Delta \rightarrow \frac{d-2}{2}$ — the stress tensor re-entering the spectrum continuously.

Hint: Expand $u^\Delta(1+v^{-\Delta})$ -type terms in the small- u , small- $1-v$ limit and match powers against the leading conformal block behavior $u^{\Delta_{n,\ell}/2}$; the systematic closed form for all n, ℓ is in the literature (Fitzpatrick–Kaplan–Poland–Simmons-Duffin), but the leading entries are a half-page of algebra.

Exercise I.2.21 (thinker). Could any *local* lattice model — finite-range couplings, finite-dimensional on-site spaces — flow to the GFF with $\Delta \neq \frac{d-2}{2}$ at criticality? Marshal the evidence from this section (existence of T on the lattice side? what happens to it in the limit?) and from chapter I.1’s locality discussion. Then reconcile with the fact that the *long-range* Ising model does reach it. What exactly about $J_{ij} \sim |i-j|^{-d-\sigma}$ evades your argument?

Hint: A local lattice model has a (lattice) energy density whose continuum limit, if the limit is a CFT, supplies a stress tensor candidate; the long-range model’s energy density is not local. This is heuristic — making it a theorem is open territory; see the discussion of stress-tensor existence in Rychkov [1601.05000](#), §1.3.2.

1.2.7 Conserved currents and their products

The stress tensor is the local operator that every local theory must have. A theory with global symmetry has, in the same spirit, one more universal piece of operator content per continuous symmetry: a conserved **current** J^μ , the source conjugate to a background gauge field as $T_{\mu\nu}$ is conjugate to the metric. The change-of-variables argument that produced (I.2.69) transcribes line for line — with an x -dependent symmetry rotation in place of the diffeomorphism. Let the symmetry act on charged operators as $\mathcal{O}_k \rightarrow e^{iq_k \alpha} \mathcal{O}_k$ with constant α ; promote $\alpha \rightarrow \alpha(x)$ and relabel the path integral. The action is not invariant under the local version, and its leading variation *defines*

³⁴For GFF as AdS boundary data see Penedones’s TASI lectures [1608.04948](#); for the long-range Ising application, Rychkov [1601.05000](#), §1.3.2. The double-twist spectrum (I.2.83) is the starting point of the large-spin analytic bootstrap (Part III.3), which quantifies the statement that *every* CFT approaches GFF behavior at large spin.

the current,

$$\delta S_E = -i \int d^d x J^\mu(x) \partial_\mu \alpha(x), \quad (\text{I.2.85})$$

while each insertion rotates by its charge, $\delta \mathcal{O}_k = i q_k \alpha(x_k) \mathcal{O}_k$. Equating the two and stripping $\alpha(x)$ yields the current's Ward identity,

$$\partial_\mu \langle J^\mu(x) \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle = \sum_k q_k \delta^d(x - x_k) \langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle, \quad (\text{I.2.86})$$

with the same two morals as before: conservation as an operator equation at separated points, and charges

$$Q[\Sigma] = \oint_\Sigma dS_\mu J^\mu \quad (\text{I.2.87})$$

that are *topological surface operators* — deformable freely until they cross a charged insertion, at which point the correlator jumps by q_k . Conservation also rigidly fixes kinematics at a fixed point: a conserved vector primary must have dimension exactly $\Delta_J = d - 1$ (and conversely, a vector primary at the unitarity-bound dimension $d - 1$ is automatically conserved — Exercise I.2.22), with two-point function

$$\begin{aligned} \langle J_\mu(x) J_\nu(0) \rangle &= C_J \frac{I_{\mu\nu}(x)}{x^{2(d-1)}}, \\ I_{\mu\nu}(x) &= \delta_{\mu\nu} - \frac{2x_\mu x_\nu}{x^2}, \\ C_J &> 0. \end{aligned} \quad (\text{I.2.88})$$

Up to this point the discussion parallels that of the stress tensor, with one index fewer. The reason conserved currents nevertheless merit a separate section — and the reason the chapter ends on them — is that the *products* of currents carry structure that no single current sees, and that structure turns out to be among the most rigid and RG-stable data in quantum field theory: a single current is fixed by kinematics, but the algebra of currents is dynamics. Three properties of current products will recur throughout these notes.

The JJ OPE knows about locality. In a local theory the product $J_\mu(x) J_\nu(0)$ expands, by Theorem I.2.3, in the theory's operator content — and the stress tensor, when it exists, appears in this OPE with a coefficient fixed by the Ward identity (I.2.69) in terms of C_J : energy density couples universally to everything charged. (In the GFF-with-current variants, the term is simply absent — the hole in the spectrum again.)

The JJJ correlator knows about anomalies. In even dimensions, the three-point function of currents contains, alongside its conformally trivial parts, a parity-odd contact structure whose coefficient is the **'t Hooft anomaly** of the symmetry — the famous triangle diagram of $d = 4$ textbooks, reborn as a statement about an exactly computable, position-space correlator of the abstract theory. We say no more here, because Part II.3 builds anomaly technology properly; but the address is worth recording now: *anomalies live in current correlators*, which is why they survive every RG flow that preserves the symmetry — correlators of conserved currents at separated points cannot jump along a continuous flow.

And already the JJ commutator hides a quantized term that no classical reasoning produces. This is the Schwinger term, the chapter's closing calculation, and a fitting one: it uses the analytic structure of Section I.2.2, the positivity of Section I.2.3, and the operator-content reasoning of this section, all at once — and it is the first appearance in these notes of a phenomenon (a central, quantized obstruction in a symmetry algebra) that Part II will industrialize.

Example I.2.7 (The Schwinger term)

The claim is the following: in any unitary theory with a nontrivial $U(1)$ symmetry, the equal-time commutator of charge density with current density *cannot vanish*,

$$[J^0(\mathbf{x}), J^i(\mathbf{y})]_{\text{equal time}} \neq 0, \quad (\text{I.2.89})$$

even though the naive canonical computation, in essentially every presentation, gives zero. Worse for the naive computation: in $d = 2$ the nonvanishing piece is a c-number with a *quantized* coefficient.

The no-go (Schwinger's positivity argument, any d). Suppose the commutator vanished. Smear the charge density against a real profile f , $J^0(f) = \int d^{d-1}x f(\mathbf{x}) J^0(0, \mathbf{x})$, and consider

$$\mathcal{C}_f \equiv \langle 0 | [J^0(f), [H, J^0(f)]] | 0 \rangle = 2 \sum_n E_n |\langle n | J^0(f) | 0 \rangle|^2 \geq 0, \quad (\text{I.2.90})$$

by inserting a complete set of states — positivity of the energy and of norms, doing the work one more time. But current conservation trades the inner commutator for the spatial current, $[H, J^0] = -i\partial_t J^0 = +i\partial_i J^i$, so $\mathcal{C}_f$ is built entirely from the equal-time $[J^0, J^i]$ — which we assumed to vanish. Then *every* state created from the vacuum by the charge density has zero amplitude, $J^0(f)|0\rangle = 0$ for all f ; and a local operator that annihilates the vacuum annihilates everything (vacuum cyclicity — Exercise I.2.24), so $J^\mu \equiv 0$. A nontrivial symmetry therefore *requires* a nonzero equal-time $[J^0, J^i]$: a “Schwinger term,” invisible to canonical manipulations, mandated by unitarity.^a

The exact computation ($d = 2$). Now compute the term, in the cleanest habitat: a 2d CFT with a $U(1)$ current — the compact boson, or the massless Dirac fermion at level $k = 1$. Conservation plus tracelessness split the current into chiral halves; take the right-mover $j(x^-)$, $x^\pm = x \pm t$, with Wightman two-point function normalized by the level,

$$\langle 0 | j(x) j(0) | 0 \rangle = \frac{k}{4\pi^2} \frac{1}{(x^- - i\epsilon)^2}, \quad (\text{I.2.91})$$

the $i\epsilon$ encoding, as always, the boundary value from the positive-energy domain. The equal-time commutator is the jump between the two boundary values — the tool of Example I.2.1, reused verbatim. With the Sokhotski formula $\frac{1}{(x \mp i\epsilon)^2} = -\frac{d}{dx} \left(\mathcal{P} \frac{1}{x} \pm i\pi \delta(x) \right)$,

$$\langle 0 | [j(x), j(0)] | 0 \rangle_{t=0} = \frac{k}{4\pi^2} \left[\frac{1}{(x - i\epsilon)^2} - \frac{1}{(x + i\epsilon)^2} \right] = -\frac{ik}{2\pi} \delta'(x). \quad (\text{I.2.92})$$

For an abelian current algebra the commutator is a pure c-number, so (I.2.92) is the full operator statement: $[j(x), j(y)] = -\frac{ik}{2\pi} \delta'(x - y)$. Assembling $J^0 = j_+ + j_-$, $J^1 = j_+ - j_-$ from the two chiralities (the left-mover's central term carries the opposite sign) gives the advertised $[J^0(x), J^1(y)] = -\frac{ik}{\pi} \delta'(x - y)$, while the charge-density commutator $[J^0, J^0]$ cancels between chiralities, as consistency demands. Smear the chiral current into charges, $Q_f \equiv \int dx f(x) j(x)$, and the result reads as algebra:

$$[Q_f, Q_g] = \frac{ik}{2\pi} \int dx f'(x) g(x), \quad (\text{I.2.93})$$

antisymmetric under $f \leftrightarrow g$ (integrate by parts) and immune to any redefinition of the charges by c-numbers: the algebra of chiral $U(1)$ transformations is realized on the Hilbert space only up to a **central extension**, with central charge k . The constant charge $f = 1$ commutes with everything — the global symmetry is intact — but the *local* structure of the symmetry is deformed, and irreparably so. The particle on a circle at $\theta = \pi$ (chapter I.1) showed a symmetry realized projectively on a two-state system; here is the field-theory version, a projective phase per mode, quantified by one integer.

The role of the spacetime dimension: a sum rule. The spectral representation of Section I.2.2, run for the conserved current, makes the dimension-dependence of all this quantitative. Conservation forces the current’s two-point function into a transverse form with a single spectral density $\rho_J(s) \geq 0$, and retracing the boundary-value computation that produced (I.2.92) — now mass level by mass level — one finds that the Schwinger term is the *total spectral weight* (Schwinger’s original form of the result):

$$\begin{aligned} [J^0(\mathbf{x}), J^i(\mathbf{y})]_{\text{equal time}} &= -ic\partial_i\delta^{d-1}(\mathbf{x}-\mathbf{y}), \\ c &\propto \int_0^\infty ds \rho_J(s) > 0. \end{aligned} \tag{I.2.94}$$

Positivity of ρ_J re-derives the no-go at a glance — a sum of positive terms cannot vanish unless every term does, i.e. unless the current creates nothing from the vacuum. And the integral diagnoses the dimensions: dimensional analysis gives the coefficient c mass dimension $d - 2$, and indeed in a CFT $\rho_J(s) \sim C_J s^{(d-4)/2}$, whose integral diverges at large s for every $d > 2$ — there the Schwinger term is cutoff-dominated, the famously “regularization-dependent” object of the 1960s literature, physical in origin but not a number you can quote. In $d = 2$, and only there, the density of a conserved current is an integrable $\rho_J(s) = k\delta(s)$ and the sum rule converges: the Schwinger term is a *pure number*, k — universal, quantized (for a compact symmetry — Part III.4 derives the integrality), and the first of the exactly known coefficients that these notes set out to collect. The reader has, in fact, just computed three things at once: the central extension of the $U(1)$ current algebra, which is the Kac–Moody level on which two-dimensional CFT is built (Part III.4); the ’t Hooft anomaly of the $U(1)$ symmetry in $d = 2$ — coupling j to a background gauge field, the central term in (I.2.92) is what obstructs gauge invariance, making this the two-dimensional sibling of the ABJ triangle; and the mechanism that generates the photon mass $e/\sqrt{\pi}$ in the Schwinger model, which Part V.4 solves using exactly this commutator. All three subjects are developments of the single computation above, which is why products of currents deserve the attention this section has given them.

^aJ. Schwinger, Phys. Rev. Lett. **3** (1959) 296; the term had been noticed by Goto and Imamura in 1955. The canonical computation misses it because the product of two currents at coincident points is singular — exactly the singularity the OPE organizes — and manipulating it as if fields were smooth drops the central piece.

Preview I.2.2 (From currents to Part II)

This section’s two findings — charges are topological surface operators; current products carry quantized, RG-inert coefficients — are the embryo of the symmetry revolution that occupies Part II. There, the topological surface operator becomes the *definition* of symmetry (no current needed: discrete symmetries get operators too, with (I.2.87) the special “Noether case”), and the

quantized coefficients in current correlators become 't Hooft anomalies, classified and matched along RG flows. When Part II.1 opens by “running Noether’s theorem backwards,” it is this section run backwards.

Exercise I.2.22 (medium). (a) Show that in a CFT, conservation $\partial^\mu J_\mu = 0$ forces $\Delta_J = d - 1$: apply the special-conformal commutator $[K_\nu, \cdot]$ to the conservation equation for a vector primary and find that consistency requires either $\Delta = d - 1$ or $\partial^\mu J_\mu \neq 0$. (b) Conversely, show the two-point function (I.2.88) with general power 2Δ in the denominator is conserved only at $\Delta = d - 1$. (c) Conclude the modern restatement of “the current does not renormalize”: conservation is not a property protected by perturbative miracles but a representation-theoretic fact about the fixed point — short multiplets stay short. (The same argument with one more index pins $\Delta_T = d$.)

Hint: For (b), brute force: $\partial^\mu [I_{\mu\nu}(x)x^{-2\Delta}] = 2(\Delta - d + 1)x_\nu x^{-2\Delta-2} \cdot (\dots)$.

Exercise I.2.23 (medium). (The sum rule, checked where it converges.) (a) In $d = 2$, show that conservation and Lorentz invariance force $\langle J_\mu(p)J_\nu(-p) \rangle = \left(\delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right) \Pi(p^2)$ with Π constant for a CFT current, and that the corresponding spectral density is $\rho_J(s) \propto k\delta(s)$: the conserved current of a 2d CFT creates exactly one massless mode. (b) Insert this density into the sum rule (I.2.94) and recover the coefficient of (I.2.92), fixing the proportionality constants in your conventions. (c) For the free Dirac fermion in $d = 4$, estimate $\rho_J(s)$ at large s from the free-field two-point function and verify that the sum rule diverges quadratically — reproducing, from pure spectral positivity, the old result that the $d = 4$ Schwinger term is cutoff-dominated.

Hint: For (a): a p -independent transverse structure has its only discontinuity at $p^2 = 0$. The massless pole is physical: it is the dual photon/compact-boson mode, and its inevitability in 2d is one route to bosonization (Part V.4).

Exercise I.2.24 (thinker). (Vacuum cyclicity, the step the no-go borrowed.) The Schwinger argument used: a local operator satisfying $A(f)|0\rangle = 0$ for all smearings f supported in some open region vanishes identically. Make this plausible with the chapter’s tools: if $A(f)|0\rangle = 0$, then $\langle 0|\mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n)A(y)|0\rangle = 0$ for y in the region; use the analyticity of Wightman functions in the times (Section I.2.2) to argue the correlator vanishes for *all* y , hence A has vanishing correlators with everything, hence $A = 0$ (as an operator on the vacuum sector). This is the germ of the Reeh–Schlieder theorem — “the vacuum is cyclic and separating for local algebras” — whose precise form, and whose startling implications for entanglement in QFT, open Part X.

Hint: A holomorphic function vanishing on an open subset of the boundary of its domain (here: an open set of real points) vanishes identically — Figure 2’s logic, run in reverse.

1.2.8 Where this leaves us

The title’s claim is earned. Local operators entered this chapter as the inventory’s oldest line item; they leave as primary data: the local operator content of a quantum field theory determines its Hilbert space, its Hamiltonian, its symmetries down to their finest quantized coefficients, and its response to the geometry it lives on — and in a conformal theory, where there is no S-matrix, correlators of local operators are not merely observables but the only observables there are. The three questions of the opening are answered, and it is worth restating by what mechanism, since each part of this claim was established in a different section.

A correlator is one analytic object viewed from several real slices: the statistical averages of a Euclidean presentation and the ordered amplitudes of quantum mechanics are connected by continuation through a domain that exists because energy is positive — and the domain of *backgrounds*

on which any of this makes sense is itself charted (Kontsevich–Segal), with Lorentzian spacetimes on its boundary. Correlators with reflection positivity *are* a theory: Hilbert space, Hamiltonian, and operators are constructed from them, by a procedure (Osterwalder–Schrader) whose sphere-adapted version (radial quantization) converts, in a CFT, the completeness of the state space into the convergent operator product expansion — collapsing the infinite tower of correlators onto the CFT data $\{\Delta_i, \lambda_{ijk}\}$ with exponentially controlled error. And the local operator content is not an amorphous list: it is required to contain the stress tensor and the symmetry currents — the operators through which geometry and symmetry act, whose existence is locality’s signature (the generalized free field showing what their absence costs), and whose products hide the quantized coefficients (Schwinger terms, anomalies) that survive everything.

What the local data does *not* do is exhaust the observables — the title says *Observables I* advisedly. The chapter leaves three loose threads, deliberately. The charges (I.2.72) and (I.2.87) are topological operators of dimension $d - 1$ — are there others, not generated by any current? (Part II’s affirmative answer reorganizes the subject.) Two theories can share all local correlators on $\mathbb{R}^d$ and differ — chapter I.1’s Exercise I.1.20 previewed it, and the missing data lives in operators supported on lines, surfaces, and boundaries: the extended operators of chapter I.3, next. And the Hilbert spaces this chapter built on planes and spheres were extracted from correlators — chapter I.4 turns the tables and takes the assignment *geometry* $\mapsto$ *states* as primary, with cutting and gluing as the organizing axiom and the state–operator correspondence as its first theorem. The inventory of chapter I.1 is being filled in, one column at a time.

Chapter exercises

Exercise I.2.25 (classic). (Thermal correlators and the KMS condition.) Compute the two-point function of the free massless scalar in $d = 4$ at temperature $T = 1/\beta$ — i.e. on the Euclidean cylinder $S^1_\beta \times \mathbb{R}^3$ — by summing images over the thermal circle. Show: (a) the Euclidean correlator is periodic in τ with period β ; (b) upon continuation to real time, periodicity becomes the KMS relation $\langle \mathcal{O}(t)\mathcal{O}(0) \rangle_\beta = \langle \mathcal{O}(0)\mathcal{O}(t - i\beta) \rangle_\beta$ — the thermal replacement for the spectral condition, with the strip $-\beta < \text{Im}t < 0$ replacing the half-plane of Figure 2. (Thermal states and KMS return in chapter I.4; the algebraic reformulation is a pillar of Part X.)

Hint: $\sum_{n \in \mathbb{Z}} \frac{1}{r^2 + (\tau + n\beta)^2}$ is summable in closed form: $\frac{\pi}{r\beta} \frac{\sinh(2\pi r/\beta)}{\cosh(2\pi r/\beta) - \cos(2\pi \tau/\beta)}$. Check the $\beta \rightarrow \infty$ limit.

Exercise I.2.26 (medium). (The OPE convergence rate is a spectral gap.) Consider the 2d Ising CFT — the critical point of the two-dimensional Ising model, already met in Section I.2.1 and Example I.2.2 — whose three lowest primaries are **1**, the spin field σ ($\Delta = 1/8$), and the energy field ϵ ($\Delta = 1$); take these data as given (Part III.4 derives them). Using Theorem I.2.3, estimate how many multiplets of the $\sigma \times \sigma$ OPE are needed to compute the four-point function $\langle \sigma\sigma\sigma\sigma \rangle$ to 1% accuracy at the fully symmetric configuration (cross-ratio $z = \bar{z} = 1/2$, where $e^{-\beta} = 3 - 2\sqrt{2} \approx 0.17$ in the optimal “ ρ -coordinate” frame). Compare with the exactly known answer (this correlator is solvable; Part III.4): how sharp is the bound?

Hint: Tail $\sim \Delta_*^{4\Delta_\sigma} e^{-\Delta_*\beta} / \Gamma(4\Delta_\sigma + 1)$ with $\beta = -\log(3 - 2\sqrt{2}) \approx 1.76$; very few multiplets are needed. The ρ -coordinate refinement is [1208.6449](#), §5.

Exercise I.2.27 (medium). Run the OS reconstruction explicitly for the generalized free field: show that the Hilbert space built by Steps 1–2 of Section I.2.3 from the Wick correlators is the Fock space over the single-particle space with inner product given by the two-point function, and identify the reconstructed Hamiltonian’s spectrum. Where, concretely, would you look in this construction for a sign that the theory is nonlocal — and fail to find one?

Hint: Gaussian correlators reconstruct Gaussian Hilbert spaces; the construction never asks whether a stress tensor exists. Locality of the *dynamics* (Hamiltonian = integral of a density) is simply never queried by the planar reconstruction.

Exercise I.2.28 (classic). (The current’s place in the operator algebra.) Work in the theory of a single free massless Dirac fermion in $d = 2$ — the continuum limit of the critical Ising-like chains of Part VIII, and the simplest 2d CFT with a $U(1)$ symmetry $\psi \rightarrow e^{i\alpha}\psi$. In Euclidean complex coordinates the chiral component has propagator $\langle\psi(z)\psi^\dagger(0)\rangle = \frac{1}{2\pi z}$, and the chiral current is the normal-ordered bilinear $j \sim :\psi^\dagger\psi:$ (point-split as in (I.2.3)). Verify by Wick contraction that the $j \times j$ OPE is

$$j(z)j(0) = \frac{k}{4\pi^2 z^2} + (\text{stress-tensor term}) + \cdots, \\ k = 1,$$

in any self-consistent normalization (state yours), confirming on this example the general claim that T appears in the JJ OPE of a local theory with strength tied to C_J . Then check the Schwinger-term computation (I.2.92) directly from fermionic Wick contractions at equal time, and observe that the central term arises from the normal-ordering subtraction — the very piece the naive canonical computation discards.

Hint: The double contraction gives the z^{-2} central term, the single contractions the operator terms.

Exercise I.2.29 (thinker). (How much of the theory is the local data, really?) Suppose two unitary CFTs have identical local CFT data: the same primaries, dimensions, and OPE coefficients, hence — by this chapter — identical correlators of local operators on $\mathbb{R}^d$. Catalogue what could still distinguish them, using only concepts already introduced in Part I (partition functions on other geometries, item 1 of Definition I.1.1; extended operators, item 4; the global form of symmetry groups, chapter I.1’s Exercise I.1.20). For one concrete handle: the torus partition function is a sum over the *full* Hilbert space — must that Hilbert space be exhausted by the states this chapter constructed from local operators? (No: chapter I.3’s defect Hilbert spaces and chapter I.4’s geometric axiomatics make the gap precise; in 2d, modular invariance ties the torus to the sphere data and the question becomes sharp — Part III.4.)

Reading

Rychkov, *EPFL Lectures on CFT in $D \geq 3$* , [1601.05000](#), chs. 1–3. The backbone of this chapter’s second half: fixed-point phenomenology and the continuous spectrum (ch. 1), radial quantization, reflection positivity, and the OPE (ch. 3). Short, opinionated, and the natural bridge to Part III.

Simmons-Duffin, [1602.07982](#), §§6–7. *TASI Lectures on the Conformal Bootstrap* — our conventions source. The state–operator correspondence done carefully via path-integral surgery, Euclidean conjugation demystified, unitarity bounds; appendix B is a compact summary of the Euclidean/Lorentzian dictionary.

Pappadopulo et al., [1208.6449](#). (Pappadopulo–Rychkov–Espin–Rattazzi.) The convergence paper behind Section I.2.5: the Cauchy–Schwarz argument, the Hardy–Littlewood spectral estimates, the exponential rate, and the ρ -coordinate optimization. Twenty readable pages; read it whole.

Kontsevich–Segal, [2105.10161](#), §2. The domain of allowable complex metrics: Definition 2.1, Theorem 2.2 (our Theorem I.2.2), and the Lorentzian-on-the-boundary picture. §§3 and 5 belong to chapter I.4.

Dedushenko, [2203.08053](#), §2.1. A compact survey of the Wightman and Osterwalder–Schrader axiom systems, useful chiefly as a guide to the original literature (including the growth-condition history this chapter only flagged).

Wilson (1969); Schwinger (1959). The primary sources for the chapter’s two named structures: the operator algebra as a definition of dynamics (Phys. Rev. **179**, 1499), and the commutator term that unitarity refuses to waive (Phys. Rev. Lett. **3**, 296). Both are short, and both are still worth reading in the original.

Books (for the shelf): Streater–Wightman, *PCT, Spin and Statistics, and All That* — the Wightman axioms and reconstruction, the spin-statistics and CPT theorems, in 200 pages of permanent value; Glimm–Jaffe, *Quantum Physics* (already on chapter I.1’s shelf) — reflection positivity and reconstruction done constructively; Haag, *Local Quantum Physics* — the algebraic tradition’s summa, background for Part X.

I.3 Observables II: defects and extended operators

Chapter I.2 closed with a warning attached to its own title: the local data, complete as it is on its own terms, does not exhaust the observables of a quantum field theory. This chapter is about the observables it misses — operators supported not at points but on lines, surfaces, and higher-dimensional submanifolds of spacetime, together with the boundaries and interfaces that are their codimension-one extreme — and its thesis is the lesson the subject absorbed between roughly 2014 and 2026: *the extended operators are part of the definition of the theory*. They are not composites of local operators, not bookkeeping for sources, and not optional decorations; they are an independent column of chapter I.1’s inventory (item 4 of Definition I.1.1), and two theories can agree on every correlator of every local operator on $\mathbb{R}^d$ and still differ — in their physics, not merely in their presentation — because they differ in their extended operators. Chapter I.1’s Exercise I.1.20 asked the reader to hold exactly this possibility in mind; this chapter makes good on it, with four-dimensional gauge theories supplying the canonical example.³⁵

Question I.3.1 — What observables live on submanifolds of spacetime? How are they defined when no Lagrangian is given — and when one is? Is the extended-operator content determined by the local data, and if not, what physics does it carry that local correlators cannot: which symmetries act on it, which phases does it diagnose, which Hilbert spaces does it create?

The reader has met extended operators before, although the textbook treatment makes them easy to mistake for technical devices. The Wilson loop appears in a gauge-theory course as a gauge-invariant functional useful for discussing confinement; boundary conditions appear as something one imposes to make a problem well-posed; the disorder operator of the Ising model, if it appears at all, appears as a trick of the duality literature. The modern view assembles these scattered objects into a single structural statement. A quantum field theory assigns data not only to points of spacetime but to embedded submanifolds of every dimension, and this assignment is constrained, organized, and physical: constrained by mutual locality (two extended operators must be insertable in the same correlator unambiguously), organized by codimension and by the symmetries that act, and physical in that the long-distance behavior of extended operators distinguishes phases that no local order parameter can tell apart. The last point is the oldest: it is Wilson’s 1974 criterion for confinement,³⁶ and the newest developments of Part II — where symmetries themselves become topological extended operators, and the objects they act on are lines and surfaces — stand directly on it.

The chapter runs as follows. We first say what an extended operator *is*, which requires confronting the same question chapter I.2 confronted for local operators: the naive constructions (integrate a local density along a curve; impose a constraint) are presentations, and the object itself must be characterized by its correlators. Doing this carefully produces a division of all extended operators into two construction types — **order**-type, built from the fields of a presentation, and **disorder**-type, defined by prescribing a singularity — and a sharper division, independent of any presentation, into **genuine** operators and operators attached to something of one dimension higher. The chapter’s centerpiece then builds the oldest disorder operator in physics, the Kramers–Wannier disorder line of the two-dimensional Ising model, entirely on the lattice, where every claim is a finite computation:

³⁵Aharony–Seiberg–Tachikawa [1305.0318](#), whose title — *Reading between the lines of four-dimensional gauge theories* — states the conclusion exactly: the missing data lives in the lines.

³⁶K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10** (1974) 2445 — the paper that introduced both lattice gauge theory and the loop observable now bearing his name, for exactly this diagnostic purpose.

the line will be constructed, proved topological, dualized, and used to diagnose both phases of the model — and at the critical point it will manufacture a fermion out of bosonic spins. With this concrete case in hand we turn to gauge theories, where the same order/disorder pair becomes the Wilson/’t Hooft pair, and where the celebrated result of Aharony–Seiberg–Tachikawa shows that choosing the mutually local set of lines is part of choosing the theory. Two further sections develop what defects do once they exist: as *probes*, they acquire conformal data of their own (one-point functions, defect-localized operators, the displacement operator) — the kinematics of defect CFT, with boundaries as the richest case; as *sources of states*, they define new Hilbert spaces — the defect (twisted) Hilbert spaces whose necessity chapter I.2’s closing already hinted at, and which chapter I.4’s cutting-and-gluing formalism will absorb as just more geometry.

Throughout, “defect” and “extended operator” are synonyms, used interchangeably as the literature does;³⁷ the convention pays off in Section I.3.5, where rotating an extended operator from “space” into “time” is the chapter’s final tool.

I.3.1 What is an extended operator?

Chapter I.2 opened by asking what a local operator is, and found that the durable answer bypasses fields: a local operator is whatever has well-defined correlation functions at a point, with the set of such objects constructed by organizing coincident-point singularities or by diagonalizing the renormalization group. The same discipline applies one dimension up, and since the rest of the chapter rests on it, we state it as a definition. Defining by correlators, rather than by any particular construction, is what makes the notion presentation-independent — the same reason chapter I.2’s Definition I.2.1 was phrased as it was — and it is what will allow the dualities met below to map extended operators built in one presentation onto extended operators built, by entirely different means, in another.

Definition I.3.1 (Extended operator)

An **extended operator** (synonymously, a **defect**) supported on a p -dimensional submanifold $\Sigma_p \subset M_d$ is an object $V(\Sigma_p)$ with well-defined correlation functions

$$\langle V(\Sigma_p) \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle \quad (\text{I.3.1})$$

jointly with the local operators and with other extended operators. The **extended-operator content** of a theory is the list of all such V , for each p , that the theory admits — item 4 of chapter I.1’s inventory (Definition I.1.1), now with the same standing as the local operator content.

As with local operators, the useful question is how such objects are *constructed* in the presentations we own. There are two constructions, and the difference between them organizes everything that follows.

³⁷There is a habit, harmless once stated, of saying *operator* when the object sits at fixed time and acts on a Hilbert space, and *defect* when it extends in the time direction and modifies the Hilbert space. In Euclidean signature the two configurations are rotations of one another, and the distinction is one of frame, not of substance; Section I.3.5 exploits the rotation. The usage follows Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#), §1.

Order-type: couple something to the submanifold

The first construction is transparent: build the extended operator out of the fields of the presentation. The mildest version inserts the integral of a local operator along Σ_p into the path integral, exponentiated so that the insertion resembles a source term in the action,

$$V_h(\Sigma_p) = \exp \left(h \int_{\Sigma_p} d^p \sigma \mathcal{O}(\sigma) \right), \quad (\text{I.3.2})$$

for some local operator $\mathcal{O}$ and coupling h . Physically this is an *impurity*: the theory in the bulk is undisturbed, but along Σ_p the action has been modified. A magnetic field applied along a line in a magnet, a charged particle's worldline in an electromagnetic field, a localized chemical potential — all are insertions of the form (I.3.2). More elaborate versions place new degrees of freedom on Σ_p — a quantum-mechanical spin sitting on a worldline (the Kondo impurity), a full p -dimensional field theory living on the defect and coupled to the bulk fields — but the principle is the same: everything is built from ingredients the presentation already supplies, locally along the defect. Operators of this construction type are called **order-type** defects. The name is retrospective, chosen for contrast with what comes next, and the Wilson line of Section I.3.3 is its most consequential instance.

Even this mild construction produces objects that are not local operators in disguise, and it is instructive to see how in the simplest possible case.

Example I.3.1 (The simplest defect: a static impurity in a free field)

Take the free massless scalar in $d = 4$ Euclidean dimensions, normalized as in chapter I.2 with $\langle \phi(x)\phi(0) \rangle = \frac{1}{4\pi^2 x^2}$, and insert the line defect (I.3.2) along the straight line $\Sigma_1 = \{(\tau, \mathbf{0})\}$ — the Euclidean worldline of a static pointlike impurity coupled linearly to ϕ :

$$V_h = \exp \left(h \int_{-\infty}^{\infty} d\tau \phi(\tau, \mathbf{0}) \right). \quad (\text{I.3.3})$$

The theory is Gaussian, so every correlator in the presence of V_h can be computed by completing the square: inserting V_h shifts ϕ by the classical field of the source. The simplest output is the one-point function of ϕ itself, which without the defect would vanish. At distance $r = |\mathbf{x}|$ from the line,

$$\langle \phi(x) \rangle_{V_h} \equiv \frac{\langle \phi(x) V_h \rangle}{\langle V_h \rangle} = h \int_{-\infty}^{\infty} d\tau \frac{1}{4\pi^2 [(x_0 - \tau)^2 + r^2]} = \frac{h}{4\pi r} : \quad (\text{I.3.4})$$

the Coulomb profile of a static charge, here derived as the response of the vacuum to a line operator. The computation is two lines long, but it carries structural lessons that generalize far beyond free fields. First, a defect gives one-point functions to operators that had none: the defect breaks the symmetries (here translations and rotations transverse to the line) that forced $\langle \phi \rangle = 0$, and the surviving symmetry — translations along the line, rotations around it, and, since $h \int \phi d\tau$ is dimensionless in $d = 4$, dilatations centered on the line — fixes the answer to be a pure power of the transverse distance, $\langle \mathcal{O} \rangle_V = a_{\mathcal{O}}/r^{\Delta}$. The coefficient $a_{\phi} = h/4\pi$ is new data, not contained in the bulk correlators; Section I.3.4 promotes such coefficients to the front rank of observables. Second, the defect requires renormalization of its own. The normalization $\langle V_h \rangle$ contains the integral $\frac{h^2}{2} \int d\tau d\tau' \langle \phi(\tau)\phi(\tau') \rangle$, which diverges at $\tau \rightarrow \tau'$ — a divergence proportional to the *length* of the line, removed by a counterterm $\delta m \int d\tau$ on the worldline, i.e. a renormalization of the impurity's rest energy. Defect couplings run, defect operators acquire

anomalous dimensions, and there is a defect renormalization group, with fixed points and all (it returns in Section I.3.4 and, as a tool for solving impurity problems, in Part III.5). Third, nothing about V_h is a local operator in disguise: no sum of local insertions reproduces an object whose correlators decay as a power of the distance *to a line*. The defect is a new entry in the inventory, even in a free theory.

Exercise I.3.1 (easy). (The impurity in general dimension.) Repeat the computation (I.3.4) for the free massless scalar in general d (two-point function $\langle \phi(x)\phi(0) \rangle = \frac{c_d}{x^{d-2}}$ with $c_d = \frac{\Gamma(d/2-1)}{4\pi^{d/2}}$): show that

$$\langle \phi(r) \rangle_{V_h} = h c_d b_d r^{3-d}, \quad b_d = \int_{-\infty}^{\infty} \frac{du}{(1+u^2)^{(d-2)/2}},$$

convergent for $d > 3$. The exponent $d-3$ agrees with $\Delta_\phi = \frac{d-2}{2}$ only at $d = 4$; explain the mismatch (the coupling h is dimensionful away from $d = 4$ and supplies the missing power), and conclude that the line defect (I.3.2) built from ϕ is scale-invariant exactly in $d = 4$, relevant (growing toward the infrared) for $d < 4$, and irrelevant for $d > 4$ — the defect renormalization group in its simplest instance.

Hint: Power counting on the worldline: the integrated operator $\int d\tau \mathcal{O}$ is a marginal defect coupling when $\Delta_{\mathcal{O}} = p = 1$. The relevant case $d = 3$ is physical: a localized magnetic field in the 3d Ising model, the “pinning defect,” flows to a nontrivial defect fixed point analyzed in Cuomo–Komargodski–Mezei 2112.10634.

Answer: $\langle \phi(r) \rangle_{V_h} = h c_d r^{3-d} b_d$ by the substitution $\tau = ru$; for $d = 4$, $c_4 = 1/4\pi^2$ and $b_4 = \pi$ reproduce (I.3.4).

Disorder-type: prescribe a singularity

The second construction is the one with no analogue among naive composites, and it is the reason extended operators cannot be derived from the local data. Instead of *adding* something along Σ_p , one *removes* Σ_p from spacetime and instructs the path integral to sum only over field configurations with a prescribed singularity — a prescribed winding, holonomy, or flux — around the excised locus. No new term is added to the action; the definition modifies the *space of configurations* rather than the weight. Operators of this construction type are called **disorder**-type defects, or disorder operators.

The prototype predates gauge theory. In a theory with a compact scalar — a field $\theta(x)$ valued in a circle, $\theta \simeq \theta + 2\pi$, like the phase of a superfluid order parameter in $d = 2$ — consider the codimension-two “vortex” operator $V_w(\Sigma_{d-2})$, defined by summing over configurations for which θ winds,

$$\oint_{\gamma} d\theta = 2\pi w, \tag{I.3.5}$$

for every small circle γ linking Σ_{d-2} once. Inside the path integral this is a constraint, not an insertion: the configurations being summed are singular along Σ_{d-2} (the field has no well-defined value there, which is why the locus is excised), and the integer w labels how the field wraps the excision. The ‘t Hooft line of Section I.3.3 is this construction transplanted to gauge fields, with (I.3.5) replaced by a quantized magnetic flux through the linking sphere; and the chapter’s centerpiece, built in the next section, is its lattice incarnation, where “prescribe a singularity” becomes a finite, checkable instruction about flipping bonds.

The two constructions look asymmetrical — one adds a source, the other forbids smoothness — but the asymmetry belongs to the presentation, not to the theory. Under the dualities of Part VI, order-type and disorder-type operators exchange roles wholesale: the vortex operator of the compact

boson is the ordinary vertex operator $e^{i\tilde{\theta}}$ of the dual boson, the 't Hooft line of one gauge theory is the Wilson line of its magnetic dual, and the Ising disorder operator about to be constructed is the order operator of the Kramers–Wannier-dual Ising model. Abstractly — at the level of the correlators (I.3.1) — there is just one set of extended operators, with two kinds of scaffolding for reaching them.

Intuition I.3.1 (Order and disorder)

An order-type defect tells the fields what to *do* along a submanifold; a disorder-type defect tells them how to *wind around* it. The first is visible in the action, the second only in the configuration space. Duality exchanges the two, because what one presentation calls a field configuration another calls a topological sector: the distinction order/disorder is a fact about presentations, while the distinction that survives presentation-independently is the set of extended operators and their mutual correlators.

Genuine operators, attached operators, and codimension

One more distinction is needed before the constructions can be put to work, and unlike order/disorder it is intrinsic. The definition of a disorder operator contains a potential ambiguity: prescribing the behavior of fields *around* Σ_p may or may not require auxiliary choices whose trace does not disappear. The disorder operator of the Ising model, as we will verify by direct computation in Section I.3.2, requires choosing a path from the operator to infinity (or to a partner operator), and correlators change sign when a $\mathbb{Z}_2$ -odd local operator crosses that path: the path is unobservable up to deformation but not removable. Such an operator is not an operator on its own; it is the endpoint of a line.

Definition I.3.2 (Genuine and attached operators)

An extended operator $V(\Sigma_p)$ is **genuine** if its correlation functions (I.3.1) depend on nothing besides Σ_p and the other insertions. It is **non-genuine**, or **attached**, if its definition requires a choice of $(p+1)$ -dimensional manifold Σ_{p+1} with $\partial\Sigma_{p+1} \supseteq \Sigma_p$, on which a (possibly invisible, possibly topological) operator must be placed. The attached operator is well-defined as the boundary of the higher-dimensional one; what fails is only its independent existence.^a

^aThe terminology is from Kapustin–Seiberg [1401.0740](#), §2, adopted as standard in Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#). The distinction sounds like pedantry and is instead load-bearing data: which operators are genuine and which attached differs between theories sharing the same local correlators, and Part II reads anomalies and gauging obstructions directly from the attachment pattern.

The reason this definition matters — the reason it is data rather than bookkeeping — is mutual locality. Two genuine operators can be inserted in a correlator at any separated locations, and the correlator is single-valued as they move. An attached operator, dragged around another operator, may return with its line swept across insertions, and the correlator may come back changed. The chapter will meet this as a computable sign in the Ising model (Section I.3.2) and as the Dirac quantization condition in gauge theory (Section I.3.3); in both cases, demanding mutual locality of the chosen operator set is what carves the possible *theories* out of the larger list of possible *operators*.

Alongside genuineness, the kinematic label that organizes extended operators is **codimension**

$q = d - p$ — the number of directions transverse to the defect — and it is codimension, not dimension, that controls most of the physics. The reason is that everything another operator can learn about a defect, it learns by going around it, and the space available for going around a p -dimensional defect in d dimensions is its **linking sphere** S^{q-1} . A local operator ($q = d$) is linked by S^{d-1} , the sphere of radial quantization; a codimension-one defect ($q = 1$) is linked by S^0 — two points, one on each side, which is why codimension-one defects are *walls* separating regions that may be in different states, different vacua, or (for interfaces) different theories; a codimension-two defect is linked by a circle, which is why codimension two is the home of winding and holonomy, as in (I.3.5). Boundaries are the degenerate extreme of the codimension-one case — a wall with theory on one side and nothing on the other — and Section I.3.4 explains why they are nonetheless the richest defects of all. Dimension, by contrast, mostly controls the defect’s own internal life: a line can carry quantum mechanics, a surface can carry a 2d field theory, and so on up.

I.3.2 The centerpiece: Kramers–Wannier and the disorder line, on the lattice

Every abstraction of the previous section has a complete, finite realization in the two-dimensional Ising model, and this section works it out in full. The construction will produce: a disorder-type extended operator built explicitly (a line defect and its endpoint operator); a direct proof that the line is *topological* — the first topological defect of these notes that is not a Noether charge; the Kramers–Wannier duality, derived rather than asserted, with the disorder operator revealed as the order operator of the dual model; the diagnosis of both phases of the model by the pair $(\langle\sigma\rangle, \langle\mu\rangle)$, a template for every phase classification in this book; mutual non-locality between order and disorder, measured by a sign; and, at the critical point, the Kadanoff–Ceva fermion — a spin- $\frac{1}{2}$ operator manufactured from commuting variables, the seed of bosonization. Chapter I.2 used the Ising model to define local operators; the same model now defines extended ones.³⁸

The model and its two expansions

Place spins $s_x = \pm 1$ on the sites x of a square lattice — for definiteness a large periodic lattice with N sites and $2N$ nearest-neighbor bonds — with

$$Z(J) = \sum_{\{s\}} \exp \left(J \sum_{\langle xy \rangle} s_x s_y \right), \quad (\text{I.3.6})$$

the two-dimensional version of chapter I.1’s Ising chain (Example I.1.3), at inverse temperature absorbed into $J > 0$. Unlike the chain, the two-dimensional model has two phases: a high-temperature (small- J) **disordered** phase with $\langle s_x \rangle = 0$ and exponentially decaying correlations, and a low-temperature (large- J) **ordered** phase in which the global $\mathbb{Z}_2$ symmetry $s_x \rightarrow -s_x$ is spontaneously broken and $\langle s_x s_y \rangle$ tends to a nonzero constant m^2 at large separation. (That the 2d model orders at low temperature, unlike the chain, is Peierls’ argument, rehearsed in chapter I.1’s Exercise I.1.11: a domain wall in $d = 2$ is a line whose energy grows with its length, and energy beats entropy at small temperature.) Everything in this section is a statement about (I.3.6) and its correlators; no continuum limit is taken until the critical point is reached at the end.

³⁸The duality is H. A. Kramers and G. H. Wannier, Phys. Rev. **60** (1941) 252, who used it to locate the critical temperature three years before Onsager’s solution. The disorder operator and the fermion are L. P. Kadanoff and H. Ceva, Phys. Rev. B **3** (1971) 3918. A modern review of disorder operators across dimensions is Fradkin 1610.05780; the operator-algebra view of the same material, which Part II adopts, is in Shao’s TASI lectures 2308.00747, §3.

The construction of the disorder operator rests on two convergent expansions of Z , one from each end of the phase diagram, and the expansions are short enough to derive here in full.

The high-temperature expansion. For Ising spins $(s_x s_y)^2 = 1$, so the bond weight linearizes:

$$e^{J s_x s_y} = \cosh J (1 + t s_x s_y), \quad t \equiv \tanh J. \quad (\text{I.3.7})$$

Insert (I.3.7) on every bond and expand the product over bonds. A term in the expansion picks, for each bond, either 1 or $t s_x s_y$; the chosen bonds form a subgraph Γ of the lattice, and the term carries $t^{|\Gamma|}$ times a product of spins, one factor of s_x for each bond of Γ ending at x . Summing over $s_x = \pm 1$ kills every term in which some site has an odd number of chosen bonds and contributes a factor 2 per site otherwise. Hence

$$Z = 2^N (\cosh J)^{2N} \sum_{\Gamma \text{ even}} t^{|\Gamma|}, \quad (\text{I.3.8})$$

where “even” means every site of Γ has even degree: the partition function is a sum over configurations of closed loops on the lattice, weighted by their total length. The same bookkeeping gives the spin correlator: the insertions $s_{x_1} s_{x_2}$ flip the parity demand at x_1 and x_2 , so

$$\langle s_{x_1} s_{x_2} \rangle = \frac{\sum_{\partial\Gamma=\{x_1, x_2\}} t^{|\Gamma|}}{\sum_{\partial\Gamma=\emptyset} t^{|\Gamma|}} : \quad (\text{I.3.9})$$

a sum over loop configurations together with one *open path* from x_1 to x_2 . At small t the shortest path dominates and $\langle s_{x_1} s_{x_2} \rangle \approx t^{|x_1 - x_2|}$: exponential decay, with $1/\xi = -\ln t$ to leading order — the two-dimensional counterpart of the chain’s exact $(\tanh J)^r$.

The low-temperature expansion. Now expand about an ordered ground state, say all spins up. A general configuration is described by which spins are flipped, and its energy cost relative to the ground state is $2J$ per **frustrated** bond — per bond whose two spins disagree. The frustrated bonds of any configuration form a collection of closed curves on the **dual lattice** — the lattice whose sites $\tilde{x}$ are the plaquette centers of the original, with one dual bond crossing each direct bond — namely the **domain walls** separating flipped from unflipped regions. The walls are closed because along any small circuit around a direct-lattice site, the spin changes sign an even number of times. Conversely, every configuration of closed dual curves arises from exactly two spin configurations (a global flip relates them). Hence, exactly,

$$Z = 2 e^{2NJ} \sum_{\omega \text{ closed}} e^{-2J|\omega|}, \quad (\text{I.3.10})$$

where ω runs over configurations of closed curves on the dual lattice and $|\omega|$ is the total wall length.

The two expansions (I.3.8) and (I.3.10) are sums over the *same* class of geometric objects — closed loops on a square lattice — with weights $t = \tanh J$ per unit length in one case and e^{-2J} per unit length in the other. Since the dual of the square lattice is again a square lattice, the two sums are literally the same function of their loop weight, and the partition functions at two different couplings are proportional whenever the weights match:

$$\tanh J^* = e^{-2J} \iff \sinh 2J^* \sinh 2J = 1. \quad (\text{I.3.11})$$

This is the **Kramers–Wannier duality**: the high-temperature Ising model at coupling J^* has the same loop expansion, hence (up to an analytic prefactor that carries no singularity) the same free

energy, as the low-temperature model at J . The map (I.3.11) exchanges the two phases — as J grows, J^* shrinks — so if the model has a single critical coupling, the singularity of the free energy must sit at the fixed point of the map,

$$\sinh 2J_c = 1, \quad J_c = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407, \quad (\text{I.3.12})$$

which is how Kramers and Wannier located the critical temperature in 1941, before any exact solution existed. The reader should pause on the logic: a relation between *observables at different couplings*, derived by reorganizing the same sum, pins down nonperturbative data. The whole of Part VI is this logic at scale.

Exercise I.3.2 (classic). (The duality, with prefactors.) Carry (I.3.8) into (I.3.10) keeping every factor: show that on a periodic N -site square lattice (where the dual lattice is again a periodic N -site square lattice),

$$\frac{Z(J)}{(2 \cosh J \sinh J)^N} = \frac{Z(J^*)}{(2 \cosh J^* \sinh J^*)^N}$$

with J^* as in (I.3.11), i.e. the duality is exact, not merely an equality of leading singularities, once the smooth prefactors are matched. Verify that (I.3.11) is an involution ($J^{**} = J$) and that $J^* \geq J_c$ when $J \leq J_c$.

Hint: Write both sides as $\sum_{\omega} w^{|\omega|}$ times prefactors, using $2 \cosh J \sinh J = \sinh 2J$ and the count of sites, bonds, and plaquettes of the periodic square lattice (N each, $2N$ bonds). On the torus there is a finite-size subtlety — the four spin structures / boundary-condition sectors mix under duality — which you may ignore in the thermodynamic limit; it returns with a vengeance, and with meaning, in Part III.4.

Answer: Matching weights $t = e^{-2J^*}$ gives, per bond, $\cosh J \cdot e^{J^*}$ worth of prefactor mismatch, which the stated normalization absorbs; involution follows since $\sinh 2J^* = 1/\sinh 2J$ is symmetric under $J \leftrightarrow J^*$.

The disorder line, constructed

Kramers–Wannier, as just derived, is a statement about partition functions. The disorder operator is what the duality does to *operators*, and it can be built with no reference to the duality at all — the duality will then be rediscovered as a statement about the built object’s correlators.

Pick two sites $\tilde{x}, \tilde{y}$ of the *dual* lattice, and any path $\tilde{\gamma}$ along dual bonds from $\tilde{x}$ to $\tilde{y}$. Each dual bond of $\tilde{\gamma}$ crosses exactly one direct bond. Define a modified partition function $Z[\tilde{\gamma}]$ by flipping the sign of the coupling on every crossed bond,

$$Z[\tilde{\gamma}] = \sum_{\{s\}} \exp \left(J \sum_{\langle xy \rangle} \varepsilon_{xy} s_x s_y \right), \quad \varepsilon_{xy} = \begin{cases} -1 & \langle xy \rangle \text{ crossed by } \tilde{\gamma}, \\ +1 & \text{otherwise,} \end{cases} \quad (\text{I.3.13})$$

and define the **disorder operator** two-point function as the normalized ratio

$$\langle \mu(\tilde{x}) \mu(\tilde{y}) \rangle \equiv \frac{Z[\tilde{\gamma}]}{Z}. \quad (\text{I.3.14})$$

The defining data is a *line* — the seam $\tilde{\gamma}$ of antiferromagnetic bonds — and the notation, which mentions only the endpoints, embodies a claim that must be proved: that the correlator depends on $\tilde{\gamma}$ only through $\partial\tilde{\gamma} = \{\tilde{x}, \tilde{y}\}$.

Here is the proof, which is one change of variables. Let $\tilde{\gamma}'$ be another dual path with the same endpoints. The union $\tilde{\gamma} \cup \tilde{\gamma}'$ is a closed dual curve, and a closed dual curve encloses a definite set

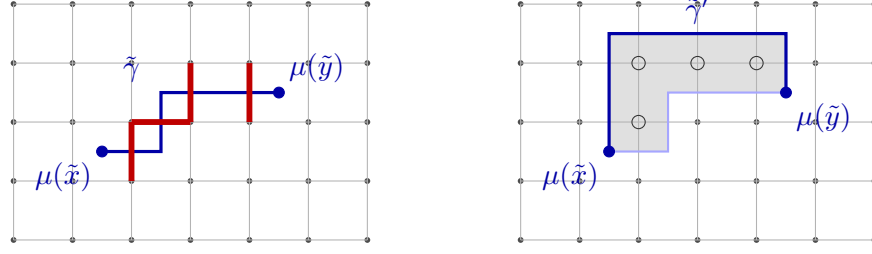


Figure 4: The disorder line. Left: a dual-lattice path $\tilde{\gamma}$ (blue) from $\tilde{x}$ to $\tilde{y}$; the couplings on the direct bonds it crosses (heavy red) are flipped, $J \rightarrow -J$. Right: deforming the path to $\tilde{\gamma}'$ is undone by flipping the spins (circled) in the enclosed region — a change of summation variable — so the correlator (I.3.14) depends only on the endpoints: the line is topological, and only its ends, the disorder operators μ , are physical.

R of direct-lattice sites (Figure 4). Change summation variables in $Z[\tilde{\gamma}']$ by flipping every enclosed spin, $s_x \rightarrow -s_x$ for $x \in R$. The bond term $s_x s_y$ changes sign exactly when one endpoint is inside R and the other outside — exactly on the bonds crossed by $\tilde{\gamma} \cup \tilde{\gamma}'$. The flip therefore converts the seam $\tilde{\gamma}'$ into the seam $\tilde{\gamma}$, term by term in the sum, and since relabeling summation variables cannot change the sum,

$$Z[\tilde{\gamma}'] = Z[\tilde{\gamma}] : \quad (\text{I.3.15})$$

the seam is *invisible*; it can be deformed freely so long as its endpoints stay put. In the language of Definition I.3.2: the line $\tilde{\gamma}$ is a *topological line defect*, and μ is a *non-genuine* operator attached to it. The line itself is a seam across which the configurations are reglued by the $\mathbb{Z}_2$ symmetry $s \rightarrow -s$. It is, in the vocabulary that Part II will make official, the *symmetry defect* of the $\mathbb{Z}_2$ symmetry — chapter I.2 found that Noether charges are topological codimension-one operators (I.2.72); here is the same statement for a discrete symmetry, with no current anywhere in sight, established by a change of variables on a lattice. The disorder operator μ is the simplest instance of the general slogan: *ends of symmetry defects are the canonical non-genuine operators*. If the closed curve $\tilde{\gamma} \cup \tilde{\gamma}'$ winds around the periodic lattice rather than bounding, the flip argument fails — there is no enclosed region — and indeed a seam wrapping a cycle is a physical modification: it computes the partition function with antiperiodic boundary conditions, an observable of the *theory on the torus*, distinct from Z . The topology of the line is trivial only up to the topology of spacetime, a refrain that returns in Section I.3.5.

Now compute (I.3.14), using the low-temperature expansion — this is where duality re-enters. In the presence of the seam, the energy bookkeeping of (I.3.10) changes in one place: a crossed bond has coupling $-J$, so it costs $2J$ when its spins *agree* and is satisfied when they disagree. Relative to the all-up state, a configuration with domain-wall curve ω (on the dual lattice, as before) now has energy cost $2J$ times the length of the *symmetric difference* $\omega \triangle \tilde{\gamma}$ — the curve that disagrees either with the walls or with the seam but not both. Therefore

$$\langle \mu(\tilde{x}) \mu(\tilde{y}) \rangle = \frac{\sum_{\omega \text{ closed}} e^{-2J|\omega \triangle \tilde{\gamma}|}}{\sum_{\omega \text{ closed}} e^{-2J|\omega|}} = \frac{\sum_{\partial\omega'=\{\tilde{x},\tilde{y}\}} e^{-2J|\omega'|}}{\sum_{\partial\omega'=\emptyset} e^{-2J|\omega'|}}, \quad (\text{I.3.16})$$

where the second equality substitutes $\omega' = \omega \triangle \tilde{\gamma}$, which runs over curve configurations with endpoints $\{\tilde{x}, \tilde{y}\}$ as ω runs over closed ones. Compare (I.3.9): the right-hand side of (I.3.16) is precisely the high-temperature representation of a *spin* two-point function, on the dual lattice, at the coupling

J^* for which $\tanh J^* = e^{-2J}$. We have derived

$$\langle \mu(\tilde{x}) \mu(\tilde{y}) \rangle_J = \langle s_{\tilde{x}} s_{\tilde{y}} \rangle_{J^*} : \quad (\text{I.3.17})$$

the disorder operator of the Ising model is the order operator of its Kramers–Wannier dual. Equation (I.3.17) is exact, at every coupling and every separation, with no continuum limit and no criticality assumed. The duality of partition functions (I.3.11) has been upgraded to a duality of operators, and the upgrade required introducing an extended operator: the dual of a perfectly ordinary local operator is the endpoint of a line. There is no way to state Kramers–Wannier as a map of local operators only — a first, fully computable demonstration that the extended operators are not optional.

Equation I.3.1 (Order–disorder duality)

$$\langle \mu(\tilde{x}) \mu(\tilde{y}) \rangle_J = \langle s_{\tilde{x}} s_{\tilde{y}} \rangle_{J^*}, \quad \sinh 2J \sinh 2J^* = 1.$$

The endpoint of the topological $\mathbb{Z}_2$ seam at coupling J has exactly the correlations of the local spin at the dual coupling J^* . At the self-dual point $J = J^* = J_c$, order and disorder correlators become equal.

Exercise I.3.3 (easy). (The whole construction, in one dimension.) For the periodic Ising chain of chapter I.1 (Example I.1.3: N sites, weight $e^{Js_i s_{i+1}}$ per bond, transfer matrix with eigenvalues $\lambda_{\pm} = 2 \cosh J, 2 \sinh J$), the “dual lattice” consists of the bond midpoints, and a disorder pair $\mu(\tilde{i})\mu(\tilde{j})$ flips the couplings on the r bonds between them. Show by transfer matrices that

$$\langle \mu\mu \rangle = \frac{Z[\text{flipped}]}{Z} \xrightarrow{N \rightarrow \infty} (\tanh J)^r = \langle s_0 s_r \rangle :$$

in $d = 1$ the disorder correlator coincides with the order correlator at the *same* coupling. Interpret: the 1d model is its own dual, the seam endpoints create the domain-wall excitations, and the chain’s lack of a phase transition is the statement that σ - and μ -type correlations always decay together.

Hint: Flipping the coupling on a bond replaces its transfer matrix $T = e^J I + e^{-J} \sigma_x$ by $T' = e^{-J} I + e^J \sigma_x = \sigma_x T$, so the flipped partition function is a trace of T ’s with a σ_x inserted at each end of the flipped stretch; diagonalize in the $|\pm\rangle$ basis where $\sigma_x |\pm\rangle = \pm |\pm\rangle$.

Answer: $Z[\text{flipped}]/Z = (\lambda_+^{N-r} \lambda_-^r + \lambda_-^{N-r} \lambda_+^r) / (\lambda_+^N + \lambda_-^N) \rightarrow (\lambda_- / \lambda_+)^r = (\tanh J)^r$ — identical to chapter I.1’s $\langle s_0 s_r \rangle$, computed there by inserting σ_z ’s. The domain-wall interpretation: σ_x inserted in the transfer-matrix product creates a wall, and $\tanh J$ per step is the wall’s propagation weight.

What the disorder operator diagnoses

The relation (I.3.17) converts known facts about the spin into new physics of the seam, and the conversion is the template for every “line operators diagnose phases” statement in these notes, including the gauge-theory one of the next section. Read (I.3.17) in each phase.

In the disordered phase $J < J_c$, the dual coupling is supercritical, $J^* > J_c$, so the dual spin orders: $\langle s_{\tilde{x}} s_{\tilde{y}} \rangle_{J^*} \rightarrow (m^*)^2 \neq 0$ at large separation. Translated: $\langle \mu\mu \rangle$ tends to a nonzero constant — the disorder operator *condenses*, $\langle \mu \rangle = m^* \neq 0$, in precisely the phase where the order operator does not. Physically: μ ’s seam reglues configurations by a spin flip, and in a disordered ensemble,

where spins carry no long-range memory of their sign, regluing costs only a local disturbance near each endpoint; the line, though arbitrarily long, is free.

In the ordered phase $J > J_c$, the dual spin is disordered and $\langle \mu\mu \rangle \sim e^{-|\tilde{x}-\tilde{y}|/\xi^*}$ decays exponentially. Physically: in an ordered background the seam is a real obstruction — the cheapest configuration in (I.3.16) is a domain wall stretched between the two endpoints, paying tension all along its length — and the correlator falls with the *length of the cheapest wall connecting the insertions*. The exponential decay of a disorder pair is the two-dimensional shadow of the area law that will diagnose confinement in Section I.3.3: there, as here, the criterion is whether an extended object's cost grows with its size, and there, as here, the growth happens in the phase where the dual charge has condensed.

The pair $(\langle \sigma \rangle, \langle \mu \rangle)$ therefore separates the phase diagram cleanly:

$$\begin{aligned} J > J_c : \quad & \langle \sigma \rangle \neq 0, \quad \langle \mu \rangle = 0 && (\text{ordered; } \mathbb{Z}_2 \text{ broken}), \\ J < J_c : \quad & \langle \sigma \rangle = 0, \quad \langle \mu \rangle \neq 0 && (\text{disordered; } \mathbb{Z}_2 \text{ unbroken}), \\ J = J_c : \quad & \langle \sigma \rangle = \langle \mu \rangle = 0, \quad \text{both correlators power-law} && (\text{critical}). \end{aligned} \quad (\text{I.3.18})$$

The disordered phase, featureless to every local order parameter, has an order parameter after all — an extended one. This is the resolution of an old embarrassment of the Landau classification (how can a phase with *no* broken symmetry be distinguished from another such phase?), and its systematic development — where the condensing objects are lines and surfaces and the “symmetries” they break are the generalized symmetries of Part II — is the generalized Landau paradigm of Part II.8. At the critical point both expectation values vanish and both two-point functions obey the same power law,

$$\langle \sigma\sigma \rangle_{J_c} \sim \langle \mu\mu \rangle_{J_c} \sim \frac{1}{|x|^{1/4}}, \quad \Delta_\sigma = \Delta_\mu = \frac{1}{8}, \quad (\text{I.3.19})$$

the equality of dimensions being forced by self-duality (I.3.17) at $J = J^* = J_c$ (the value $\frac{1}{8}$ itself is Onsager-era input, derived three ways in Part III.4). In the continuum limit at J_c , then, the Ising CFT contains two scalar primaries of dimension $\frac{1}{8}$: the genuine local operator σ , and the non-genuine μ , attached to the topological $\mathbb{Z}_2$ line. They are not related by any local similarity transformation — one is mutually local with everything, the other is not, as we now verify.

Exercise I.3.4 (medium). (The disorder condensate, to leading order.) Deep in the disordered phase ($J \rightarrow 0$), compute $\langle \mu(\tilde{x})\mu(\tilde{y}) \rangle$ directly from the definition (I.3.14), by expanding both $Z[\tilde{\gamma}]$ and Z to second order in J . A naive guess is that every crossed bond suppresses the ratio, which would make the correlator decay exponentially in the seam *length* and contradict path independence. Show instead that each crossed bond contributes $\frac{\cosh(-J)}{\cosh J} = 1$, and that all corrections organize into *endpoint* contributions plus terms exponentially small in the separation, so that $\langle \mu\mu \rangle \rightarrow \langle \mu \rangle^2 \neq 0$. Then check consistency with duality: $\langle \mu \rangle^2 = (m^*)^2$ with m^* the spontaneous magnetization at $J^* \rightarrow \infty$, which tends to 1, and indeed your direct computation gives $\langle \mu\mu \rangle \rightarrow 1$ as $J \rightarrow 0$.

Hint: At $J = 0$ exactly, $Z[\tilde{\gamma}] = Z = 2^N$: flipping couplings is invisible when spins are uncorrelated, so $\langle \mu\mu \rangle = 1$ on the nose. Work at small J and watch the high-temperature graphs of (I.3.8): a graph contributes a sign $(-1)^{\#(\Gamma \cap \tilde{\gamma})}$, and closed graphs cross the seam an even number of times unless they wind around an endpoint.

Mutual locality, and a fermion from spins

The operator content assembled so far — genuine σ , attached μ — carries one more computable structure: the two are mutually non-local, and the failure is a clean $\mathbb{Z}_2$ monodromy. Consider

a correlator with both insertions, $\langle s_z \mu(\tilde{x}) \mu(\tilde{y}) \rangle$, defined by (I.3.13) with a spin inserted. The path-independence argument (I.3.15) acquires an exception: deforming $\tilde{\gamma}$ across the site z flips the summation variable s_z , which now sits *in the correlator*, not only in the weight. The correlator with the deformed seam equals *minus* the original:

$$\langle s_z \mu \mu \rangle_{\tilde{\gamma}'} = - \langle s_z \mu \mu \rangle_{\tilde{\gamma}} \quad \text{when } \tilde{\gamma}' - \tilde{\gamma} \text{ sweeps across } z. \quad (\text{I.3.20})$$

Equivalently: transporting σ around one endpoint μ along a closed loop — which forces the seam to be swept across σ once — returns the correlator with a factor (-1) . The pair (σ, μ) is **mutually semi-local** with monodromy -1 . This is the lattice prototype of the linking phases that govern which line operators may coexist in gauge theory (Section I.3.3); there as here, the phase is computed by an entirely kinematic argument — no dynamics, only the bookkeeping of what crosses what.

Kadanoff and Ceva noticed what the monodromy is good for. Place the two mutually semi-local operators at nearest-neighbor positions — a spin s_x and an adjacent disorder operator $\mu(\tilde{x})$ — and call the composite

$$\psi(x, \tilde{x}) \sim s_x \mu(\tilde{x}). \quad (\text{I.3.21})$$

Rotate the plane by 2π about the composite. The seam attached to μ is dragged once completely around, and in the course of the rotation it sweeps across the partner spin s_x exactly once: by (I.3.20), the configuration returns to itself with a factor (-1) . An operator that changes sign under a 2π rotation has half-integer spin. The Ising model — commuting variables, not a Grassmann number in sight — contains a *fermion*, assembled from an order and a disorder operator, with the minus sign of Fermi statistics supplied by the topology of the attached line. At the critical point the composite becomes a conformal field of dimensions $(h, \bar{h}) = (\frac{1}{2}, 0)$ (and its conjugate $(0, \frac{1}{2})$), with

$$\langle \psi(x) \psi(0) \rangle \sim \frac{1}{x}, \quad (\text{I.3.22})$$

the free-fermion correlator: the continuum limit of the critical Ising model is, in this precise operator-by-operator sense, a theory of free Majorana fermions.³⁹ Fermionization, bosonization, the Jordan–Wigner transformation — the entire technology by which two-dimensional physics trades spins for fermions — is the systematic exploitation of (I.3.21), and the reader who tracks where the Jordan–Wigner “string” goes will find it is exactly the disorder seam (Exercise I.3.5).

Idea I.3.1 (Extended operators are made, and make, new physics)

The disorder line was constructed from the symmetry of the theory (a $\mathbb{Z}_2$ seam), proved topological by a change of variables, and found to carry, at its ends, operators that (i) serve as the order parameter of the symmetry-*unbroken* phase, (ii) realize the dual model’s local operators, and (iii) combine with local operators into fields of new spin and statistics. None of this data is visible in, or derivable from, the correlators of genuine local operators alone — yet all of it is exact, lattice-computable physics. “The definition of a theory includes its extended operators” is not a definition being defended; it is a description of what the Ising model was

³⁹Kadanoff–Ceva, op. cit., established (I.3.22) on the lattice. The statement that the $\sigma\mu$ composite has spin $\frac{1}{2}$, argued here by monodromy, is visible in the exact lattice correlators; the free-fermion description is completed by the Onsager/Schultz–Mattis–Lieb solution and is derived in the continuum in Part III.4, with the global structure — in what sense the Ising CFT and the Majorana CFT are and are not the same theory — in Parts II and VIII (Seiberg–Shao [2307.02534](#) is the careful modern treatment).

found to contain.

The construction generalizes in every direction, and the generalizations are best signposted now, while the mechanism is fresh. For a $\mathbb{Z}_N$ or $U(1)$ symmetry, the seam reglues by a symmetry element g and the endpoint operators carry a g -dependent monodromy — disorder operators exist for every global symmetry of every lattice model. In higher dimensions d , the seam is a codimension-one sheet $\tilde{\Sigma}_{d-1}$ of flipped bonds, its boundary a codimension-two disorder operator: in the 3d Ising model the disorder operator is a *line* (the boundary of a flipped-bond membrane), whose condensation-versus-tension dichotomy again diagnoses the phases, and whose gauge-theory counterpart is the 't Hooft line met momentarily. And when the symmetry being reglued is a *gauge* redundancy rather than a global symmetry, the seam itself becomes invisible in a stronger sense but its possible boundaries reorganize: that story is the next section's. The single lattice mechanism — flip couplings across a hypersurface, watch the boundary — underlies all of it.

Exercise I.3.5 (medium). (Kramers–Wannier and the disorder operator in the quantum chain.) The transfer-matrix limit of the 2d Ising model (chapter I.1's Example I.1.3 logic, run with a spatial dimension kept) is the transverse-field Ising chain: N sites, Pauli operators X_j, Z_j , Hamiltonian

$$H = -g \sum_{j=1}^N X_j - \sum_{j=1}^N Z_j Z_{j+1},$$

periodic identification $N+1 \equiv 1$; the ordered phase is $g < 1$, the disordered $g > 1$, the critical point $g = 1$, and the $\mathbb{Z}_2$ symmetry is $\eta = \prod_j X_j$. (a) Define dual-site variables

$$\tilde{Z}_{j+\frac{1}{2}} = \prod_{i \leq j} X_i, \quad \tilde{X}_{j+\frac{1}{2}} = Z_j Z_{j+1},$$

and check they obey the Pauli algebra on the dual sites (commuting at different sites, anticommuting on-site). (b) Show $X_j = \tilde{Z}_{j-\frac{1}{2}} \tilde{Z}_{j+\frac{1}{2}}$ and hence $H(g) = g \tilde{H}(1/g)$ with $\tilde{H}$ the same Hamiltonian in dual variables: Kramers–Wannier as an exact rewriting, self-dual at $g = 1$. (c) Identify $\mu_{j+\frac{1}{2}} = \tilde{Z}_{j+\frac{1}{2}}$: a half-infinite *string* of spin flips — the quantum-chain incarnation of the seam, manifestly attached to a line (the string), manifestly the order parameter of the $g > 1$ phase (where $X_j \approx +1$ and the string averages to a nonzero value). (d) On the periodic chain, show the string cannot close consistently: a single μ is not an operator on the untwisted Hilbert space, only μ pairs are — and a single μ instead lives in the Hilbert space twisted by η (Section I.3.5). (e) Combine with a Jordan–Wigner transformation to exhibit the Kadanoff–Ceva fermion as $\psi_j \sim Z_j \prod_{i < j} X_i$, and verify $\{\psi_i, \psi_j\} \propto \delta_{ij}$: the string makes commuting variables anticommute.

Hint: For (d): formally, $\prod_{i \leq N} X_i = \eta$, so the “closed-up” string is the symmetry operator itself, not an endpoint operator; the obstruction to ending it is the $\mathbb{Z}_2$ charge it would leak. Each part is a short computation; the content is in reading them.

Preview I.3.1 (The seam at the self-dual point: a non-invertible line)

At $J = J_c$ the model equals its dual, so Kramers–Wannier becomes a *symmetry of one theory* — and, since it maps the genuine σ to the attached μ , it cannot be implemented by any unitary operator. It is implemented by a topological line $\mathcal{D}$ with the fusion rule $\mathcal{D} \times \mathcal{D} = 1 + \eta$: a **non-invertible symmetry**, the first ever identified, and on the lattice an explicit $2^N \times 2^N$ matrix commuting with the critical chain Hamiltonian. Part II.5 builds this properly (Shao [2308.00747](#),

§3, is the source); it is mentioned here because the reader now owns every ingredient: the η seam, the duality, and the order–disorder exchange.

I.3.3 Lines in gauge theory: the spectrum is part of the definition

Gauge theory in four dimensions is where the chapter’s thesis was sharpened into a precise statement, and the logic tracks the Ising construction step for step: an order-type line (Wilson), a disorder-type line (’t Hooft), a mutual-locality constraint (Dirac), and the discovery that consistent maximal choices of lines are *plural* — distinct theories, agreeing on all local correlators, distinguished only between the lines. The section develops each step in turn. Throughout, “gauge theory” means the quantum theory built from a gauge *algebra* $\mathfrak{g}$ together with whatever additional global data the analysis turns out to demand; part of the point is to watch that demand emerge.

The Wilson line

The Wilson line answers a physical question: what does the gauge theory do to an external charge? Take Maxwell theory first — a $U(1)$ gauge field with Euclidean action $S = \frac{1}{4e^2} \int d^4x F_{\mu\nu} F_{\mu\nu}$ — and introduce a probe particle of charge $q \in \mathbb{Z}$, infinitely heavy so that it has no dynamics of its own: it simply sits on a prescribed worldline C . The coupling of a charge to the gauge field is $q \int_C A$, so integrating out the probe leaves behind the insertion

$$W_q(C) = \exp \left(iq \oint_C A_\mu dx^\mu \right), \quad (\text{I.3.23})$$

the **Wilson line** of charge q . For a nonabelian algebra the exponential must be path-ordered and traced in a chosen representation R ,

$$W_R(C) = \text{tr}_R \text{P exp} \left(i \oint_C A \right), \quad (\text{I.3.24})$$

because the probe now carries internal color states that the gauge field rotates as it moves; the trace sums over the probe’s internal states, and gauge invariance of the closed loop follows from cyclicity. The label of a Wilson line is therefore a representation, and which representations are allowed is exactly the question this section builds toward.

Equation (I.3.24) is the order-type construction (I.3.2) in gauge-theory clothing — a coupling integrated along the defect — and, like the Ising impurity, it serves as a probe of the vacuum. The diagnostic works as follows. Take C to be a rectangle: a charge–anticharge pair created at some time, held at separation R for a Euclidean time $T \gg R$, then annihilated. The path integral with this insertion computes the amplitude for the static pair to persist, so

$$\langle W(C_{R \times T}) \rangle \sim e^{-TV(R)}, \quad T \rightarrow \infty, \quad (\text{I.3.25})$$

with $V(R)$ the energy of the static pair: the interquark potential, extracted from a line operator’s expectation value. The two characteristic large-loop behaviors,

$$\langle W(C) \rangle \sim \begin{cases} e^{-c \text{Perimeter}(C)} & \text{perimeter law: } V(R) \rightarrow \text{const, liberated charges,} \\ e^{-\varsigma \text{Area}(C)} & \text{area law: } V(R) = \varsigma R, \text{ confinement, string tension } \varsigma, \end{cases} \quad (\text{I.3.26})$$

distinguish phases of the gauge theory — Wilson’s criterion. A perimeter-law contribution is always present (it is the probe’s self-energy, the divergence met in Example I.3.1, removed by a worldline

counterterm), so the content of (I.3.26) is whether, after that subtraction, the loop feels its area. The parallel with the Ising disorder pair is exact: an extended object whose cost grows with its extent (wall tension there, string tension here) signals one phase; an extended object that gets free of its extent signals the other. Which object does which depends on the phase, and having *both* an order line and a disorder line in hand is what makes the diagnostics complete — the burden of Section I.3.3.

Example I.3.2 (The Wilson loop in free Maxwell theory)

The one gauge theory where (I.3.25) can be evaluated from start to finish is the free Maxwell field, and we do it once in full, because the evaluation separates what is special to free fields from what is structural. The theory is Gaussian, so

$$\langle W_q(C) \rangle = \exp \left(-\frac{q^2}{2} \oint_C \oint_C dx^\mu dy^\nu \langle A_\mu(x) A_\nu(y) \rangle \right), \quad (\text{I.3.27})$$

with the photon propagator in Feynman gauge $\langle A_\mu(x) A_\nu(0) \rangle = \frac{e^2 \delta_{\mu\nu}}{4\pi^2 x^2}$ (the gauge-dependent pieces integrate to zero around closed loops). Evaluate on the $R \times T$ rectangle, $T \gg R$. The double line integral splits into self-pairings of each side and cross-pairings. The self-pairing of each long side is the divergent self-energy — $\frac{e^2 q^2}{4\pi^2} T \int \frac{ds}{s^2}$, linear in T with a short-distance divergent coefficient, the worldline mass counterterm again. The physics is in the cross-pairing of the two long sides, which are traversed in opposite directions (relative minus sign):

$$-2 \times \frac{e^2 q^2}{2} \int_0^T dt \int_0^T dt' \frac{(-1)}{4\pi^2 [(t-t')^2 + R^2]} \xrightarrow{T \gg R} + \frac{e^2 q^2 T}{4\pi^2} \int_{-\infty}^{\infty} \frac{ds}{s^2 + R^2} = \frac{e^2 q^2 T}{4\pi R}. \quad (\text{I.3.28})$$

Assembling via (I.3.25) (and absorbing the self-energies into the counterterm),

$$V(R) = -\frac{e^2 q^2}{4\pi R} : \quad (\text{I.3.29})$$

the Coulomb attraction of a static pair, recovered as the area-free, perimeter-renormalized content of a line operator's expectation value. Free Maxwell theory is in the perimeter-law class — charges can be separated at finite cost — as expected for a theory of free photons. The reader should note which features of the computation were specific to free fields (the Gaussian closing of the exponential) and which were not (the worldline counterterm; the extraction of $V(R)$; the perimeter/area alternative): in strongly coupled Yang–Mills the former fails utterly while the latter organizes the numerical and theoretical study of confinement to this day.

Exercise I.3.6 (medium). (Cusps.) The rectangle has four corners, and the cross-pairings of adjacent sides produce, besides the finite potential, a logarithmic divergence localized at each corner. Extract it: show that two straight segments meeting at an angle θ contribute to $-\ln\langle W \rangle$ a term $\frac{e^2 q^2}{4\pi^2} \Gamma_{\text{cusp}}(\theta) \ln(L/a)$ with a a short-distance cutoff, and compute Γ_{cusp} at this (lowest) order. Smooth loops have no such divergence: the renormalization of a line operator depends on the *geometry* of its support — length divergences always, angle divergences at cusps. (The cusp anomalous dimension became a central object of gauge-theory integrability and of the modern amplitudes program, Part IX.)

Hint: Parametrize the two segments from the common corner and use the same propagator; the $\ln$ arises from the

region where both points approach the corner. $\Gamma_{\text{cusp}}(\theta) \propto \theta \cot \theta - 1$ in Euclidean signature at this order.

The 't Hooft line, and mutual locality

The Wilson line used only the order-type construction, and the Ising analysis instructs us to seek its disorder partner. In Maxwell theory the recipe of Section I.3.1 applies verbatim: excise a line L from spacetime and sum over gauge fields with prescribed singular behavior around it — prescribed flux through the linking two-sphere,

$$\int_{S^2_{\text{link}}} F = 2\pi m, \quad m \in \mathbb{Z} : \quad (\text{I.3.30})$$

the **'t Hooft line** of magnetic charge m , the worldline of an infinitely heavy magnetic monopole that the theory does not contain as a dynamical particle.⁴⁰ No term is added to the action; the configuration space is enlarged to include monopole sectors around L . The construction is the vortex defect (I.3.5) with the winding of a scalar replaced by the flux of a gauge field, and the codimension adjusted to match: in $d = 4$ a line has codimension three, its linking sphere is the *two*-sphere, and flux through S^2 is therefore the data that can be prescribed.

Wilson and 't Hooft lines are mutually constrained, and the constraint is the gauge-theory version of the Ising monodromy (I.3.20). Transport a Wilson line of charge q around a closed loop γ that links the 't Hooft line of charge m once. The Wilson insertion is $e^{iq \oint_{\gamma} A}$, and as γ is swept around the monopole worldline it spans a closed surface through which the flux is (I.3.30); the holonomy changes by

$$\exp \left(iq \int_{S^2} F \right) = e^{2\pi i q m}. \quad (\text{I.3.31})$$

For integer q and m this phase is 1 and the two insertions are mutually local: Maxwell theory with charges in $\mathbb{Z}$ admits both families as genuine lines. But the phase (I.3.31) is the general currency, and it becomes a nontrivial constraint the moment fractional pairings appear — as they do for every nonabelian algebra with a nontrivial center. For dyonic lines carrying both charges, (q, m) and (q', m') , the same sweep produces the **Dirac–Schwinger–Zwanziger pairing**

$$\langle (q, m), (q', m') \rangle = q m' - m q', \quad (\text{I.3.32})$$

and single-valuedness of correlators demands that the pairing be an integer (in the units where (I.3.31) is $e^{2\pi i(qm' - mq')}$, an element of $\mathbb{Z}$) for every pair of lines the theory contains. Mutual locality is not automatic; it is a constraint on the *set* of lines, and sets obeying it are in general far from unique.

$SU(2)$ versus $SO(3)$: one algebra, three theories

Now run the constraint in the simplest nonabelian setting, following Aharony–Seiberg–Tachikawa.⁴¹ Take pure Yang–Mills theory with algebra $\mathfrak{su}(2)$ and no matter. Local operators — gauge-invariant

⁴⁰G. 't Hooft, *On the phase transition towards permanent quark confinement*, Nucl. Phys. B **138** (1978) 1, which introduced both the operator and the order/disorder classification of gauge-theory phases used in this section. The quantization of m in (I.3.30) is Dirac's: F is the curvature of a $U(1)$ bundle on S^2_{link} , and its flux is 2π times the bundle's Chern number.

⁴¹[1305.0318](#), §1, whose presentation this subsection compresses; the general-group analysis is their §§2–6. The underlying observation that the line spectrum distinguishes gauge groups goes back to Kapustin [hep-th/0501015](#) and is implicit in 't Hooft's classification.

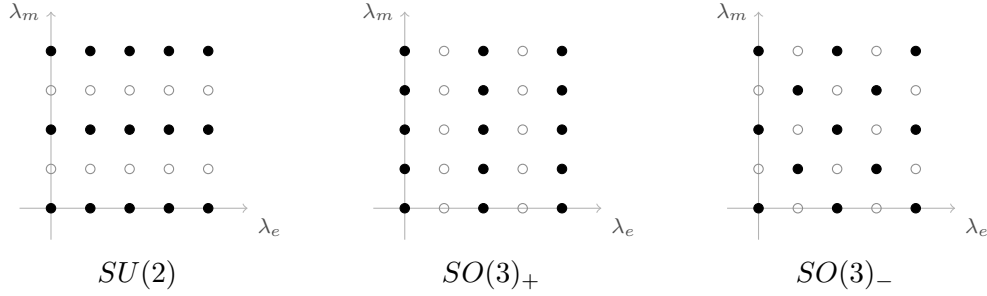


Figure 5: The line operators of the three theories with gauge algebra $\mathfrak{su}(2)$, after Aharony–Seiberg–Tachikawa. Each panel shows the electric–magnetic charge lattice (λ_e, λ_m) ; filled dots are the lines the theory contains, open circles those it excludes. All three sets are closed under fusion, mutually local under the pairing (I.3.32), and maximal — and no two are the same set.

polynomials in the field strength and its derivatives — are blind to the global form of the gauge group: their correlators on $\mathbb{R}^4$ are identical whether “the gauge group” is taken to be $SU(2)$ or $SO(3) = SU(2)/\mathbb{Z}_2$. If the theory were its local correlators, the choice would be a convention. It is not, and the lines show it.

Charge labels: electric charge under the center of $SU(2)$ is a weight modulo roots, $z_e \in \mathbb{Z}_2$ — integer spin representations have $z_e = 0$, half-integer spin (the fundamental) $z_e = 1$ — and magnetic charge likewise, $z_m \in \mathbb{Z}_2$, by running (I.3.30) with the dual group’s weights. A line class is a pair $(z_e, z_m) \in \mathbb{Z}_2 \times \mathbb{Z}_2$, four classes in all, and the pairing (I.3.32) reduces mod 2:

$$\langle (z_e, z_m), (z'_e, z'_m) \rangle = z_e z'_m - z_m z'_e \bmod 2. \quad (\text{I.3.33})$$

A consistent theory must pick a set of classes that (i) is closed under fusion (bringing two lines together gives a line, so the set is a subgroup), (ii) is mutually local — the pairing (I.3.33) vanishes on the set — and (iii) is *maximal*: any class mutually local with everything in the set is in the set. Maximality is the statement that there is no obstruction to including a consistent line, so excluding it would amount to declaring extra structure that is not there; relaxing it leads not to new theories but to incomplete descriptions of these same ones (the systematic account of why, via one-form symmetries and gauging, is Part II.4’s). The subgroup $\{(0, 0)\}$ alone fails (iii), since everything pairs to zero with it. The enumeration is then a one-line exercise in $\mathbb{Z}_2 \times \mathbb{Z}_2$: the maximal mutually local subgroups are exactly the three order-two subgroups,

$$\begin{aligned} \{(0, 0), (1, 0)\} : & \quad SU(2) && \text{Wilson line in the fundamental; no fundamental 't Hooft line,} \\ \{(0, 0), (0, 1)\} : & \quad SO(3)_+ && \text{fundamental 't Hooft line; no fundamental Wilson line,} \\ \{(0, 0), (1, 1)\} : & \quad SO(3)_- && \text{only the dyonic line carries nontrivial charge,} \end{aligned} \quad (\text{I.3.34})$$

— *three* distinct quantum field theories built on the single algebra $\mathfrak{su}(2)$, sharing every local correlator on $\mathbb{R}^4$, distinguished by which extended operators exist (Figure 5). The names record that the Wilson lines of $SU(2)$ are those of the simply connected group (all representations), while both $SO(3)$ theories admit only integer-spin Wilson lines, differing from each other in their magnetic sector. This is the resolution — in the affirmative, with an exact count — of chapter I.1’s Exercise I.1.20.

That the three theories differ is so far a statement about which correlators exist. The differences reach physics with no visible connection to global structure, along three channels that the rest of these notes will use repeatedly: phase diagnostics, the periodicity of θ , and compactification.

Phase diagnostics. In a confining $SU(2)$ vacuum the fundamental Wilson line shows an area law. In $SO(3)_+$ there *is* no fundamental Wilson line whose area law could be checked: the available diagnostics are the 't Hooft line and the dyon, and which of them shows perimeter law distinguishes vacua — e.g. the two vacua of $\mathcal{N} = 1$ supersymmetric $\mathfrak{su}(2)$ theory, exchanged by $\theta \rightarrow \theta + 2\pi$, are distinguished in $SO(3)_\pm$ by which line condenses, while in $SU(2)$ they look alike to every line [1305.0318](#). Phase classification is relative to the theory's line content: “confining” is not a property of a gauge algebra but of a theory, tested by the lines that theory possesses.

The range of θ . The θ -angle weights instanton sectors by $e^{i\theta\ell}$, and in $SU(2)$ the instanton number ℓ is an integer, making θ 2π -periodic. In the $SO(3)$ theories there are gauge bundles with $\ell \in \frac{1}{2}\mathbb{Z}$ (on spin manifolds), so naively θ has period 4π — but the Witten effect intervenes: shifting θ by 2π shifts the electric charge of every magnetic line by its magnetic charge,

$$\theta \rightarrow \theta + 2\pi : \quad (\lambda_e, \lambda_m) \longrightarrow (\lambda_e + \lambda_m, \lambda_m), \quad (\text{I.3.35})$$

which maps the set $SO(3)_+$ into the set $SO(3)_-$ and back: the two $SO(3)$ theories are exchanged, not preserved, by a 2π shift, and each is 4π -periodic on its own.⁴² The periodicity of a coupling constant — local-physics data, one would have said — is set by the line spectrum.

Compactification. Put the theories on $\mathbb{R}^3 \times S^1$. A line wrapping the S^1 becomes, from the three-dimensional viewpoint, a *local* operator, so theories differing in their lines differ in their 3d local operator content — and at that point not even local correlators agree anymore. The differences are quantitative and dramatic: the supersymmetric Witten index (the counting of vacua on a torus) of $\mathcal{N} = 1$ $\mathfrak{su}(2)$ theory differs between $SU(2)$ and the $SO(3)_\pm$ theories [1305.0318](#). Two theories indistinguishable by local experiments in infinite flat space are distinguished by local experiments in a periodic box. For a subject — lattice gauge theory, for instance — that lives in periodic boxes, the global form is not philosophy.

Idea I.3.2 (The line spectrum is part of the definition)

Specifying a gauge theory requires specifying its gauge algebra *and* its set of line operators — a maximal mutually local sublattice of electric–magnetic charges. Distinct maximal choices are distinct theories: identical on local correlators in $\mathbb{R}^d$, they differ in phase diagnostics, in the periodicity of parameters, and in their physics upon compactification. What was true for the Ising model (the dual operator lives at the end of a line the local data never sees) is true for Yang–Mills, with “which lines exist” promoted from a curiosity to a defining datum. Reading between the lines is reading the definition.

Exercise I.3.7 (classic). (The enumeration, and its $SU(N)$ generalization.) (a) Verify that (I.3.34) lists *all* subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2$ that are isotropic for the pairing (I.3.33) and maximal, and that maximality is doing work: exhibit a mutually local set that is consistent but incomplete, and the lines that complete it. (b) For algebra $\mathfrak{su}(N)$ with N prime, the classes live in $\mathbb{Z}_N \times \mathbb{Z}_N$ with pairing $z_e z'_m - z_m z'_e \bmod N$. Show the maximal isotropic subgroups are: the electric one $\{(z_e, 0)\}$ (theory $SU(N)$), and the N subgroups generated by $(k, 1)$ for $k = 0, \dots, N-1$ (theories $(SU(N)/\mathbb{Z}_N)_k$, permuted cyclically by $\theta \rightarrow \theta + 2\pi$) — $N+1$ theories per algebra, more than the two global forms of the group. (c) For non-prime N there are intermediate quotients and additional discrete labels; count the theories for $N = 4$.

⁴²The Witten effect — E. Witten, *Dyons of charge $e\theta/2\pi$* , Phys. Lett. B **86** (1979) 283 — states that in a θ -vacuum a monopole of magnetic charge m carries electric charge $q = n + \theta m/2\pi$: the θ -term, evaluated on a monopole background, acts as an electric source. Part V.3 derives it; here we use only the bookkeeping consequence (I.3.35).

Hint: For (b): an isotropic subgroup of order N (the maximal order, by a counting argument mirroring Lagrangian subspaces of a symplectic form) either contains $(1, 0)$ -type elements only, or contains an element with $z_m \neq 0$, which for prime N generates everything of the form (kz, z) . For (c): $SU(4)$, $SU(4)/\mathbb{Z}_2$ with two discrete θ -like choices, $SU(4)/\mathbb{Z}_4$ with four — see [1305.0318](#), §6, to check your count.

Answer: (a) $\{(0, 0)\}$ and any single nontrivial class form consistent but non-maximal sets; each completes uniquely to one of the three. (b) Isotropy of $\langle(k, 1), (k', 1)\rangle = k - k' \bmod N$ forces a single generator; $N + 1$ in all. (c) Seven: $1 + 2 + 4$.

Exercise I.3.8 (medium). (Wrapped lines as local order parameters.) On $\mathbb{R}^3 \times S^1_\beta$, the wrapped fundamental Wilson line of $SU(2)$ — the **Polyakov loop** $P(\mathbf{x}) = \text{tr } P \exp(i \oint_{S^1} A)$ — is a genuine local operator of the 3d theory. (a) Interpreting S^1_β as thermal time (chapter I.2's Exercise I.2.25), show that $\langle P(\mathbf{x}) \rangle = e^{-\beta F_q}$ with F_q the free-energy cost of a single static quark, so that $\langle P \rangle = 0$ diagnoses confinement and $\langle P \rangle \neq 0$ deconfinement at temperature $1/\beta$. (b) Identify the symmetry of the thermal $SU(2)$ theory that P is charged under — the center symmetry acting by a $\mathbb{Z}_2$ -valued aperiodic gauge transformation around the circle — and recast (a) as an ordinary Landau symmetry-breaking statement *for the wrapped line*: deconfinement is the spontaneous breaking of center symmetry, with P its local order parameter. (c) Explain why the $SO(3)_+$ theory, which lacks the fundamental Wilson line, lacks this symmetry and this order parameter — the thermal phase structure must be probed differently, e.g. by the wrapped 't Hooft line. (The symmetry of (b) is a one-form symmetry in four-dimensional disguise; Part II.4 will say this properly, and Part V.2 uses it.)

Hint: (a) is the standard thermal interpretation of a static source: the wrapped line inserts $e^{-\beta H}$ -evolution with one infinitely heavy charge added. (b) The aperiodic transformation multiplies P by -1 but preserves the action and all local operators.

Note I.3.1. The scope of these statements has two edges that should be marked. First, this section's lines were *probe* operators in theories whose dynamical fields are blind to the center; dynamical fundamental matter (as in QCD) makes the fundamental Wilson line screenable and breaks the area/perimeter dichotomy — phase diagnostics in such theories need the finer tools of Part II (in QCD proper, strictly speaking, only crossovers and symmetries remain). Second, the count (I.3.34) is for $\mathbb{R}^4$ or S^4 ; on manifolds with more topology the same theories require still more data (how to sum the $SO(3)$ bundles, the discrete θ -like parameters of [1305.0318](#) §§3, 6), foreshadowing the general principle of chapter I.4 that a theory is an assignment to *all* backgrounds at once.

The chapter's first two installments — Ising and Yang–Mills — carry the same moral with different machinery, and it is the machinery overlap that should be memorized: a disorder-type extended operator, a topological line/surface implementing a symmetry, a monodromy pairing constraining coexistence, condensation-versus-tension as the phase diagnostic. Surface operators in four dimensions (the boundaries of flipped-coupling membranes; Gukov–Witten operators [hep-th/0612073](#) in gauge theory), vortex lines in three-dimensional scalar theories, and the higher-form generalizations all run on these rails; Part II turns the rails into the subject.

I.3.4 Defects as probes: the kinematics of defect CFT

So far the chapter has treated extended operators as things to *define* and to *count*. This section treats them as things to *measure with*, in the setting where measurement is sharpest: a conformal

field theory hosting a scale-invariant defect. The motivation is the same as for studying CFTs at all. Just as a generic theory is an RG flow between fixed points (chapter I.1), a generic *defect* in a critical system — an impurity, a boundary, a wall — flows, as it is viewed from farther and farther away, to a scale-invariant defect in the CFT, and the fixed-point data organizes everything universal about the impurity. The Kondo problem is the classic instance: the screening of a magnetic impurity by conduction electrons is, in modern language, a defect RG flow between two defect fixed points of a free-fermion bulk. What follows is the kinematics of such fixed points: which correlators are fixed by symmetry, and which numbers — the **defect CFT data** — a solution of the defect must supply.⁴³

Conformal defects and their one-point functions

A flat p -dimensional defect $\mathcal{D}$ in $\mathbb{R}^d$ breaks the conformal group, and the definition of a conformal defect is fixed by asking how much of it can survive: a defect cannot preserve the translations and special conformal transformations that move it, but nothing forces it to break the transformations that map its worldvolume to itself. Demanding the maximum is the right definition for the same reason fixed points were the right landmarks in chapter I.1 — it is the symmetry situation that infrared limits of impurities produce.

Definition I.3.3 (Conformal defect)

A **conformal defect** of dimension p in a d -dimensional CFT is an extended operator on a flat $\Sigma_p \subset \mathbb{R}^d$ (or its conformal images) preserving

$$SO(d+1, 1) \longrightarrow SO(p+1, 1) \times SO(q), \quad q = d - p : \quad (\text{I.3.36})$$

the conformal group of its own worldvolume times the rotations around it. A **defect CFT** is a bulk CFT together with such a defect.

The first factor says the defect supports a p -dimensional conformal structure; the second says the transverse directions remember their rotational symmetry.

The first consequence of (I.3.36) repeats what Example I.3.1 found in the free theory: bulk operators acquire one-point functions. For a scalar bulk primary $\mathcal{O}$ of dimension Δ , invariance under defect translations and rotations forces $\langle \mathcal{O} \rangle_{\mathcal{D}}$ to depend only on the transverse distance $r = |x_{\perp}|$, and invariance under dilatations fixes the power:

$$\langle \mathcal{O}(x) \rangle_{\mathcal{D}} = \frac{a_{\mathcal{O}}}{r^{\Delta}}. \quad (\text{I.3.37})$$

(Invariance under the defect's special conformal transformations then holds automatically — Exercise I.3.9 — so (I.3.37) is the complete answer, not the leading term of one.) The numbers $a_{\mathcal{O}}$, one for each bulk primary, are the first sheet of defect CFT data. They are the response of the critical system to the impurity, they are measurable (in simulations and, for boundary cases, in experiments), and they are constrained but not determined by the bulk data $\{\Delta_i, C_{ijk}\}$. Spinning bulk operators can also acquire one-point functions, but only if their index structure can be built

⁴³The systematic treatment in general dimension and codimension is Billò–Gonçalves–Lauria–Meineri [1601.02883](#), which this section follows; boundaries were done first, by Cardy in 2d (Nucl. Phys. B **240** (1984) 514) and by McAvity–Osborn in general d (cond-mat/9505127). Northe's lectures [2411.03381](#), §§7–8, are the gentlest systematic entry.

from the available invariants; the stress tensor's is fixed up to a single coefficient, conventionally also called a_T , which controls the defect's gravitational dressing and enters the defect analogues of the c -theorems (Part IV).

The second consequence is a new operator spectrum. Local operators can live *on* the defect — the defect-localized excitations met on the lattice as modifications near the seam — and they form a p -dimensional conformal theory in their own right, with dimensions $\hat{\Delta}$ and OPE coefficients of their own, organized by the $SO(p+1, 1) \times SO(q)$ symmetry in which the transverse rotations $SO(q)$ act as an internal global symmetry. A bulk operator brought close to the defect expands in the defect spectrum,

$$\mathcal{O}(x_{\parallel}, x_{\perp}) \sim \sum_{\hat{\mathcal{O}}} \frac{b_{\mathcal{O}\hat{\mathcal{O}}}}{r^{\Delta-\hat{\Delta}}} \hat{\mathcal{O}}(x_{\parallel}), \quad r = |x_{\perp}| \rightarrow 0, \quad (\text{I.3.38})$$

the **bulk-to-defect OPE**, convergent within correlators for the same Hilbert-space reasons as the bulk OPE of chapter I.2 (cut along a cylinder around the defect instead of a sphere around a point; chapter I.4 supplies the geometry). The coefficient of the defect identity in (I.3.38) is the one-point coefficient, $b_{\mathcal{O}\hat{1}} = a_{\mathcal{O}}$. The defect spectrum $\{\hat{\Delta}\}$ with the couplings $\{b_{\mathcal{O}\hat{\mathcal{O}}}\}$ is the second sheet of defect data, and it is *new*: no theorem reconstructs it from the bulk CFT data, and the same bulk CFT can host many conformal defects of the same dimension (the Ising CFT's three boundary conditions, below, are the simplest exhibit).

One more piece of the spectrum is universal. A defect breaks the bulk translations transverse to it, and chapter I.2's logic — symmetries broken by a state or a background still constrain it through Ward identities — localizes the breaking in a defect operator. Performing a transverse translation $\delta x^i = \epsilon^i$ only near the defect, the action changes only by a term supported on the defect, and the change defines the **displacement operator** D^i : the stress-tensor Ward identity (I.2.69) acquires a defect-localized source,

$$\partial_{\mu} T^{\mu i}(x) = -\delta_{\mathcal{D}}^{(q)}(x_{\perp}) D^i(x_{\parallel}), \quad i \text{ transverse}, \quad (\text{I.3.39})$$

read as an operator equation inside correlators, exactly as (I.2.69) was. The identity fixes everything kinematic about D^i : it transforms as a vector of $SO(q)$, and its dimension is *exactly*

$$\hat{\Delta}_D = p + 1, \quad (\text{I.3.40})$$

protected for the same representation-theoretic reason that $\Delta_T = d$ and $\Delta_J = d - 1$ are (chapter I.2, Exercise I.2.22): the dimension is tied to the delta function's by the identity (I.3.39), which a conformal defect cannot deform away. The normalization of D^i is fixed by (I.3.39) too, so the coefficient of its two-point function,

$$\langle D^i(x_{\parallel}) D^j(0) \rangle_{\mathcal{D}} = C_D \frac{\delta^{ij}}{|x_{\parallel}|^{2(p+1)}}, \quad (\text{I.3.41})$$

is itself an observable: the defect's rigidity, the cost of bending it. A defect is *topological* precisely when $D^i = 0$ — deforming it costs nothing, which connects this section's kinematics back to the seams and charges of the chapter's first half: the Ising line defect $\tilde{\gamma}$ has vanishing displacement; the conformal defects of this section, generically, do not.

Example I.3.3 (The free-scalar line defect as a defect CFT)

Return to Example I.3.1 — the line $V_h = e^{h \int d\tau \phi}$ in the free massless scalar, $d = 4$, $p = 1$, $q = 3$ — and read off its defect CFT data. The defect coupling is exactly marginal (Exercise I.3.1),

so there is a one-parameter *family* of conformal line defects labeled by h , a “defect conformal manifold.” The one-point data: $a_\phi = h/4\pi$ from (I.3.4), and, by Wick’s theorem, $a_{\phi^2} = (h/4\pi)^2$ — in a Gaussian theory all one-point coefficients are powers of the one. The displacement operator can be written in closed form: shifting the line by $\delta x^i(\tau)$ changes the defect action by $h \int d\tau \partial_i \phi \delta x^i$, so

$$D^i(\tau) = h \partial_\perp^i \phi(\tau, \mathbf{0}), \quad (\text{I.3.42})$$

a defect-localized primary of dimension $2 = p + 1$, as (I.3.40) demands — in an interacting theory $\partial_\perp \phi$ would have an anomalous dimension, and the displacement identity is what forbids one here for this particular combination. Its norm follows from two derivatives of the propagator restricted to the line:

$$\langle D^i(\tau) D^j(0) \rangle = h^2 \partial_\perp^i \partial_\perp^j \frac{1}{4\pi^2 [(\tau - \tau')^2 + (x_\perp - x'_\perp)^2]} \Big|_{\perp=0} = \frac{h^2}{2\pi^2} \frac{\delta^{ij}}{\tau^4}, \quad (\text{I.3.43})$$

so $C_D = h^2/2\pi^2$: the rigidity of the defect grows with the coupling, vanishing smoothly as $h \rightarrow 0$ where the defect becomes trivial (hence topological in the trivial sense). Every number in this example is exact; the same list $(a_\phi, \hat{\Delta}, b_{\phi\phi}, C_D)$ is what one must compute, by bootstrap or ϵ -expansion or simulation, for interacting defects.

The interacting cousin of this example is one of the lighthouse examples of these notes. In the 3d Ising CFT the same construction — a magnetic field localized on a line, $e^{h \int d\tau \sigma}$ — has $\Delta_\sigma \approx 0.518 < 1 = p$, so the coupling is strongly relevant: the line accumulates magnetization, h grows toward the infrared, and the flow ends at a nontrivial conformal line defect, the **pinning defect**, with no tunable parameter left. Its data (a_ϵ , the defect spectrum, C_D) has been computed in the ϵ -expansion and at large N and measured on the lattice; it is a new solved corner of the 3d Ising universality class, invisible to the bulk bootstrap.⁴⁴ Experimentally, the pinning defect is a single frozen impurity in a critical uniaxial magnet — the kind of object condensed-matter physics has measured for decades, now organized by the same kinematics (I.3.37)–(I.3.41) as the Wilson lines of $\mathcal{N} = 4$ super Yang–Mills.

Exercise I.3.9 (easy). Derive (I.3.37): using the embedding of the defect conformal algebra in the bulk one (the generators preserved by a flat p -plane are: translations and Lorentz transformations along the plane, transverse rotations, dilatation, and the special conformal transformations with parameter along the plane), show that the one-point function of a scalar primary must take the stated form, and that a bulk primary with transverse spin (a nontrivial $SO(q)$ representation built only from transverse indices) can have a one-point function only if an invariant tensor is available — e.g. none for a transverse vector, one (δ^{ij} , plus x -dependent combinations) for a symmetric tensor like T^{ij} .

Hint: Dilatation plus parallel translations give $a_\phi r^{-\Delta}$; check parallel special conformal transformations explicitly, using $K_\parallel = IP_\parallel I$ with I the inversion, which maps the flat defect to itself.

Exercise I.3.10 (medium). A defect breaks enough symmetry that the two-point function of bulk primaries is no longer fixed: show that for two scalar bulk insertions in the presence of a flat p -dimensional conformal defect ($1 \leq q \leq d-1$), the surviving symmetry leaves exactly *two* independent

⁴⁴Cuomo–Komargodski–Mezei 2112.10634 construct and solve the pinning defect in the $O(N)$ models; the defect-bootstrap program around such systems is reviewed with the numerics in Part III.5’s reading. For this chapter the example’s role is conceptual: a one-line modification of a known CFT produces new universal data.

cross-ratios, so $\langle \mathcal{O}(x_1)\mathcal{O}(x_2) \rangle_{\mathcal{D}} = (\text{prefactor}) \times f(u, v)$ with f undetermined by symmetry. Count: $2d$ coordinates, minus the dimension of $SO(p+1, 1) \times SO(q)$ orbits. The function f admits two expansions — bulk OPE then one-point functions (I.3.37), or two bulk-to-defect OPEs (I.3.38) — whose equality is the **defect crossing equation**, the engine of the defect bootstrap (Part III.5).

Hint: Convenient invariants: with both points at transverse distances r_1, r_2 and transverse angle θ , take $u = \frac{x_{12}^2}{r_1 r_2}$ and $v = \cos \theta$. For $q = 1$ (a boundary) there is no angle and only one cross-ratio survives.

Boundaries: the richest defects

Among all defects, the codimension-one ones with nothing on the other side — **boundaries** — carry the most structure, for a reason visible already in the counting of Exercise I.3.10: the lower the codimension, the more of the bulk symmetry the defect can absorb into its own worldvolume, and at $q = 1$ the defect worldvolume supports a $(d-1)$ -dimensional conformal theory coupled to the bulk through a single transverse direction. The boundary version of the defect data — one-point functions $\langle \mathcal{O} \rangle_{\alpha} = a_{\mathcal{O}}^{\alpha} / (2y)^{\Delta}$ at distance y from a boundary condition α , boundary-localized operators, the bulk-to-boundary OPE — was worked out by Cardy and others in the 1980s, decades before the general- q kinematics, under the name **boundary conformal field theory** (BCFT); the modern defect technology is in large part that theory's generalization.

Two dimensions is where boundaries are completely understood, and the structure found there is the template for chapter I.4. In a 2d CFT on the upper half plane, a conformal boundary condition must conserve energy at the wall: no flux of the stress tensor through the real axis, which in the complex coordinates of the plane reads

$$T(z) = \bar{T}(\bar{z}) \quad \text{at } z = \bar{z}. \quad (\text{I.3.44})$$

This **gluing condition** halves the symmetry — of the two Virasoro algebras of Part III.4's 2d CFTs, only the diagonal survives, which is the $q = 1$ case of (I.3.36) — and its solutions, the conformal boundary conditions, turn out to be rigidly classified objects. For the Ising CFT there are exactly *three*:⁴⁵

$$\alpha \in \{+, -, f\} : \quad \langle \sigma \rangle_{\pm} = \pm \frac{A}{(2y)^{1/8}}, \quad \langle \sigma \rangle_f = 0, \quad \langle \epsilon \rangle_{\pm} = + \frac{B}{(2y)}, \quad \langle \epsilon \rangle_f = - \frac{B}{(2y)}, \quad (\text{I.3.45})$$

with A, B fixed positive constants — and they are exactly the continuum limits of the three things one can do to Ising spins at the edge of a lattice: freeze them up, freeze them down, or set them free. The matching of an abstract classification to an exhaustive list of microscopic boundary manipulations is the boundary version of universality, and it is this rigidity that makes boundaries “richest”: the boundary conditions of a rational CFT are a finite, fusion-closed set, in canonical bijection with the bulk primaries, and they generate, through their boundary-condition-changing operators and annulus amplitudes, a second solvable algebra on top of the bulk OPE. Boundary flows are organized by their own monotonicity theorem (the g -theorem: the boundary entropy g , the universal constant in the annulus partition function, decreases along boundary RG flows — e.g. a small boundary magnetic field on the f boundary drives a flow $f \rightarrow +$, with $g_f = 1 > g_+ = 1/\sqrt{2}$), so

⁴⁵J. L. Cardy, *Boundary conditions, fusion rules and the Verlinde formula*, Nucl. Phys. B **324** (1989) 581 — the classification of boundary conditions in rational CFTs and the discovery that they are counted by, and fuse like, the chiral primaries. We cite, not derive: the derivation needs the torus/annulus technology of Part III.4 and is given there; Northe [2411.03381](#), §7.6, has it in full pedagogical detail.

the boundary sector has its own theory space, its own landmarks, and its own geography, mirroring chapter I.1's picture one dimension down.⁴⁶

For this chapter the structural point outranks the formulas: a boundary condition is not auxiliary data needed to “set up” a theory on a half space — it is an extended operator, with a spectrum, couplings, and RG flows of its own, subject to classification exactly as local operators are. When chapter I.4 sets up the cutting-and-gluing formalism, boundary conditions will reappear as the *states* the formalism assigns to cut edges, and the Ising triple (I.3.45) will become a basis of a three-dimensional state space.

Exercise I.3.11 (classic). (Free-field boundaries by images.) For the free massless scalar on the half space $y \geq 0$ in $d = 4$ with Neumann ($\partial_y \phi|_0 = 0$) or Dirichlet ($\phi|_0 = 0$) boundary conditions, the propagator is the flat-space one plus or minus an image term,

$$\langle \phi(x)\phi(x') \rangle_{N/D} = \frac{1}{4\pi^2} \left[\frac{1}{(x-x')^2} \pm \frac{1}{(x-\bar{x}')^2} \right],$$

with $\bar{x}'$ the mirror image. (a) Verify the boundary conditions and compute the one-point function of $:\phi^2:$ (defined by chapter I.2's point-splitting (I.2.3)): $a_{:\phi^2:}^{N/D} = \pm \frac{1}{16\pi^2}$ in the conventions $\langle :\phi^2: \rangle = a/(2y)^2$. The same bulk theory supports boundaries with one-point data of either sign — data the bulk does not know. (b) Identify the boundary operator spectra: the boundary values of ϕ (Neumann) versus $\partial_y \phi$ (Dirichlet), with $\hat{\Delta} = 1$ and 2 respectively. (c) Integrate the Ward identity (I.3.39) across the wall to show that the displacement of a boundary is the boundary value of the normal-normal component of the stress tensor, $D = T^{yy}|_{y=0}$, whose dimension $p+1 = d$ is then automatic; evaluate it in the Dirichlet theory, where $T^{yy}|_{y=0} = \frac{1}{2}:\partial_y \phi^2:$ has dimension 4, as (I.3.40) requires.

Hint: For (a), point-split along the boundary direction so the image term contributes its coincident-point value $\pm 1/(4\pi^2(2y)^2)$. For (c), only the $(\partial_y \phi)^2$ term of T^{yy} survives at a Dirichlet wall, since $\phi| = 0$ kills the parallel derivatives there.

Answer: (a) The direct term cancels against the normal-ordering subtraction, leaving exactly the image contribution: $a = \pm 1/16\pi^2$.

I.3.5 Defect Hilbert spaces

One entry of the chapter's question box remains: which Hilbert spaces do defects create? The answer completes the chapter's accounting and hands the baton to chapter I.4, and it begins with the rotation flagged in the opening: in Euclidean signature, an extended operator can be positioned at a fixed time — where it acts as an *operator* on the ordinary Hilbert space — or extended *along* the time direction, where it is not an operator on anything but a modification of the arena: a static impurity threading all of space-time history. In the second frame the natural objects are the energy eigenstates of the modified system: the theory with a defect crossing the spatial slice has its own space of states, the **defect Hilbert space** $\mathcal{H}_{\mathcal{D}}$ (for symmetry defects, also called the **twisted sector**).

The lattice makes this concrete with no formalism at all. Take the transverse-field Ising chain of Exercise I.3.5 and thread the $\mathbb{Z}_2$ seam through a fixed spatial position — say the bond $(N, 1)$ — for all time. In the classical 2d picture this is a column of flipped vertical bonds; in the Hamiltonian

⁴⁶Affleck–Ludwig, Phys. Rev. Lett. **67** (1991) 161, for g and the flow statement. Everything in this paragraph is developed properly in Part III.5; the g -theorem's entropic proof belongs to Part X's methods.

picture it is a flipped sign on one term,

$$H_\eta = -g \sum_{j=1}^N X_j - \sum_{j=1}^{N-1} Z_j Z_{j+1} + Z_N Z_1 : \quad (\text{I.3.46})$$

the chain with antiperiodic boundary conditions for the spins. The defect Hilbert space is the same 2^N -dimensional space, but “twisted” names the right concept: what is defined intrinsically is the pair (space, Hamiltonian), and H_η is not unitarily equivalent to H by any *local* change of variables near the seam — conjugating by X_k moves the flipped bond by one site (Exercise I.3.12), so the seam’s *position* is unphysical, in keeping with its topological nature, but removing it entirely requires the global flip $\prod X_j$, which is the symmetry operator itself, and the seam stays. Twisted boundary conditions are a distinct superselected arena, with its own spectrum: the lattice realization of a line defect wrapping a spatial circle.

The continuum counterpart assigns the twisted sector an operator meaning through the state–operator correspondence, and this is where the chapter’s non-genuine operators find their Hilbert-space home. Chapter I.2’s radial quantization mapped local operators at the origin to states on a sphere surrounding it. Now place at the origin the *endpoint* of a line defect — a disorder operator μ — with the line running off to infinity. Every sphere around the origin is punctured once by the line: the states live on a sphere *with a defect point marked*, i.e. in the defect Hilbert space of the line. Conversely every state of the twisted sector maps to an operator at the end of the line:

$$\text{genuine local operators} \leftrightarrow \text{states in } \mathcal{H}_{\mathcal{S}^{d-1}}, \quad \text{end-of-line operators} \leftrightarrow \text{states in } \mathcal{H}_{\mathcal{D}}. \quad (\text{I.3.47})$$

The disorder operator μ , inadmissible as a state of the ordinary Ising Hilbert space (its seam must puncture any surrounding circle), is a perfectly good state — the ground state, it turns out — of the η -twisted sector. For the critical Ising model the full content of $\mathcal{H}_\eta$ is known exactly (Part III.4 derives it from modular covariance of the torus partition function; Shao 2308.00747, §3.1, tabulates it): the conformal primaries are

$$\mathcal{H}_\eta : \quad \mu \text{ with } (h, \bar{h}) = \left(\frac{1}{16}, \frac{1}{16}\right), \quad \psi, \bar{\psi} \text{ with } \left(\frac{1}{2}, 0\right), \left(0, \frac{1}{2}\right). \quad (\text{I.3.48})$$

The twisted sector contains exactly the non-genuine operators the lattice construction produced — the disorder operator, at the duality-mandated dimension $\frac{1}{8} = \frac{1}{16} + \frac{1}{16}$, and the Kadanoff–Ceva fermions, whose half-integer Lorentz spin $|h - \bar{h}| = \frac{1}{2}$ is *allowed* here precisely because the cylinder identification that forces integer spin on genuine operators is twisted by η : the monodromy minus sign of (I.3.20)/(I.3.21), recast as a spin selection rule for the sector. Nothing about the list (I.3.48) is deducible from the genuine operator content $(1, \epsilon, \sigma)$ without the defect: the twisted Hilbert space is new data, the Hilbert-space face of the chapter’s thesis.

Intuition I.3.2 (Defect Hilbert spaces complete the operator–state dictionary)

Operators attached to a line are states on a sphere punctured by the line. The dictionary local operators $\leftrightarrow$ sphere states (chapter I.2) was the $\mathcal{D} = \emptyset$ row of a table whose other rows are indexed by the theory’s defects; each defect $\mathcal{D}$ contributes a sector $\mathcal{H}_{\mathcal{D}}$, with its own spectrum, its own spin selection rules, and — on the torus and in finite volume — its own contribution to the counting of states. A theory’s Hilbert-space content, like its operator content, is not exhausted until the defects are in.

Counting is in fact where defect Hilbert spaces first force themselves on a calculation. The torus partition function with the seam wrapping a *spatial* cycle is a twisted trace — insert the symmetry operator η , since crossing the seam in time means applying the symmetry — while wrapping the seam around the *time* cycle computes the untwisted trace over the twisted sector:

$$Z[\eta \text{ in time}] = \text{tr}_{\mathcal{H}} \left(\eta e^{-\beta H} \right), \quad Z[\eta \text{ in space}] = \text{tr}_{\mathcal{H}_{\eta}} e^{-\beta H_{\eta}}, \quad (\text{I.3.49})$$

and the two configurations are related by the Euclidean rotation exchanging the cycles. Consistency of this exchange — one set of numbers computed two ways — is a strong constraint tying the twisted spectra to the action of symmetries on the untwisted one; in two dimensions, where the exchange is the modular transformation of Part III.4, it determines lists like (I.3.48) outright. The reader has seen this engine before: chapter I.1’s particle on a circle equated a Hamiltonian trace to a winding sum (Example I.1.2) — the same identity, with the seam’s winding sectors in place of the particle’s. And the gauge-theory version closed Section I.3.3: on $\mathbb{R}^3 \times S^1$ the theories with different line spectra developed different local physics because wrapped lines contribute states/operators that unwrapped presentations never see. Defect sectors are not exotica; they are where a finite-volume system stores a sizable fraction of its information.

Exercise I.3.12 (easy). In the twisted chain (I.3.46), conjugate H_{η} by the single-site flip X_1 and show the antiferromagnetic bond moves from $(N, 1)$ to $(1, 2)$; iterate to move it anywhere. Conclude that the spectrum of H_{η} is independent of where the seam sits (the lattice form of “the line is topological”), and show that conjugating by the *full* string $X_1 X_2 \cdots X_N = \eta$ returns the seam to its starting bond — consistently, since η commutes with H_{η} . Why does no product of *fewer* than N flips remove the seam?

Hint: Count antiferromagnetic bonds mod 2: each X_k toggles the sign of the two bonds adjacent to site k , so the parity of flipped bonds around the circle is invariant — a $\mathbb{Z}_2$ flux through the ring that local moves cannot change.

Exercise I.3.13 (thinker). (The twisted Ising chain, solved, and matched to (I.3.48).) The critical chain $g = 1$ is solvable by the Jordan–Wigner transformation of Exercise I.3.5(e), in both the untwisted and twisted sectors. (a) Show that the $\mathbb{Z}_2$ twist of the spins maps, through the string, to a change of fermion boundary conditions, and that the combined bookkeeping (fermion parity sectors versus periodic/antiperiodic fermions) gives four sector pairings, of which the spin chain uses two in $\mathcal{H}$ and two in $\mathcal{H}_{\eta}$. (b) Diagonalize the critical fermion chain with both boundary conditions; using as given the cylinder dictionary $E = \frac{2\pi}{N} \left(h + \bar{h} - \frac{c}{24} - \frac{\bar{c}}{24} \right) + N\epsilon_0$ (derived in chapter I.4; $c = \bar{c} = \frac{1}{2}$ here), match the lowest twisted-sector levels to $(h, \bar{h}) = (\frac{1}{16}, \frac{1}{16})$, $(\frac{1}{2}, 0)$, and $(0, \frac{1}{2})$. (c) Locate the momentum quantum numbers and verify the half-integer Lorentz spin of the ψ states. This is (I.3.48), derived from 2×2 matrices.

Hint: Antiperiodic fermions have modes at $k \in \frac{2\pi}{N}(\mathbb{Z} + \frac{1}{2})$, periodic at $k \in \frac{2\pi}{N}\mathbb{Z}$ including the zero mode whose occupation splits degenerate pairs. The $\frac{1}{16}$ arises as the Casimir-energy difference between the two mode quantizations; the matching numerics are unambiguous already at modest N if you prefer to check by computer. Seiberg–Shao 2307.02534, §§2–3, performs closely related bookkeeping with all the global subtleties displayed.

I.3.6 Where this leaves us

The thesis was that the definition of a theory includes its extended operators, and the chapter has now seen it confirmed at every level of the inventory at once. At the level of operators: the Ising model carries, beyond its local spins, a topological line whose endpoints realize the dual model’s order parameter — an object constructed, proved topological, and dualized in three lattice

computations — and Yang–Mills theory with a fixed algebra and fixed local correlators splits into several theories according to which electric–magnetic line lattice it admits. At the level of data: a conformal defect carries one-point coefficients, a defect-operator spectrum, and a displacement norm, none of it reconstructible from bulk CFT data, all of it physical down to the measured properties of a pinned impurity in a critical magnet. At the level of states: each defect contributes a twisted Hilbert space, housing exactly the operators (disorder fields, fermions from bosons) that the genuine-operator dictionary excludes, and contributing inescapably to finite-volume counting. The phrase “part of the definition” is thus meant three ways at once — operators, data, states — and in each the extended sector is independent input, constrained by mutual locality and maximality but not derived from anything local.

Two of the chapter’s running characters are about to be promoted. The topological seam — a symmetry, realized as a deformable extended operator — is the protagonist of Part II, where “symmetries are topological operators” becomes the definition and everything in this chapter (monodromies, attached endpoints, twisted sectors, the charge lattices of gauge theory) reappears as representation theory of those operators; the reader who wants the immediate continuation should step from here directly to Part II.1. And the assignment of Hilbert spaces to geometries-with-defects — spheres with marked points, circles with seams — is one more instance of the pattern that chapter I.4 takes as primary: a quantum field theory assigns data to geometric objects, and cutting and gluing is the organizing rule. Boundaries, which this chapter treated as the richest defects, will return there in their other role, as the states assigned to cuts. The inventory of chapter I.1 has one column left to formalize, and it is geometry’s.

Chapter exercises

Exercise I.3.14 (classic). (The disorder operator one dimension up: Wegner’s duality.) In the 3d Ising model, the construction of Section I.3.2 produces a disorder *line*: flip the couplings on all bonds pierced by a membrane $\tilde{\Sigma}$ of dual plaquettes, with the observable depending only on $\partial\tilde{\Sigma}$. (a) Run the path-independence argument: deforming the membrane at fixed boundary is a change of summation variables. (b) Run the low-temperature expansion in the presence of the membrane and show that the dual description is a $\mathbb{Z}_2$ gauge theory on the dual lattice (domain walls become the gauge theory’s plaquette excitations) — Wegner’s duality, the first appearance of a gauge theory as a dual. (c) Identify the image of the disorder line in the gauge theory: the Wilson loop of the $\mathbb{Z}_2$ gauge field. The order/disorder pair of the 3d Ising model is the (matter, Wilson-loop) pair of $\mathbb{Z}_2$ gauge theory, and the confinement/deconfinement transition of the latter is the ordinary magnetic transition of the former, read through duality. (Part II.2 constructs $\mathbb{Z}_2$ gauge theory in its own right; Part V.2 uses this dictionary for the phases of gauge theories.)

Hint: F. J. Wegner, J. Math. Phys. **12** (1971) 2259. For (b): low-temperature configurations of the 3d Ising model are labeled by closed domain-wall *surfaces* on the dual lattice; match them to the closed-surface (zero-flux) configurations of dual-lattice $\mathbb{Z}_2$ plaquette variables, with e^{-2J} per unit area mapping to the dual gauge coupling exactly as in (I.3.11).

Exercise I.3.15 (medium). (Interfaces, and defects between presentations.) An **interface** is a codimension-one defect with possibly different theories on the two sides; a boundary is an interface with the empty theory on one side. (a) Show that the folding trick — reflect one side through the wall — converts an interface between theories $\mathcal{T}_1$ and $\mathcal{T}_2$ into a boundary condition of the product theory $\mathcal{T}_1 \times \mathcal{T}_2$, so the classification of interfaces is a special case of the classification of boundaries (one reason boundaries are “richest”). (b) For the free massless scalar in 2d, construct

the one-parameter family of conformal interfaces gluing $\partial\phi_1 = \lambda \partial\phi_2$ (with the dual condition on the other chirality), check energy conservation across the wall, and identify the two degenerate limits as the Dirichlet and Neumann boundaries of the doubled theory. (c) A *topological* interface between a theory and itself implements a symmetry (Section I.3.4: $D^i = 0$ means freely deformable); which λ in (b) gives a topological interface, and which symmetry of the compact boson does it implement?

Hint: (b) Energy conservation is $T_1 - \bar{T}_1 = T_2 - \bar{T}_2$ at the wall. (c) $\lambda = \pm 1$; the sign choice is the boson's $\mathbb{Z}_2$ reflection. Conformal interfaces of $c = 1$ theories are an exactly understood zoo, and Part VI.4 uses them to *generate* dualities.

Exercise I.3.16 (medium). (The vortex line as a disorder operator.) For the compact boson in $d = 3$ — field $\theta \simeq \theta + 2\pi$, action $\frac{K}{2} \int (\partial\theta)^2$ — define the vortex line $V(\Sigma_1)$ by the winding prescription (I.3.5). (a) Solve for the minimal-action configuration around a straight vortex line and show its action diverges logarithmically in both the infrared and ultraviolet — the famous vortex logarithms — so that $\langle V \rangle$ vanishes on $\mathbb{R}^3$ but vortex *pairs* have power-law correlators governed by K . (b) Show the monodromy of the charge- n vertex operator $e^{in\theta}$ around V is $e^{2\pi inw}$ — trivial, consistently with both being genuine in this theory — and contrast with the half-charge operator $e^{i\theta/2}$, which is not a legal operator of the periodic theory precisely because its monodromy around the unit vortex would be -1 . Periodicity of the target, the spectrum of allowed vertex operators, and the quantization of vortex winding form one consistency loop: mutual locality once again decides the operator content.

Hint: (a) $\theta = w\varphi$ (the transverse angle) solves the equation of motion; $S = \pi K w^2 \int dz \int_a^L dr/r$ per unit length. (b) is a direct evaluation of $e^{in\theta}$'s phase as θ winds.

Exercise I.3.17 (thinker). (Extended operators under duality: a checklist revisited.) Chapter I.1's Exercise I.1.4 asked what one could compare across a proposed duality without solving either side; this chapter has substantially enlarged the list. For the particle–vortex duality of chapter I.1 (the $O(2)$ Wilson–Fisher point as $|\phi|^4$ versus the abelian Higgs model), match as much of the extended-operator content as you can, qualitatively: what does the vortex line of one description map to? The Wilson line of the dynamical gauge field? The boundary conditions? Which monodromy pairings must agree, and what does agreement test that the local-operator dictionary does not? Write your answers as conjectures with stated confidence — this is how duality checks are assembled in practice — and keep the list for Part VI, which grades it.

Hint: The conserved-current matching $j^\mu \leftrightarrow \epsilon^{\mu\nu\rho} \partial_\nu a_\rho / 2\pi$ is the anchor: charge of one side = vorticity of the other. Whatever ends a line on one side must map to something ending the image line on the other — endability is duality-invariant data (it returns as such in Part II.4).

Exercise I.3.18 (thinker). (How much theory space does a defect carry?) Example I.3.3 found a line of defect fixed points; the pinning defect found an isolated one. Formulate, by analogy with chapter I.1's theory-space picture, the space of *defects in a fixed bulk CFT*: points (conformal defects), flows (defect RG), landmarks (the trivial defect; the “factorized” defects that cut the theory in two). What plays the role of the c -theorem? Look up nothing; reason from the structures in this chapter — then compare your guesses against the known answers (the g -theorem in $d = 2$; the defect entropy/ b -theorem family in higher d , Part IV's methods applied one codimension down) when they arrive in Parts III.5 and IV.

Reading

Aharony–Seiberg–Tachikawa, 1305.0318, §1. The chapter's gauge-theory spine: line operators, mutual locality, the three $\mathfrak{su}(2)$ theories, the consequences for θ and for compactification.

The introduction is self-contained and readable now; §§2–6 generalize to arbitrary groups and add the discrete θ -parameters.

Gaiotto–Kapustin–Seiberg–Willett, [1412.5148](#), §§1–2. The founding paper of generalized symmetries, cited here for its preliminaries: the operator/defect dictionary, genuine versus attached operators, and charges as topological operators — the bridge from this chapter to Part II. (Kapustin–Seiberg [1401.0740](#), §2, introduced the genuine/non-genuine distinction.)

Shao, *TASI lectures*, [2308.00747](#), §3. The Ising model’s lines, endpoints, and twisted sectors in continuum language, including the non-invertible duality line previewed here; the companion lattice constructions are in §3.3. Seiberg–Shao [2307.02534](#) does the chain with complete care about the global issues (and is where “the lattice $\mathbb{Z}_2$ seam” meets precise continuum statements).

Billò–Gonçalves–Lauria–Meineri, [1601.02883](#). Defect CFT kinematics in general dimension: one-point functions, bulk-to-defect OPE, and (§5) the displacement operator and its Ward identities. Sections 1–2 and 5 correspond to this chapter’s Section I.3.4.

Northe, [2411.03381](#), §§7–8. BCFT and topological defects in 2d, with exercises and solutions: the gluing condition, boundary states, the Cardy classification, Verlinde lines. The natural bridge from this chapter to Parts III.4–III.5.

Classics. Kramers–Wannier, Phys. Rev. **60** (1941) 252 (the duality); Kadanoff–Ceva, Phys. Rev. B **3** (1971) 3918 (disorder operators and the fermion); Wegner, J. Math. Phys. **12** (1971) 2259 ($\mathbb{Z}_2$ gauge theory and its duality); Wilson, Phys. Rev. D **10** (1974) 2445 (the loop and the area law); ’t Hooft, Nucl. Phys. B **138** (1978) 1 (disorder lines in gauge theory and the phase classification). Each is short, physical, and the original source of one section of this chapter.

Fradkin, [1610.05780](#). *Disorder operators and their descendants*: a modern review of the disorder-operator idea across dimensions and models, from Ising to dualities of 2+1d gauge theories; generous with the condensed-matter realizations this chapter only gestured at.

Books (for the shelf): Recknagel–Schomerus, *Boundary Conformal Field Theory and the World-sheet Approach to D-Branes* — BCFT done completely, with the string-theory payoff; the standard reference for the boundary-state formalism that Part III.5 will use.

A Conventions

Units and signature. We set $\hbar = c = 1$ throughout. The Lorentzian metric signature is **mostly-plus**, $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ (and its obvious truncations in other dimensions). Consequently a free scalar has

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2, \quad p^2 = p_\mu p^\mu = -m^2 \text{ on shell.} \quad (\text{A.1})$$

Srednicki and Weinberg use this signature; Peskin–Schroeder does not. Much of the symmetry and CFT literature works directly in Euclidean signature, where the issue is moot.

Wick rotation. Euclidean time is $t = -i\tau$ with τ real, so that $e^{iS_L} = e^{-S_E}$ and a standard kinetic term rotates as

$$i \int dt \frac{1}{2} \dot{q}^2 \longrightarrow - \int d\tau \frac{1}{2} \left(\frac{dq}{d\tau} \right)^2, \quad (\text{A.2})$$

i.e. S_E is positive for a standard kinetic term. Thermal physics at temperature $T = 1/\beta$ lives on a Euclidean time circle of circumference β : $Z(\beta) = \text{tr } e^{-\beta H}$.

Topological terms stay imaginary. A term in the action with no dependence on the metric acquires no factor of i under Wick rotation: it enters the Euclidean weight as a pure phase. For the θ -term of quantum mechanics on a circle (chapter I.1),

$$i \frac{\theta}{2\pi} \int dt \dot{q} \longrightarrow i \frac{\theta}{2\pi} \oint dq = i\theta w, \quad w \in \mathbb{Z}, \quad (\text{A.3})$$

and the Euclidean weight is $e^{-S_E + i\theta w}$. Forgetting this i (i.e. writing $e^{-\theta w}$) is a classic error; the same rule governs $\theta F\tilde{F}$ in 4d and Chern–Simons terms in 3d.

Euclidean operators and conjugation (from chapter I.2). Euclidean Heisenberg evolution is $\mathcal{O}(\tau, \mathbf{x}) = e^{H\tau} \mathcal{O}(0, \mathbf{x}) e^{-H\tau}$, so a Lorentzian-Hermitian scalar satisfies $\mathcal{O}(\tau, \mathbf{x})^\dagger = \mathcal{O}(-\tau, \mathbf{x})$: Euclidean conjugation reflects through the quantization surface. Operators with spin additionally pick up a factor $-i$ per Lorentzian time index under Wick rotation ($\mathcal{O}_E^{0\dots} = -i \mathcal{O}_L^{0\dots}$, etc.), so that Euclidean correlators are $SO(d)$ -covariant; conjugation then carries a time-reflection matrix per index. See Simmons-Duffin [1602.07982](#), §7.1.

Radial quantization conventions (from chapter I.2). Following Simmons-Duffin [1602.07982](#): Euclidean signature; the dilatation generator D is Hermitian with $[D, \mathcal{O}(0)] = \Delta \mathcal{O}(0)$, $[P_\mu, \mathcal{O}(x)] = \partial_\mu \mathcal{O}(x)$, and $[K_\mu, \mathcal{O}(0)] = 0$ for a primary; algebra $[D, P_\mu] = P_\mu$, $[D, K_\mu] = -K_\mu$, $[K_\mu, P_\nu] = 2\delta_{\mu\nu} D - 2M_{\mu\nu}$. Conjugation in radial quantization is inversion: $\mathcal{O}(x)^\dagger = x^{-2\Delta} \mathcal{O}(x^\mu/x^2)$ for a Hermitian scalar, whence $P_\mu^\dagger = K_\mu$, $D^\dagger = D$.

Stress tensor (from chapter I.2). Euclidean: $T_{\mu\nu} = \frac{2}{\sqrt{g}} \frac{\delta S_E}{\delta g^{\mu\nu}}$; Lorentzian (mostly-plus): $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}}$, giving $T_{00} \geq 0$ for the standard scalar. Ward identity with our signs:

$$\partial_\mu \langle T^{\mu\nu}(x) \mathcal{O}_1(x_1) \cdots \rangle = - \sum_i \delta^d(x - x_i) \partial_{x_i}^\nu \langle \mathcal{O}_1(x_1) \cdots \rangle. \quad (\text{A.4})$$

Improvements $T_{\mu\nu} \rightarrow T_{\mu\nu} + \xi(\delta_{\mu\nu} \partial^2 - \partial_\mu \partial_\nu) \mathcal{O}$ are identically conserved and change no charges; “traceless” always means “for the improved representative.”

Symmetry and CFT conventions. When the chapters reach the symmetry material (Part II) we follow the conventions of Shao's TASI lectures [2308.00747](#); for CFT material (Part III), Simmons-Duffin [1602.07982](#). Conventions for p -form gauge fields, flux normalization, and Dirac matrices will be pinned here when first needed.